

Title Page

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Protocol Title:

A Phase 3, Randomized, Multicenter, Double-Blind Study to Investigate the Efficacy and Safety of Retatrutide Once Weekly Compared with Placebo in Adult Participants with Type 2 Diabetes and Inadequate Glycemic Control with Diet and Exercise Alone (TRANSCEND-T2D-1)

Protocol Number: J1I-MC-GZBY

Amendment Number: a

Compound: Retatrutide (LY3437943)

Brief Title:

Effect of Retatrutide Compared with Placebo in Adult Participants with Type 2 Diabetes and Inadequate Glycemic Control with Diet and Exercise Alone (TRANSCEND-T2D-1)

Study Phase: 3

Acronym: TRANSCEND-T2D-1

Sponsor Name: Eli Lilly and Company

Legal Registered Address: Indianapolis, Indiana, USA 46285

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Approval Date: Protocol Amendment (a) Electronically Signed and Approved by Lilly on date provided below.

Document ID: VV-CLIN-133434

Medical Monitor Name and Contact Information will be provided separately.

Protocol Amendment Summary of Changes Table

DOCUMENT HISTORY	
Document	Date
Original Protocol	29-Nov-2023

Amendment [a]**Overall rationale for the amendment**

The overall rationale for the changes implemented in the amendment is to

- add new AESI of “malnutrition and inadequate fluid intake” and update the title and text for AESI of “ketoacidosis due to inadequate nutritional intake and other types of metabolic acidosis and ketosis”
- provide more detailed guidance on recommending, monitoring, and managing nutritional intake and hydration
- update criteria for dose modification, and
- include adjudication of urgent visit for heart failure.

The changes are included in the table below.

Minor editorial or formatting changes are not included in this table.

Section # and Name	Description of Change	Brief Rationale
1.1. Synopsis 3. Objectives, Endpoints, and Estimands	Updated the wording for the Additional Secondary Objective from “relationships between retatrutide and safety, tolerability, and efficacy measures” to “relationships between retatrutide <u>exposure</u> and safety, tolerability, and efficacy measures”.	Clarification
1.1. Synopsis 3. Objectives, Endpoints, and Estimands	Additional note added to clarify the intended purpose of each estimand.	Clarification
	Added a clarification to the definition of “population” for treatment regimen and efficacy estimand attributes.	
1.3. Schedule of Activities	Added language for monitoring and routinely assessing the need to adjust or discontinue antihypertensive medications, diuretics, laxatives, and anti-emetics for concomitant medications.	For consistency with recommendations in Sections 6.9.1 and 10.10
	Added language for monitoring for and encouraging adequate nutritional intake and hydration throughout the study as part of the participant education.	For consistency with recommendations in Sections 5.3.3 and 5.3.4
	Added additional evaluations for “Lifestyle considerations, diabetes education (including BG meter and SMBG training), and meals and nutrition” to all visits, expanding beyond the existing evaluation at Visit 2.	To incorporate additional counseling and monitoring for patient safety
	Added an additional evaluation of antihypertensive and lipid-lowering medications at ED visit.	To assess the “change in antihypertensive and lipid-lowering medications” endpoint at ED visit
5.1. Inclusion Criteria	Added a note in Inclusion Criterion 8 to address the usage of gender-neutral terms (AMAB and AFAB) in the document.	For clarification and consistency with TRANSCEND T2D protocols
5.2. Exclusion Criteria	Removed the word “clinically significant” from Exclusion Criterion 22.	Clarification to avoid ambiguity

Section # and Name	Description of Change	Brief Rationale
	Defined study intervention in Exclusion Criterion 36 as retatrutide, placebo, or an active comparator.	Clarification
5.2. Exclusion Criteria 8.3.3.8. Thyroid Malignancies and C-Cell Hyperplasia	Updated calcitonin units from “ng/L” to “pg/mL (Conventional) and pmol/L (SI)” in Exclusion Criterion 24 and in Section 8.3.3.8.	To align with central lab reporting
5.3.3 Meals and Nutrition 5.3.4. Monitoring Nutritional Needs	Updated the section title from “Meals and Dietary Restrictions” to “Meals and Nutrition”. Added more detailed guidance on monitoring and managing nutritional and hydration status throughout the study.	To facilitate monitoring and management of participants with or at risk of poor nutrition or dehydration
6.6. Dose Modification	Added the word “in a blinded fashion” for all dose adjustments managed through IWRS.	Clarification
	Added the option of dose modification for participants with poor nutrition or dehydration.	To facilitate the management of participants with or at risk of poor nutrition or dehydration
6.6.2. Management of Participants with Poor Nutrition or Dehydration 10.11. Appendix 11: Management of Poor Nutrition and Dehydration	Added new section with accompanying flow chart in Appendix.	To facilitate the management of participants with or at risk of poor nutrition or dehydration
6.6.3. Management of Participants with BMI ≤ 20 kg/m ²	Added recommendation for monitoring weight after dose reduction for BMI ≤ 20 kg/m ² .	To provide additional guidance regarding the management of participants with BMI ≤ 20 kg/m ²
6.9.1. General Considerations 10.10. Appendix 10: Considerations for Concomitant Medications	Added language for monitoring and routinely assessing the need to adjust or discontinue concomitant medications, such as antihypertensive medications, diuretics, lipid-lowering agents, laxatives, and anti-emetics.	To provide more detailed recommendations on the management of concomitant medications

Section # and Name	Description of Change	Brief Rationale
6.9.2. Prohibited or Restricted Use Medications	Added additional language to clarify that the use of subcutaneous/intramuscular or implantable/injectable contraceptives is prohibited except when initiated and used continuously for at least 18 months prior to study entry.	To provide clarification in response to regulatory feedback
7.1. Discontinuation of Study Intervention	Added new discontinuation criterion.	To facilitate the management of participants with poor nutrition or dehydration
8.2.6.1. Close Hepatic Monitoring	Updated the second bullet point under close hepatic monitoring actions to include “Subsequently, the frequency of monitoring may be lowered to once every 1 to 2 weeks, if the participant’s clinical condition and lab results stabilize.”	To provide clarification on continued hepatic monitoring
8.2.6.3. Study Interruption and Discontinuation	Updated the laboratory criteria for study drug interruption/discontinuation and monitoring in participants with elevated baseline hepatic values. The previous criterion “ALT or AST $\geq 2\times$ baseline or ≥ 250 U/L (whichever occurs first) and TBL $\geq 2\times$ ULN” has been revised to “ALT or AST $\geq 2\times$ baseline or ≥ 250 U/L (whichever occurs first) and TBL $\geq 2\times$ ULN or INR ≥ 1.5 .”	To incorporate INR as an additional parameter for hepatic monitoring
8.3.2. Pregnancy	Subsection Female participants who become pregnant: The last paragraph was revised to clarify that any female participant who becomes pregnant while participating in the study will discontinue study intervention “ or and ” <u>will be withdrawn from the study.</u>	Correction
8.3.3.1. Major Adverse Cardiovascular Events	Added “or urgent visit” to “hospitalization for heart failure.”	To expand adjudication criteria, and include clinically relevant cases related to heart failure apart from hospitalization

Section # and Name	Description of Change	Brief Rationale
8.3.3.17. Ketoacidosis Due to Inadequate Nutritional Intake and Other Types of Metabolic Acidosis and Ketosis	Updated the section title from “Metabolic Acidosis and Ketosis” to “Ketoacidosis Due to Inadequate Nutritional Intake and Other Types of Metabolic Acidosis and Ketosis” Updated the AESI wording.	To align the protocol with updated IB and risk profile
8.3.3.19. Malnutrition and Inadequate Fluid Intake	Added new AESI.	To align the protocol with updated IB and risk profile
10.2 Appendix 2: Clinical Laboratory Tests	Added “Magnesium” to Clinical Chemistry Panel. Added new subsection 10.2.2 to describe centrally available laboratory tests related to assessment of nutritional status and micronutrient levels.	To facilitate monitoring and management of participants with or at risk of poor nutrition or dehydration
	For LDL-C test, revised the comment to state “LDL will be directly measured if triglycerides exceed maximum value for calculation.”	For clarity as central laboratories have different cutoffs for triglyceride max

Section # and Name	Description of Change	Brief Rationale
10.6 Appendix 6: Liver Safety: Suggested Actions and Follow-up Assessments	Updated “Hepatic Evaluation Testing” section to include • “Note that testing for some analytes may not be available from the central lab or local lab in certain regions due to local availability of assay or regulatory requirements. If testing is not available, this will not be considered a protocol deviation”. • “If local testing is performed in lieu of central testing, the results should be recorded in the appropriate local lab CRF. For example, if immediate participant management circumstances preclude collection of central lab samples – e.g., emergency room visit or hospitalization”. Hepatic D virus (HDV) testing to be assayed by local laboratory instead of central laboratory.	To provide clarification on analyte testing for continued hepatic monitoring

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1. Protocol Summary

1.1. Synopsis

Protocol Title: A Phase 3, Randomized, Multicenter, Double-Blind Study to Investigate the Efficacy and Safety of Retatrutide Once Weekly Compared with Placebo in Adult Participants with Type 2 Diabetes and Inadequate Glycemic Control with Diet and Exercise Alone (TRANSCEND-T2D-1)

Brief Title: Effect of Retatrutide Compared with Placebo in Adult Participants with Type 2 Diabetes and Inadequate Glycemic Control with Diet and Exercise Alone (TRANSCEND-T2D-1)

Regulatory Agency Identifier Number:

IND: 145825

Rationale:

Study J1I-MC-GZBY (GZBY) is designed to determine the comparative benefits and risks of retatrutide versus placebo, in participants with type 2 diabetes (T2D) who have inadequate glycemic control with diet and exercise alone, are naive of insulin therapy except for gestational diabetes or ≤ 14 days use for acute treatment and have not been treated with any oral glucose-lowering medication or noninsulin injectables within 90 days prior to study entry. Additionally, the study will compare the effects of retatrutide and placebo on body weight and overall safety profile.

Objectives, Endpoints, and Estimands:

Objectives	Endpoints
Primary	
To demonstrate the superiority of retatrutide 4 mg, 9 mg, and/or 12 mg versus placebo in glycemic control	Change from baseline in HbA1c (%) at Week 40
Key Secondary (controlled for type 1 error)	
To demonstrate that retatrutide 4 mg, 9 mg, and/or 12 mg are superior to placebo in additional measures of glycemic control	- Incidence of participant achieving HbA1c $< 7.0\%$ (53 mmol/mol) at Week 40 (Y/N) - Incidence of participant achieving HbA1c $\leq 6.5\%$ (48 mmol/mol) at Week 40 (Y/N) - Incidence of participant achieving HbA1c $< 5.7\%$ (39 mmol/mol) at Week 40 (Y/N) - Change from baseline in fasting serum glucose (mg/dL) (central laboratory) at Week 40

Objectives	Endpoints
To demonstrate that retatrutide 4 mg, 9 mg, and/or 12 mg are superior to placebo in weight management	- Percentage change from baseline in body weight at Week 40 - Change from baseline in body weight at Week 40 - Incidence of participant achieving weight reduction of $\geq 5\%$, $\geq 10\%$, and $\geq 15\%$ from baseline at Week 40 (Y/N)
To demonstrate that retatrutide 4 mg, 9 mg, and/or 12 mg are superior to placebo in additional measures of glycemic control and weight management	Incidence of participant achieving HbA1c $\leq 6.5\%$ (48 mmol/mol) and $\geq 10\%$ weight reduction from baseline at Week 40 (Y/N)
To demonstrate that retatrutide 9 mg and/or 12 mg are superior to placebo in measures of lipid parameters	Percentage change from baseline in lipid parameters at Week 40 - non-HDL cholesterol - triglycerides
To demonstrate that retatrutide 9 mg and/or 12 mg are superior to placebo in measures of blood pressure	Change from baseline in systolic blood pressure (mmHg) at Week 40
Additional Secondary	
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on additional measures of glycemic control	Change from baseline in daily average 7-point SMBG (mg/dL) at Week 40
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on additional measures of weight management	From baseline at Week 40 - incidence of participant achieving weight reduction of $\geq 20\%$ and $\geq 25\%$ (Y/N) - change in waist circumference (cm) - change in BMI (kg/m²)
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on additional measures of blood pressure	Change from baseline in diastolic blood pressure (mmHg) at Week 40
To compare retatrutide 4 mg with placebo on additional measures of blood pressure	Change from baseline in systolic blood pressure (mmHg) at Week 40
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on measures of lipid parameters	Percentage change from baseline in lipid parameters at Week 40: - total cholesterol - HDL-cholesterol - LDL-cholesterol - VLDL-cholesterol - apolipoprotein B
To compare retatrutide 4 mg with placebo on measures of lipid parameters	Percentage change from baseline in lipid parameters at Week 40: - non-HDL cholesterol - triglycerides
To compare retatrutide 9 mg and 12 mg with placebo on biomarker of inflammation	Percentage change from baseline in hsCRP at Week 40

Objectives	Endpoints
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on patient-reported outcomes	- Change from baseline to Week 40 in - EQ-5D-5L health state utilities and visual analog scale score - APPADL total score - IWQOL-Lite-CT domain and total scores - DTSQs - DTSQc at Week 40 - DID-EQ at Week 4 subscale and global items scores. - Change from baseline at Week 20 and change from baseline at Week 40 in - CoEQ domain and item (hunger and fullness) scores - FCQ-T-r domain and total scores
To characterize the PK and PD of retatrutide and the relationships between retatrutide exposure and safety, tolerability, and efficacy measures	Population PK and PD parameters
To describe the safety of retatrutide in participants with T2D	Summary of safety data, including frequency of - SAEs - TEAEs - discontinuations due to AEs

Abbreviations: AE = adverse event; APPADL = Ability to Perform Physical Activities of Daily Living; BMI = body mass index; CoEQ = Control of Eating Questionnaire; DID-EQ = Diabetes Injection Device – Experience Questionnaire; DTSQc = Diabetes Treatment Satisfaction Questionnaire-Change Version; DTSQs = Diabetes Treatment Satisfaction Questionnaire-Status Version; FCQ-T-r = Food Craving Questionnaire-Trait reduced; HbA1c = hemoglobin A1c; HDL = high-density lipoprotein; hsCRP = high-sensitivity C-reactive protein; IWQOL-Lite-CT = Impact of Weight on Quality of Life-Lite Clinical Trials Version; LDL = low-density lipoprotein; PD = pharmacodynamic(s); PK = pharmacokinetic(s); SAE = serious adverse event; SMBG = self-monitoring of blood glucose; T2D = type 2 diabetes; TEAE = treatment-emergent adverse event; VLDL = very-low-density lipoprotein; Y/N= yes/no.

Primary estimands

There will be 2 estimands for the primary objective planned in the study. The 2 estimands, treatment regimen estimand and efficacy estimand, address intercurrent events (ICEs) using the treatment policy strategy and the hypothetical strategy, respectively. The treatment regimen estimand will be the primary estimand, with the efficacy estimand considered supportive.

Treatment policy strategy

The occurrence of the ICE is considered irrelevant in defining the treatment effect of interest; the values for the variable of interest are used regardless of whether the ICE occurs.

Hypothetical strategy

A scenario is envisaged in which the ICE would not occur. The value of the variable to reflect the clinical question of interest is the value that the variable would have taken in the hypothetical scenario defined.

Treatment regimen estimand

The clinical question of interest is: *What is the treatment difference in the change in HbA1c from baseline at Week 40 between retatrutide 4 mg, 9 mg, and/or 12 mg compared with placebo in individuals who meet the eligibility criteria regardless of adherence to study intervention or initiation of additional antihyperglycemic medications?*

Rationale for the treatment regimen estimand

The estimand aims to evaluate the efficacy of retatrutide that reflects the real-life behavior of the target population.

Treatment regimen estimand attributes

- **Population** - the target population is the group of individuals who meet the eligibility criteria. For the analysis, the intent-to-treatment principle will be applied.
- **Endpoint** - change from baseline in HbA1c at Week 40.
- **Treatment condition** - the randomly assigned treatment with allowance for potential dose interruptions and dose modification regardless of adherence to study intervention or initiation of additional antihyperglycemic medications.
- **Intercurrent events** - no ICEs are defined since treatment adherence and the initiation of additional antihyperglycemic medications are a part of the treatment condition.
- **Population-level summary and treatment effect of interest** - difference in mean changes from baseline in HbA1c at Week 40 with retatrutide 4 mg, 9 mg, and/or 12 mg compared to placebo.

Efficacy estimand

The clinical question of interest is: *What is the treatment difference in the change in HbA1c from baseline at Week 40 between retatrutide 4 mg, 9 mg, and/or 12 mg compared with placebo in individuals who meet the eligibility criteria if they would remain on their randomly assigned study intervention for 40 weeks and would not initiate additional antihyperglycemic medications?*

Rationale for the efficacy estimand

This estimand aims to evaluate the efficacy of retatrutide under the ideal condition that all participants would adhere to the randomly assigned study intervention without being confounded by the initiation of additional antihyperglycemic medications.

Efficacy estimand attributes

- **Population** - the target population is the group of individuals who meet the eligibility criteria. For the analysis, the intent-to-treatment principle will be applied.
- **Endpoint** - change from baseline in HbA1c at Week 40.
- **Treatment condition** - the randomly assigned treatment with allowance for potential dose interruptions and dose modifications.
- **Intercurrent events** - ICEs include permanent discontinuation of study intervention and initiation of additional antihyperglycemic medications and will be handled by the hypothetical strategy. The potential outcome of interest is the response in the efficacy measurement if participants would remain on their randomly assigned study intervention for

40 weeks and would not initiate additional antihyperglycemic medications. Dose modifications and interruptions will not be considered an ICE since they are part of the treatment condition.

- **Population-level summary and treatment effect of interest** - difference in mean change from baseline in HbA1c at Week 40 with retatrutide 4 mg, 9 mg, and/or 12 mg compared to placebo.

Overall Design:

Study GZBY is a Phase 3, multicenter, randomized, parallel-arm, double-blind, placebo-controlled study to investigate the efficacy and safety of retatrutide 4 mg, 9 mg, or 12 mg compared with placebo in participants with T2D and inadequate glycemic control with diet and exercise alone, are naïve to insulin use and have not been treated with any antihyperglycemic medication during the 90 days preceding Visit 1.

Brief Summary:

The purpose of this study is to investigate the efficacy and safety of retatrutide 4 mg, 9 mg, or 12 mg administered QW for 40 weeks compared with placebo for mean change from baseline in HbA1c in participants with T2D and inadequate glycemic control with diet and exercise alone.

This study will consist of 3 periods:

- a screening period: up to 4 weeks
- a treatment period: 40 weeks, including dose escalation and maintenance, and
- a safety follow-up period: 4 weeks.

Study Population:

In general, an individual may take part in this study if they

- are at least 18 years of age or the legal age of consent in the jurisdiction in which the study is taking place at the time of signing the informed consent, whichever is older.
- have a clinical diagnosis of T2D based on the World Health Organization classification or other locally applicable diagnostic standards.
- have HbA1c $\geq 7.0\%$ (53 mmol/mol) to $\leq 9.5\%$ (80 mmol/mol), as determined by the central laboratory at Visit 1 (screening).
- are naïve to insulin therapy except for gestational diabetes or ≤ 14 days use for acute treatment and have not used any oral or injectable antihyperglycemic medications during the 90 days preceding Visit 1, or between Visit 1 and Visit 3 (randomization).

Number of Participants:

Approximately 480 participants will be randomly assigned in a 1:1:1:1 ratio to retatrutide 4 mg (approximately 120), retatrutide 9 mg (approximately 120), retatrutide 12 mg (approximately 120), and placebo (approximately 120), respectively.

Intervention Groups and Duration:

Intervention Name	Retatrutide	Placebo
Dosage Level(s)	2 mg 4 mg ^a 6 mg 9 mg ^a 12 mg ^a	Not applicable
Route of Administration	Subcutaneous injection QW	Subcutaneous injection QW

Abbreviation: QW = once weekly.

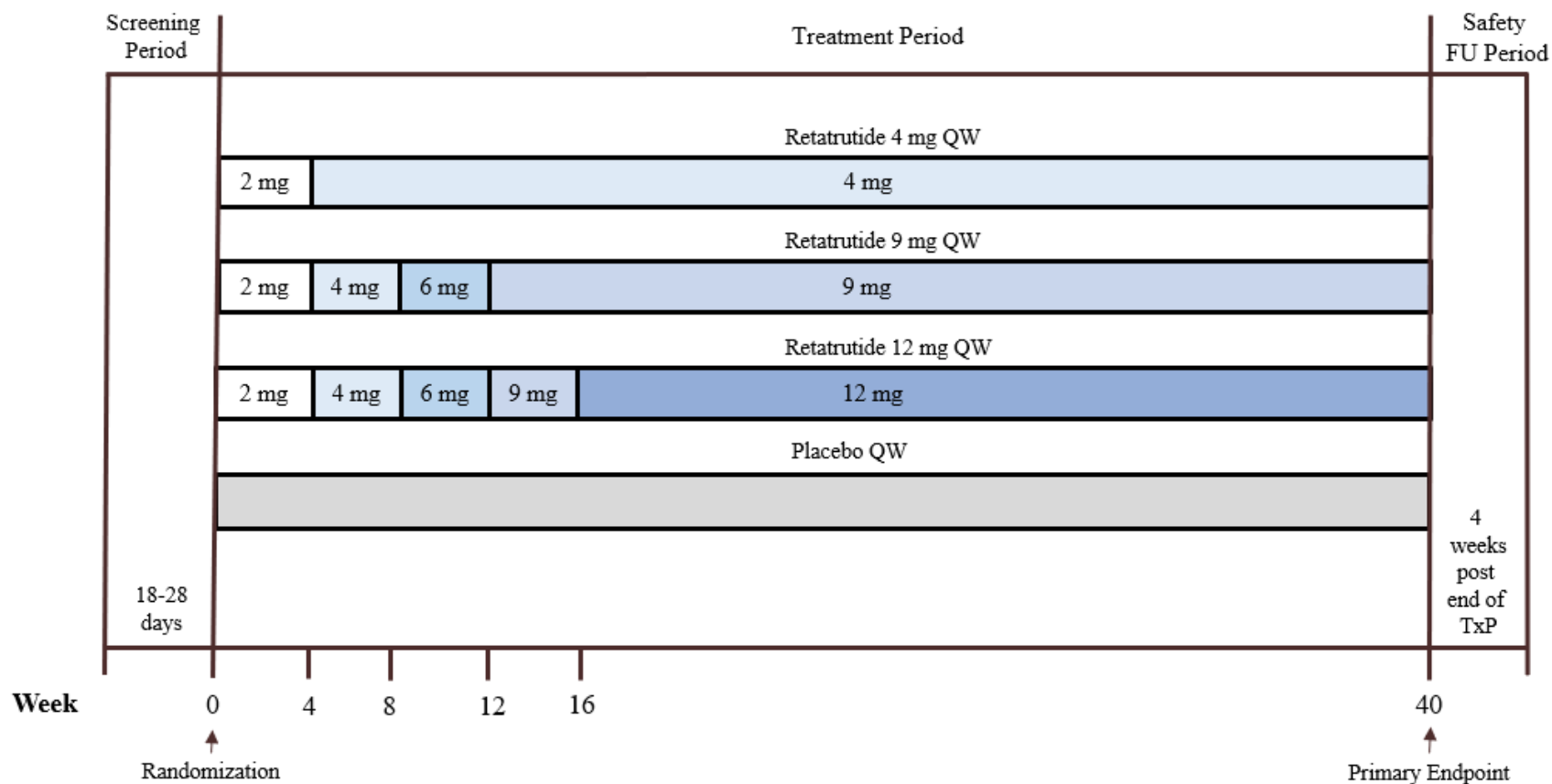
^a Retatrutide 4 mg, 9 mg, and 12 mg are planned maintenance doses for the treatment period.

Ethical Considerations of Benefit/Risk:

The safety and efficacy profile seen to date for retatrutide supports the overall benefit risk for participants in this study. Considering the measures taken to minimize risk to participants in this study, the potential risks identified in association with retatrutide are outweighed by the anticipated benefits that may be afforded to participants living with T2D.

Data Monitoring Committee: No

1.2. Schema



Abbreviations: FU = follow-up; QW = once weekly; TxP = treatment period.

1.3. Schedule of Activities (SoA)

The SoA described below should be followed for all participants enrolled in Study J1I-MC-GZBY (GZBY). However, for those participants whose participation in this study is affected by exceptional circumstances, such as pandemics or natural disasters, please refer to Section 10.14 for additional guidance.

Visit details

Visit procedures may be conducted over more than 1 day if all activities are completed within the allowed visit tolerance period of each visit.

Fasting visits

On all fasting visits, study participants are required to report to the site in a fasting condition, after at least 8 hours without eating, drinking (except water), or performing any significant physical activity. Additionally, the participant should not take their study intervention before the fasting visit. If a participant is adversely affected by the fasting condition, they are allowed to eat; however, specific study procedures need to be completed while fasting.

If a participant attends a fasting visit in the non-fasting state, reschedule the visit at a time within the allowable visit interval.

Screening visits

Visit 2 may occur as soon as the criteria for Visit 1 are confirmed. At Visit 2, screening laboratory results will be reviewed to confirm eligibility. Visit 3 to occur no sooner than 18 days after Visit 1.

PK only visits

The only procedure performed for these visits is PK sample collection.

Early discontinuation (ED) visit

For early discontinuation that occurs before the last visit in treatment period, see the activities listed for ED in the SoA.

Visit 801 (safety follow-up period)

A posttreatment safety follow-up visit will occur approximately 4 weeks following the last treatment period visit. Participants who have discontinued study intervention early and are only willing to come back for Visit 11 (Week 40) are not required to complete a safety follow-up visit if they have been off of study intervention for at least 28 days (see Section 7.1.1).

Fundus photography

Baseline fundus photography should be performed on participants who remain eligible based on screening activities performed at Visit 1, including laboratory results. The overall ophthalmological evaluation of the baseline fundus photographs must be completed prior to randomization to confirm eligibility. Fundus photography and further ophthalmological assessment at

- Visit 9 (Week 24) will have a 4-week visit interval tolerance.
- Visit 11 (Week 40) will have -4 weeks to +2 weeks visit interval tolerance and all ophthalmological assessments should be completed preferably prior to start of new antihyperglycemic medication.

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
Informed consent	X																The ICF must be signed before any tests or procedures are performed. See Section 10.1.3 for additional details.	
Inclusion/exclusion criteria; confirmation of eligibility	X	X	X														Confirm inclusion and exclusion criteria prior to randomization and administration of first dose of study intervention.	
Demographics	X																Includes ethnicity (where permissible), year of birth, sex, and race.	
Preexisting conditions and medical history, including relevant surgical history	X																Collect all ongoing conditions and relevant past surgical and medical history.	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
Prespecified medical history	X																	
Prior treatments for interest	X																Data regarding prior use of GLP-1 RA and GIP/GLP-RA are collected. See Section 6.9.	
Concomitant medications	X	X	X	X		X	X		X		X	X	X	X	X	X	Monitor closely and routinely assess the need to adjust or discontinue antihypertensive medications, diuretics, laxatives, and anti-emetics. Refer to Sections 6.9.1 and 10.10 for additional details.	
Evaluation of antihypertensive and lipid-lowering medications							X					X		X	X		As applicable for participants receiving these medications. Evaluate the changes in treatment intensity, within each therapeutic area, from	

	Period I - Screening		Period II - Treatment Period													Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5	
Fasting visit			X	X		X	X		X		X	X	X	X	X	X	
Visit Detail					PK only			PK only		PK only							
																	randomization until the time of evaluation.
AEs	X	X	X	X		X	X	X	X		X	X	X	X	X	X	Any events that occur after signing the informed consent are considered AEs as defined in Section 8.3.1 and Section 10.3. Additional data are collected for certain AEs. See Section 8.3.3.
Substance use (alcohol, caffeine, tobacco, nicotine replacement)			X														
Physical evaluation																	
Height	X																See Section 10.12.
Weight	X		X	X		X	X		X		X	X	X	X	X	X	Measure in kg. See Section 10.12.

	Period I - Screening		Period II - Treatment Period													Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5	
Fasting visit			X	X		X	X		X		X	X	X	X	X	X	
Visit Detail					PK only			PK only		PK only							
Waist circumference			X	X		X	X		X		X	X	X	X	X	X	Measure in cm. See Section 10.12.
Physical examination	X																Excludes pelvic, rectal, and breast examinations unless clinically indicated. See Section 8.2.1.
Symptom-directed physical assessment		X				X			X		X						Conducted at the discretion of the PI or qualified personnel.
Vital signs (triplicate)	X	X	X	X		X	X		X		X	X	X	X	X	X	Measured in triplicate after participant has been sitting at least 5 minutes. Vital signs should be taken before ECG tracing. See Sections 8.2.2 and 10.12.
ECG 12-lead (triplicate; central)			X									X		X	X	X	Collect triplicate ECGs before all blood samples. ECG may be repeated at the

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
																	investigator’s discretion at any visit. See Section 8.2.3.	
Fundus photography and further ophthalmological assessment		X										X		X	X		Eligibility (see exclusion criterion 13, Section 5.2) MUST be confirmed based on fundus photographs and any additional ophthalmological evaluation needed. Perform follow-up fundus photography and any additional ophthalmological evaluation will be performed at Week 24 (±4 weeks) only if retinopathy or DME were present at baseline. All participants will repeat fundus photography and any additional ophthalmological evaluation	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
																	at Week 40 (-4 weeks to +2 weeks) preferably prior to start of new antihyperglycemic medication. See Section 8.2.8.	
Participant diary (electronic)																		
Dispense eDiary		X															Includes hypoglycemia, weekly dosing for retatrutide or placebo, and 4-point and 7-point SMBG, as applicable.	
eDiary review			X	X		X	X		X		X	X	X	X	X	X	eDiary review to include SMBG data, weekly dosing for retatrutide or placebo, and hypoglycemia events.	
Remind participants about 7-point SMBG		X									X		X				Site staff will remind participants of collection times at specified visits and approximately 2 weeks before	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
																	subsequent visits. See Section 5.3.2.	
Review 7-point SMBG			X									X		X			Participant will collect two 7-point SMBGs on non-consecutive days within 2 weeks prior to each visit specified. See Section 5.3.2.	
eDiary return																X	eDiary can be returned earlier if that visit is the last visit of the study for the participant.	
Patient-reported outcomes (electronic)																		
EQ-5D-5L			X											X	X			
APPADL			X											X	X			
IWQOL-Lite-CT			X											X	X			
DTSQs			X											X	X			
DTSQc														X	X			

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
DID-EQ				X											X		At ED if discontinuation is at or before Week 4.	
CoEQ			X								X			X	X			
FCQ-T-r			X								X			X	X			
PHQ-9	X		X				X					X		X	X	X	Collect AEs before the collection of PHQ-9.	
Clinician-administered assessments (electronic)																		
C-SSRS Screening/Baseline	X																.	
C-SSRS Since Last Assessed			X	X		X	X		X		X	X	X	X	X	X	Collect AEs before the collection of the C-SSRS.	
Participant education																		
Lifestyle considerations, diabetes education (includes BG meter and SMBG)	X	X	X	X		X	X		X		X	X	X	X	X		All training outlined in Section 5.3 should be repeated as needed to encourage participant compliance. Monitor for and encourage	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
training), and meals and nutrition																	adequate nutritional intake and hydration throughout the study. See Sections 5.3.3 and 5.3.4.	
Dispense BG meter supplies as needed (includes collection of unused supplies)		X				X			X		X							
Participants watch the KwikPen injection training video and site staff demonstrates the demo device	X																To be repeated at Visit 2, if needed.	
Observe participant practicing KwikPen demo on injection pad		X															Repeat as needed at Visit 3.	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
Observe participants administer study intervention			X														Repeat as needed. All assessments should be completed before the administration of the first dose of study intervention.	
Laboratory tests and sample collections																		
Hematology	X		X				X					X		X	X	X		
Clinical chemistry	X		X				X					X		X	X	X		
Glucose			X	X		X	X		X		X	X	X	X	X	X	See “Fasting visits” in the SoA preamble.	
Hemoglobin A1c (HbA1c)	X		X	X		X	X		X		X	X	X	X	X	X		
Lipids			X				X					X		X	X	X		
Lipase			X				X					X		X	X	X		
Pancreatic amylase			X				X					X		X	X	X		

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
Insulin			X				X					X		X	X	X		
C-peptide			X				X					X		X	X	X		
Apo B			X											X	X			
hsCRP			X											X	X			
Calcitonin	X						X					X		X	X	X		
Cystatin-C	X		X				X					X		X	X	X		
eGFR	X		X				X					X		X	X	X	CKD-EPI cystatin-C (2012).	
UACR			X									X		X	X	X		
GAD antibodies			X															
Serum pregnancy	X																Collect for WOCBP. See Section 10.4.	
Urine pregnancy (local)			X				X					X		X	X		The result must be available before the first dose of study intervention for WOCBP.	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
																	Perform additional pregnancy tests if a menstrual period is missed, if there is clinical suspicion of pregnancy, or as required by local law or regulation.	
FSH	X																Perform as needed to confirm postmenopausal status. See Section 10.4.	
HCV screening tests	X																If HCV antibody test is positive, perform an HCV RNA test. See exclusion criterion 19, Section 5.2.	
HBV screening tests	X																For participants who are positive for HBcAb, testing for HBV DNA will be performed. See exclusion criterion 19, Section 5.2.	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
Immunogenicity samples (collect pre-dose)			X	X			X					X		X	X	X	Collect additional unscheduled samples if systemic drug hypersensitivity reactions (immediate or nonimmediate; see Section 10.2.1) or injection site reactions occur (severe or recurrent; see Section 8.3.3.13). Collect immunogenicity and PK samples for immunogenicity before dosing.	
PK samples for immunogenicity (pre-dose)				X			X					X		X	X	X		
PK (postdose)			X		X			X		X							Week 0 (Visit 3): Collect PK sample anytime on the same day after administration of the	

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
																	dose at the site. Weeks 7, 15, and 19: collect only at the time windows of 1 to 24 hr, 24 to 96 hr, OR 120 to 168 hr post dose, as assigned by IWRS.	
Stored samples																		
Genetics sample			X														Collect sample at or after the specified visit.	
Exploratory biomarker samples			X				X					X		X	X		Collect samples pre-dose.	
Randomization and dosing-related activities																		
Register visit with IWRS	X	X	X	X		X	X		X		X	X	X	X		X		
Randomization via IWRS			X															

	Period I - Screening		Period II - Treatment Period														Period III - Safety Follow-up	Comments
Visit number	1	2	3	4	-	5	6	-	7	-	8	9	10	11	ED	801	Visit 2 may occur as soon as criteria for Visit 1 are confirmed. Visit 3 to occur no sooner than 18 days after Visit 1.	
Weeks from Visit 3 (randomization)	18-28 days		0	4	7	8	12	15	16	19	20	24	32	40		4 wks post end of TxP	Unscheduled visits may occur as needed.	
Visit interval tolerance (days)	-		-	±3		±3	±3		±3		±5	±5	±5	±5		±5		
Fasting visit			X	X		X	X		X		X	X	X	X	X	X		
Visit Detail					PK only			PK only		PK only								
Dispense study intervention via IWRS			X	X		X	X		X		X	X	X				Site to confirm package assignment and dispense to participant for at home-dosing.	
Dispense supplies to participant (for at-home dosing)			X	X		X	X		X		X	X	X				Dispense as needed.	
Participant returns unused study intervention/supplies				X		X	X		X		X	X	X	X	X			
Assess study intervention compliance				X		X	X		X		X	X	X	X	X			

Abbreviations: AEs = adverse events; APPADL = Ability to Perform Physical Activities of Daily Living; Apo B = apolipoprotein B; BG = blood glucose; CKD-EPI = Chronic Kidney Disease-Epidemiology Collaboration; CoEQ = Control of Eating Questionnaire; C-SSRS = Columbia-Suicide Severity Rating Scale; DME = diabetic macular edema; DTSQc = Diabetes Treatment Satisfaction Questionnaire-Change Version; DTSQs = Diabetes Treatment Satisfaction Questionnaire-Status Version; DID-EQ = Diabetes Injection Device – Experience Questionnaire; ECG = electrocardiogram; ED = early discontinuation; eGFR = estimated glomerular filtration rate; FCQ-T-r = Food Craving Questionnaire - Trait reduced; FSH = follicle-stimulating hormone; GAD = glutamic acid decarboxylase; GLP-1 RA = glucagon-like peptide-1 receptor agonist; HBcAb = hepatitis B core antibody; HBV = hepatitis B virus; HCV = hepatitis C virus; hsCRP = high-sensitivity C-reactive protein; ICF = informed consent form; IWQOL-Lite-CT = Impact of Weight on Quality of Life-Lite Clinical Trials Version; IWRS = interactive-web response system; PHQ-9 = Patient Health Questionnaire-9; PI = principal investigator; PK = pharmacokinetics; RNA = ribonucleic acid; SMBG = self-monitoring of blood glucose; SoA = Schedule of Activities; TxP = treatment period; UACR = urinary albumin/creatinine ratio; WOCBP = women of childbearing potential.

2. Introduction

Retatrutide, a novel synthetic molecule, which shows potent agonist action at the GIPR, GLP-1R, and GCGR is being developed for improving glycemic control and for chronic weight management. It has a chemical structure and pharmacologic profile that is distinct from the GLP-1 RAs and from GIP/GLP-1 RAs due to the additional effects on GCGRs, which is unique among the approved incretin mimetics. It is administered QW by SC injection. Preclinical and Phase 1 and 2 clinical data have shown improved glycemic control, weight reduction, and improvement in other key cardiometabolic parameters relevant for patients with T2D (Coskun et al. 2022; Urva et al. 2022; Rosenstock et al. 2023).

2.1. Study Rationale

Study GZBY is designed to determine the comparative benefits and risks of retatrutide versus placebo, in participants with T2D who have inadequate glycemic control with diet and exercise alone, are naive of insulin therapy except for gestational diabetes or ≤ 14 days use for acute treatment and have not been treated with any oral glucose-lowering medication or noninsulin injectables within 90 days prior to study entry. Additionally, the study will compare the effects of retatrutide and placebo on body weight and overall safety profile.

2.2. Background

Type 2 diabetes

T2D is a metabolic condition characterized by impaired glycemic control caused by increased insulin resistance and progressive beta-cell failure and consequently inadequate insulin secretion (Zheng et al. 2018). T2D is frequently associated with comorbidities such as overweight or obesity, hypertension, increased blood lipoprotein concentrations, and a higher risk for macro- and microvascular complications. Achieving an HbA1c target of $<7\%$ has been shown to reduce microvascular complications of T2D when instituted early in the course of disease and more stringent goals may be appropriate and beneficial for certain individuals (ADA 2023). Following a person-centered holistic approach, ADA standards of care in diabetes (ADA 2023) also outline the importance of selection of a treatment regimen that supports weight management goals.

Unmet needs for diabetes treatment

With the prevalence of T2D increasing worldwide, new antihyperglycemic agents offering significant improvements in glycemic control and additional weight loss as well as other potential health benefits, which would result in improved health outcomes, are highly desired.

Injectable incretin-based treatments are commonly used in combination with oral antihyperglycemic medication to achieve and maintain glucose control (ADA 2023). These treatment options offer improved glycemic control, low risk of hypoglycemia, and the potential for clinically relevant weight reduction. However, a proportion of patients do not reach individualized treatment targets despite high levels of compliance with the treatment regimens. Therefore, there is an unmet need for additional treatment options that enhance glycemic control, promote weight reduction, and preserve an acceptable benefit/risk profile.

Clinical data for retatrutide

A detailed description of the chemistry, pharmacology, efficacy, and safety of retatrutide is provided in the IB. Currently, two Phase 2 studies of retatrutide have been completed.

Phase 2 study in patients with T2D

Study J1I-MC-GZBD was a 36-week, Phase 2, multicenter, randomized, double-blind study of QW retatrutide compared with placebo and dulaglutide (1.5 mg) in participants with T2D. Four maintenance doses of retatrutide were evaluated in the study, 0.5 mg, 4 mg, 8 mg, and 12 mg with the primary outcome of change from baseline in HbA1c (Rosenstock et al. 2023).

Key efficacy and safety data at 36 weeks

- HbA1c change from baseline was -0.3% for placebo, -1.4% for dulaglutide 1.5 mg, and -1.3% to -2.2% for retatrutide (4 mg to 12 mg).
- HbA1c targets of <7% and ≤6.5% were achieved by 60% and 43% of participants treated with dulaglutide 1.5 mg, respectively, and 59% to 82% and 45% to 82% of participants treated with retatrutide (4 mg to 12 mg), respectively.
- Body weight change from baseline was -3% for placebo, -2% for dulaglutide 1.5 mg and -7.9% to -16.9% for retatrutide (4 mg to 12 mg).
- Body weight reduction of at least 5%, 10%, and 20% were achieved by 17%, 2%, and 0% of participants treated with dulaglutide 1.5 mg, respectively, and 61% to 94%, 22% to 75%, and 9% to 40% of participants treated with retatrutide (4 mg to 12 mg), respectively.
- The overall safety profile of retatrutide was consistent with that of the GLP-1 RA class in people with T2D, with most common TEAEs being GI related (including nausea, vomiting, diarrhea, and constipation). The majority of GI AEs were mild to moderate in severity and occurred mostly during the dose escalation period and decreased over time.

Data from Phase 2 studies demonstrated the efficacy and safety of multiple retatrutide doses administered QW and support continued development of retatrutide for T2D.

2.3. Benefit/Risk Assessment

Detailed information about the known and expected benefits and risks and reasonably expected AEs of retatrutide may be found in the IB.

2.3.1. Risk Assessment

Risks associated with retatrutide

The identified risks and potential risks associated with retatrutide are similar to those of marketed GLP-1 RAs and the GIP/GLP-1 RA, tirzepatide.

The most commonly reported TEAEs observed in the retatrutide clinical studies, in healthy participants or participants with T2D, are GI effects, including nausea, vomiting, and diarrhea.

Other risks and potential risks associated with retatrutide include

- Acute pancreatitis

- Increases in HR and decreases in systolic BP
- Hypoglycemia when used with sulfonylureas or insulin
- Thyroid C-cell tumors (only observed in rodents with long-acting GLP-1 RAs and GIP/GLP-1 RA), and
- Skin burning sensation and related AEs.

Sections 5.1, 5.2, and 8.2 address mitigation, management, and monitoring of the known risks and potential risks associated with retatrutide.

Study procedures

Blood draws

Risks of blood draws for clinical laboratory assessments include the potential for some pain and a small risk of bruising or infection at the site.

ECG collection

Collection of ECGs may cause mild discomfort during placement and removal of leads from the skin. Some participants may experience local irritation or redness at the sites of lead attachment.

Retinal assessment

Dilating drops, if used for pupil dilation for the eye examination, may sting temporarily and may cause temporary blurred vision, difficulty focusing, and light sensitivity. Pharmacologic dilation may be contraindicated in some participants with eye conditions, for example, closed angle glaucoma.

2.3.2. Benefit Assessment

Data from Phase 2 studies of retatrutide supports an expectation of improved glycemic control and weight reduction in patients with T2D. Improvements in some cardiometabolic risk factors, including BP and serum lipids, may also be expected in this patient population.

Participants may also benefit from receiving personal health information, diabetes education, routine safety assessments, and frequent engagement with healthcare providers during the study, which provide opportunities for coaching and support.

2.3.3. Overall Benefit Risk Conclusion

The safety and efficacy profile seen to date for retatrutide supports the overall benefit risk for participants in this study. Considering the measures taken to minimize risk to participants in this study, the potential risks identified in association with retatrutide are outweighed by the anticipated benefits that may be afforded to participants living with diabetes.

3. Objectives, Endpoints, and Estimands

Objectives	Endpoints
Primary	
To demonstrate the superiority of retatrutide 4 mg, 9 mg, and/or 12 mg versus placebo in glycemic control	Change from baseline in HbA1c (%) at Week 40
Key Secondary (controlled for type 1 error)	
To demonstrate that retatrutide 4 mg, 9 mg, and/or 12 mg are superior to placebo in additional measures of glycemic control	- Incidence of participant achieving HbA1c <7.0% (53 mmol/mol) at Week 40 (Y/N) - Incidence of participant achieving HbA1c ≤6.5% (48 mmol/mol) at Week 40 (Y/N) - Incidence of participant achieving HbA1c <5.7% (39 mmol/mol) at Week 40 (Y/N) - Change from baseline in fasting serum glucose (mg/dL) (central laboratory) at Week 40
To demonstrate that retatrutide 4 mg, 9 mg, and/or 12 mg are superior to placebo in weight management	- Percentage change from baseline in body weight at Week 40 - Change from baseline in body weight at Week 40 - Incidence of participant achieving weight reduction of ≥5%, ≥10%, and ≥15% from baseline at Week 40 (Y/N)
To demonstrate that retatrutide 4 mg, 9 mg, and/or 12 mg are superior to placebo in additional measures of glycemic control and weight management	Incidence of participant achieving HbA1c ≤6.5% (48 mmol/mol) and ≥10% weight reduction from baseline at Week 40 (Y/N)
To demonstrate that retatrutide 9 mg and/or 12 mg are superior to placebo in measures of lipid parameters	Percentage change from baseline in lipid parameters at Week 40 - non-HDL cholesterol - triglycerides
To demonstrate that retatrutide 9 mg and/or 12 mg are superior to placebo in measures of blood pressure	Change from baseline in systolic blood pressure (mmHg) at Week 40
Additional Secondary	
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on additional measures of glycemic control	Change from baseline in daily average 7-point SMBG (mg/dL) at Week 40
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on additional measures of weight management	From baseline at Week 40 - incidence of participant achieving weight reduction of ≥20% and ≥25% (Y/N) - change in waist circumference (cm) - change in BMI (kg/m²)
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on additional measures of blood pressure	Change from baseline in diastolic blood pressure (mmHg) at Week 40

Objectives	Endpoints
To compare retatrutide 4 mg with placebo on additional measures of blood pressure	Change from baseline in systolic blood pressure (mmHg) at Week 40
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on measures of lipid parameters	Percentage change from baseline in lipid parameters at Week 40: • total cholesterol • HDL-cholesterol • LDL-cholesterol • VLDL-cholesterol • apolipoprotein B
To compare retatrutide 4 mg with placebo on measures of lipid parameters	Percentage change from baseline in lipid parameters at Week 40: • non-HDL cholesterol • triglycerides
To compare retatrutide 9 mg and 12 mg with placebo on biomarker of inflammation	Percentage change from baseline in hsCRP at Week 40
To compare retatrutide 4 mg, 9 mg, and 12 mg with placebo on patient-reported outcomes	• Change from baseline to Week 40 in ○ EQ-5D-5L health state utilities and visual analog scale score ○ APPADL total score ○ IWQOL-Lite-CT domain and total scores ○ DTSQs • DTSQc at Week 40 • DID-EQ at Week 4 subscale and global items scores • Change from baseline at Week 20 and change from baseline at Week 40 in ○ CoEQ domain and item (hunger and fullness) scores ○ FCQ-T-r domain and total scores
To characterize the PK and PD of retatrutide and the relationships between retatrutide exposure and safety, tolerability, and efficacy measures	Population PK and PD parameters
To describe the safety of retatrutide in participants with T2D	Summary of safety data, including frequency of • SAEs • TEAEs • discontinuations due to AEs

Abbreviations: AE = adverse event; APPADL = Ability to Perform Physical Activities of Daily Living; BMI = body mass index; CoEQ = Control of Eating Questionnaire; DID-EQ = Diabetes Injection Device – Experience Questionnaire; DTSQc = Diabetes Treatment Satisfaction Questionnaire-Change Version; DTSQs = Diabetes Treatment Satisfaction Questionnaire-Status Version; FCQ-T-r = Food Craving Questionnaire-Trait reduced; HbA1c = hemoglobin A1c; HDL = high-density lipoprotein; hsCRP = high-sensitivity C-reactive protein; IWQOL-Lite-CT = Impact of Weight on Quality of Life-Lite Clinical Trials Version; LDL = low-density

lipoprotein; PD = pharmacodynamic(s); PK = pharmacokinetic(s); SAE = serious adverse event; SMBG = self-monitoring of blood glucose; T2D = type 2 diabetes; TEAE = treatment-emergent adverse event; VLDL = very-low-density lipoprotein; Y/N= yes/no.

Primary estimands

There will be 2 estimands for the primary objective planned in the study. The 2 estimands, treatment regimen estimand and efficacy estimand, address ICEs using the treatment policy strategy and the hypothetical strategy, respectively. The treatment regimen estimand will be the primary estimand, with the efficacy estimand considered supportive.

Treatment policy strategy

The occurrence of the ICE is considered irrelevant in defining the treatment effect of interest; the values for the variable of interest are used regardless of whether the ICE occurs.

Hypothetical strategy

A scenario is envisaged in which the ICE would not occur. The value of the variable to reflect the clinical question of interest is the value that the variable would have taken in the hypothetical scenario defined.

Treatment regimen estimand

The clinical question of interest is: *What is the treatment difference in the change in HbA1c from baseline at Week 40 between retatrutide 4 mg, 9 mg, and/or 12 mg compared with placebo in individuals who meet the eligibility criteria regardless of adherence to study intervention or initiation of additional antihyperglycemic medications?*

Rationale for the treatment regimen estimand

The estimand aims to evaluate the efficacy of retatrutide that reflects the real-life behavior of the target population.

Treatment regimen estimand attributes

- **Population** - the target population is the group of individuals who meet the eligibility criteria. For the analysis, the intent-to-treatment principle will be applied.
- **Endpoint** - change from baseline in HbA1c at Week 40.
- **Treatment condition** - the randomly assigned treatment with allowance for potential dose interruptions and dose modification regardless of adherence to study intervention or initiation of additional antihyperglycemic medications.
- **Intercurrent events** - no ICEs are defined since treatment adherence and the initiation of additional antihyperglycemic medications are a part of the treatment condition.
- **Population-level summary and treatment effect of interest** - difference in mean changes from baseline in HbA1c at Week 40 with retatrutide 4 mg, 9 mg, and/or 12 mg compared to placebo.

Efficacy estimand

The clinical question of interest is: *What is the treatment difference in the change in HbA1c from baseline at Week 40 between retatrutide 4 mg, 9 mg, and/or 12 mg compared with placebo in individuals who meet the eligibility criteria if they would remain on their randomly assigned*

study intervention for 40 weeks and would not initiate additional antihyperglycemic medications?

Rationale for the efficacy estimand

This estimand aims to evaluate the efficacy of retatrutide under the ideal condition that all participants would adhere to the randomly assigned study intervention without being confounded by the initiation of additional antihyperglycemic medications.

Efficacy estimand attributes

- **Population** - the target population is the group of individuals who meet the eligibility criteria. For the analysis, the intent-to-treatment principle will be applied.
- **Endpoint** - change from baseline in HbA1c at Week 40.
- **Treatment condition** - the randomly assigned treatment with allowance for potential dose interruptions and dose modifications.
- **Intercurrent events** - ICEs include permanent discontinuation of study intervention and initiation of additional antihyperglycemic medications and will be handled by the hypothetical strategy. The potential outcome of interest is the response in the efficacy measurement if participants would remain on their randomly assigned study intervention for 40 weeks and would not initiate additional antihyperglycemic medications. Dose modifications and interruptions will not be considered an ICE since they are part of the treatment condition.
- **Population-level summary and treatment effect of interest** - difference in mean change from baseline in HbA1c at Week 40 with retatrutide 4 mg, 9 mg, and/or 12 mg compared to placebo.

4. Study Design

4.1. Overall Design

Study GZBY is a Phase 3, multicenter, randomized, parallel-arm, double-blind, placebo-controlled study to investigate the efficacy and safety of retatrutide 4 mg, 9 mg, or 12 mg compared with placebo in participants with T2D and inadequate glycemic control with diet and exercise alone, are naïve to insulin use and have not been treated with any antihyperglycemic medication during the 90 days preceding Visit 1.

Approximately 480 participants will be randomly assigned in a 1:1:1:1 ratio to retatrutide 4 mg (approximately 120), retatrutide 9 mg (approximately 120), retatrutide 12 mg (approximately 120), and placebo (approximately 120), respectively.

This study will consist of 3 periods:

- a screening period: up to 4 weeks
- a treatment period: 40 weeks, including dose escalation and maintenance, and
- a safety follow-up period: 4 weeks.

4.2. Scientific Rationale for Study Design

Choice of primary endpoint

The primary endpoint of HbA1c is a commonly used endpoint in clinical trials evaluating the efficacy and safety of antihyperglycemic agents.

Choice of comparator

Regulatory authorities require the safety profile of new drugs to be well-characterized, including a sufficient number of placebo exposures. Per ICH E 1 (EMA 1995), a placebo-controlled design is necessary to directly compare the AE rate in the drug-treated group with the background event rate in the patient population being studied.

Double-blind design

A double-blind design was selected to minimize participant and investigator bias in assessments for efficacy, safety, and study intervention tolerability.

Duration of the study treatment

The planned duration of the study treatment for 40 weeks is considered sufficient and appropriate for participants to optimize dosing of retatrutide and to assess the full effects and benefit/risk of each treatment dose of retatrutide on glycemic control.

Collection of race and ethnicity data

In this study, collection of demographic information includes race and ethnicity. The scientific rationale is based on the need to assess variable response in safety and/or efficacy based on race or ethnicity. This question can be answered only if all the relevant data are collected.

4.3. Justification for Dose

Retatrutide doses of 4 mg, 9 mg, and 12 mg QW will be evaluated in this study for assessment of efficacy and safety for glycemic control in patients with T2D.

These doses and associated escalation scheme were selected based on assessment of safety, efficacy, (improvement in glycemic control and body weight reduction), tolerability data, and PK/PD exposure–response modeling of data from Phase 1 and Phase 2 studies.

Participants randomly assigned to retatrutide will start with a 2 mg QW dose and have a dose escalation every 4 weeks until the 4 mg, 9 mg or 12 mg dose is reached. This dose escalation scheme is designed to minimize GI tolerability concerns.

The maintenance doses of 4 mg, 9 mg, and 12 mg are expected to provide clinically relevant glycemic control relative to placebo with increasing efficacy with increasing dose.

The selected dose levels and dose range will support a robust dose–exposure–response analysis of multiple safety and efficacy measures to support further evaluation of the benefit/risk of the selected retatrutide doses.

4.4. End of Study Definition

The end of the study is defined as the date of the last visit of the last participant in the study or last scheduled procedure shown in the SoA for the last participant in the trial globally.

5. Study Population

Participant eligibility for enrollment in the study is based on the criteria listed in this section. The inclusion and exclusion criteria used to determine eligibility should only apply at screening or other specified visits, and not continuously throughout the study.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1. Inclusion Criteria

Participants are eligible to be included in the study only if all of the following criteria apply:

Age

1. Participant must be at least 18 years of age or the legal age of consent in the jurisdiction in which the study is taking place at the time of signing the informed consent, whichever is older.

Type of participant and disease characteristics

2. Have a clinical diagnosis of T2D based on the World Health Organization classification or other locally applicable diagnostic standards (see Section 10.8).
3. Have HbA1c $\geq 7.0\%$ (53 mmol/mol) to $\leq 9.5\%$ (80 mmol/mol) despite diet and exercise treatment, as determined by the central laboratory at Visit 1 (screening).
4. Are naïve to insulin therapy except for gestational diabetes or ≤ 14 days use for acute treatment and have not used any oral or injectable antihyperglycemic medications during the 90 days preceding Visit 1, or between Visit 1 and Visit 3 (randomization).
5. Are motivated and willing to make themselves available for the duration of the study and are able to follow study procedures as required, including
 - store provided study intervention as directed
 - perform SMBG per protocol
 - maintain a study eDiary
 - complete patient questionnaires, and
 - self-inject study intervention.

Note: Persons with physical limitations who are not able to perform the injections must have the assistance of an individual trained to inject the intervention.

Weight

6. Are of stable weight (self-reported change ≤ 5 kg or 11 pounds) for at least 90 days prior to Visit 1 and agree to not initiate an intensive diet or exercise program during the study with the intent of reducing body weight other than the lifestyle and/or dietary measures for diabetes treatment.
7. Have a BMI ≥ 23.0 kg/m² at Visit 1.

Sex and contraceptive/barrier requirements

8. Males, WOCBP, or WNOCBP may participate in this trial.

Contraceptive use by participants should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies. For the contraception requirements of this protocol, see Section 10.4.

Note: the word “male” is used to describe an individual assigned male at birth (AMAB) and the word “female” is used to describe an individual assigned female at birth (AFAB), for simplicity.

Informed consent

9. Are capable of giving signed informed consent as described in Section 10.1.3, which includes compliance with the requirements and restrictions listed in the ICF and in this protocol.

5.2. Exclusion Criteria

Participants are excluded from the study if any of the following criteria apply:

Medical conditions***Diabetes-related***

10. Have T1D.
11. Have a history of ketoacidosis or hyperosmolar state or coma within the last 6 months prior to Visit 1, or between Visit 1 and Visit 3.
12. Have a history of severe hypoglycemia or hypoglycemia unawareness within the last 6 months prior to Visit 1, or between Visit 1 and Visit 3.
13. Are currently receiving or planning to receive treatment for diabetic retinopathy and/or macular edema, for example, laser photocoagulation or intravitreal injections of anti-vascular endothelial growth factor inhibitors.

Note: All participants will have a baseline fundus photography assessment. In the presence of any degree of diabetic retinopathy and/or macular edema, further ophthalmologic assessment will follow local standards of care to confirm eligibility.

Obesity-related

14. Have a prior or planned surgical treatment for obesity

Exceptions: The following are allowed if they occurred >1 year before screening (Visit 1):

- liposuction
- cryolipolysis, or
- abdominoplasty.

15. Have a prior or planned endoscopic (for example, mucosal ablation or gastric artery embolization) and/or device-based (for example, lap band, intragastric balloon, or endoluminal duodenal-jejunal bypass liner) therapy for obesity.

Exception: Prior device-based therapy is acceptable if device removal was more than 6 months prior to screening (Visit 1).

Cardiovascular

16. Have New York Heart Association Functional Classification IV congestive heart failure.
17. Have had any of the following CV conditions within 90 days prior to Visit 1, or between Visit 1 and Visit 3:
- acute myocardial infarction
 - cerebrovascular accident (stroke), or
 - hospitalization for congestive heart failure.

Gastrointestinal

18. Have a known clinically significant gastric emptying abnormality, for example, severe diabetic gastroparesis or gastric outlet obstruction, or chronically take drugs that directly affect GI motility.

Hepatic

19. Have acute or chronic hepatitis, including a history of autoimmune hepatitis, signs or symptoms of any other liver disease other than MASLD, or any of the following as determined by the central laboratory during screening:
- ALT or AST level $>5.0 \times$ ULN for the reference range
 - ALP level $\geq 1.5 \times$ ULN for the reference range
 - TBL level $\geq 1.5 \times$ ULN for the reference range, except for cases of known Gilbert's syndrome
 - hepatitis B infection, defined as
 - positive hepatitis B core antibody and positive for HBV DNA, or
 - positive hepatitis B surface antigen, and
 - positive hepatitis C antibody and positive for HCV RNA.

Renal

20. Have an eGFR <15 mL/min/1.73 m² as determined by the central laboratory at Visit 1 (calculated by Chronic Kidney Disease-Epidemiology cystatin-c equation).

Hematologic

21. Have any clinically relevant hematological condition that may interfere with HbA1c measurement, for example, hemolytic anemias or sickle cell disease.

Malignancy

22. Have a history of an active or untreated malignancy or are in remission from a malignancy for less than 5 years.

Exceptions: basal or squamous cell skin cancer, in situ carcinomas of the cervix, or in situ or Grade 1 (for example, Gleason 6 or lower) prostate cancer.

Endocrine

23. Have family (first-degree relative) or personal history of medullary thyroid carcinoma or multiple endocrine neoplasia syndrome Type 2.
24. Have a serum calcitonin level of
- a. ≥ 20 pg/mL (≥ 5.9 pmol/L) at Visit 1, if eGFR ≥ 60 mL/min/1.73 m², and
 - b. ≥ 35 pg/mL (≥ 10.2 pmol/L) at Visit 1, if eGFR < 60 mL/min/1.73 m²
- as determined by the central laboratory at Visit 1.

Psychobehavioral

25. Have a history of unstable major depressive disorder or other severe psychiatric disorder, such as schizophrenia, bipolar disorder, or other serious mood or anxiety disorder, within the last 2 years.

OR

In the investigator's opinion, have any significant mental health disorder that may put the individual at higher risk of study participation.

26. Have a PHQ-9 score of 15 or more at Visit 1 or Visit 3.
27. Have any lifetime history of a suicide attempt.
28. Are, in the judgment of the investigator, actively suicidal and therefore deemed to be at significant risk for suicide.
29. Have answered "yes" to either Question 4 or Question 5 on the "Suicidal Ideation" portion of the C-SSRS, **and** the ideation occurred within the past month prior to Visit 1 or Visit 3.

OR

Have answered "yes" to any of the suicide-related behaviors on the "Suicidal Behavior" portion of the C-SSRS, **and** the behavior occurred within the past month prior to Visit 1 or Visit 3.

General

30. Had chronic or acute pancreatitis any time.
31. Have had a transplanted organ or awaiting an organ transplant.
- Exception:** corneal transplants (keratoplasty).

Prior/concomitant therapy

32. Within 90 days prior to Visit 1, or between Visit 1 and Visit 3

- are receiving or have received chronic (>14 days) systemic glucocorticoid therapy, excluding topical, intraocular, intranasal, interphalangeal, or inhaled preparations, or
 - have evidence of a significant, active autoimmune abnormality (for example, lupus or rheumatoid arthritis) that is likely to require, in the opinion of the investigator, such treatment during the study.
33. Have taken prescribed or over-the-counter medication, or alternative remedies, including herbal or nutritional supplements, intended to promote body weight reduction within 90 days prior to Visit 1, or between Visit 1 and Visit 3 (see Section 10.9 for examples).
34. Have started SC/intramuscular or implantable/injectable contraceptives such as Depo-Provera®, Nexplanon®, within 18 months prior to screening (Visit 1).

Note: Intrauterine devices are acceptable.

Prior/concurrent clinical trial experience

35. Are currently enrolled in any other clinical study involving an IP or any other type of medical research judged not to be scientifically or medically compatible with this study.
36. Have previously completed or withdrawn from this study or any other study investigating retatrutide after receiving at least 1 dose of study intervention (retatrutide, placebo, or an active comparator).
37. Have participated in a clinical study and received treatment, whether active or placebo within the last 30 days or duration equal to 5 times the half-life of study intervention, whichever is longer.

Other exclusion criteria

38. Have any condition, unwillingness, or inability not covered by any of the other exclusion criteria, that, in the opinion of the investigator, might
- jeopardize the safety of the participant; for example, allergies, intolerance, or hypersensitivity to GLP-1 RAs, GIP/GLP-1 RAs, or retatrutide or excipient or contraindication
 - compromise compliance with the protocol; for example, recreational drug use or alcohol abuse, or
 - confound data interpretation.
39. Are investigator site personnel directly affiliated with this study and/or their immediate families. Immediate family is defined as a spouse, parent, child, or sibling, whether biological or legally adopted.
40. Are Eli Lilly and Company employees or are employees of any third party involved in the study who requires exclusion of their employees.
41. Are pregnant or intend to become pregnant or to breastfeed during the study.

5.3. Lifestyle Considerations

Throughout the study, participants will have access to qualified medical staff who will provide diabetes management counseling, which will include instructions on diet and exercise and education about the signs, symptoms, and treatment of hypoglycemia.

Participants should be counseled to continue their usual exercise habits and follow a healthy meal plan, with an emphasis on intake of healthy carbohydrate, protein, fat, and fluid.

Dietary counseling will be provided throughout the study, as needed. Per inclusion criterion 6 (Section 5.1), participants should not initiate during the study an organized diet or exercise weight reduction program other than the lifestyle and/or dietary measures for diabetes treatment.

For certain assessments, the participants will be required to come to the site in a fasting state, except for water, for at least 8 hours, as specified in the SoA.

5.3.1. Diabetes Education

Diabetes education is to be performed by personnel who are qualified to educate participants on symptoms and management of hyperglycemia and hypoglycemia and diabetes management. All trainings should be repeated as needed to ensure participant compliance.

5.3.2. Self-Monitoring of Blood Glucose (SMBG)

At Visit 2, participants will receive a study-approved glucometer and related testing supplies for use during the study.

Site personnel will train the participant on correct use of the glucometer for SMBG and reporting of hypoglycemia. Participants will be instructed to monitor BG any time they have symptoms suggestive of hypoglycemia and will collect 4-point and 7-point SMBGs as outlined below.

Four-point SMBG

Participants will be instructed to perform 4-point SMBG measurements once weekly during the treatment period and to record all results in the eDiary according to instructions; these results will be used for assessment of severe persistent hyperglycemia (Section 6.9.4). Four-point SMBG profiles will consist of measurements obtained on a single day

- before 3 main meals, and
- at bedtime.

During the weeks when the 7-point SMBG profiles are to be collected, participants will not be required to collect 4-point SMBG profiles, and the 4-point BG values included in the 7-point profiles may be used for assessment of severe persistent hyperglycemia.

Seven-point SMBG

Participants will also be instructed to collect two 7-point SMBG profiles on non-consecutive days within 2 weeks prior to specified visits and record all results in the eDiary according to instructions preceding prespecified visits as shown in the SoA (Section 1.3); these results will be used for the efficacy analysis. Seven-point SMBG profiles will consist of measurements obtained on a single day

- before 3 main meals

- approximately 2 hours after 3 main meals, and
- at bedtime.

5.3.3. Meals and Nutrition

Participants will have access to qualified medical staff who will provide diabetes management counseling regarding a healthy diet.

Counseling considerations

The focus of counseling should facilitate balanced and sufficient food and beverage choices for adequate nutritional intake and hydration. Counseling should include instruction on customized nutrient-dense food and beverage choices to reflect personal preferences, cultural traditions, and budgetary considerations.

Adequate nutritional intake and hydration recommendations may be individualized based on the participant's health status, including age and comorbidities, appetite levels, and GI tolerability. Recommendations should focus on prioritizing intake of healthy carbohydrate, protein, fat, and fluid.

Recommendations may include the following:

	General daily intake recommendations
Fluids (that is, water or decaffeinated and non-sugary beverages)	>2000 to 3000 mL
Energy	a minimum of 800 to 1000 kcal (recommended energy intakes should be individualized based on factors like age, sex, and activity level).
Protein	0.8 to 1.5 g/kg, consider adjustment for participants with CKD (in some cases, calculation may be based on ideal body weight)
Carbohydrates	45% to 65% of energy intake
Fiber	>21 g (women) >30 g (men)
Fat	20% to 35% of energy intake

Consider the following principles to guide participant education in counseling sessions:

- the importance of meeting nutritional and hydration recommendations
- decreased appetite and increased fullness after eating typically drive a reduction in food intake. Additional voluntary food restriction, when appetite is reduced, is unnecessary and may be harmful
- avoid skipping meals
- eat smaller, more frequent (every 3 to 4 hours) meals or snacks
- consider approximately ½ plate from fruits or vegetables, ¼ plate from whole grains, and ¼ plate from protein (from an 8-inch to 10-inch plate)
- include fiber-rich foods, and
- limit foods high in solid fats, added sugar, and salt.

In addition, the following strategies are encouraged to facilitate adequate nutritional intake and hydration:

- multivitamin and/or targeted micronutrient supplementation, as clinically appropriate
- meal replacements (for example, protein- and micronutrient-rich shakes or bars) that can be sipped or eaten slowly over several hours, particularly for participants with markedly reduced appetite and poor intake, and
- more frequent nutritional counseling, either in person or virtually, if needed (contact the Lilly Medical team with any questions for implementation).

5.3.4. Monitoring Nutritional Needs

Retatrutide has the potential for significant body weight reduction. It acts, in part, by reducing appetite leading to a reduction in food intake. There is a potential that some participants will have significantly reduced caloric intake.

Individual risk factors for poor nutrition or dehydration should be taken into consideration, including, but not limited to, age, medical history/comorbidities, concomitant medications, clinical conditions (for example, infection, renal function status), and GI symptoms (for example, nausea, vomiting, diarrhea). Clinically monitor the nutritional and hydration status of participants, and assess for any signs or symptoms of poor nutrition or dehydration, including

- vital signs
- targeted physical exam
- recent central laboratory data, and
- review of relevant adverse events and other signs or symptoms, such as fatigue, skin tenting, and dark-colored urine.

Obtain additional laboratory tests, as clinically appropriate, via the central laboratory. Use local laboratory tests as needed, for example, for acute management or if laboratory test is not available centrally. See Section 10.2 for more information on clinical laboratory tests.

By recognizing early signs of poor nutrition and dehydration, preventive actions can be taken to reduce the risk of potential complications.

Monitor for the following which could be risk factors/signs for poor nutrition or dehydration

Potential concern	Early signs include, but are not limited to
Poor nutrition	carbohydrate intake <50 g/day protein intake <60 g/day (minimum protein intake requirements may be different for participants with CKD) fat intake <15 g/day caloric intake <800 to 1000 kcal/day skipping multiple meals
Dehydration	low fluid intake (<2000 mL/day) volume loss (for example, diuretic or laxative use, diarrhea, vomiting) elevated pulse rate decreased or low BP high blood urea nitrogen to creatinine (BUN/Cr) ratio

If there are concerns for poor nutrition or dehydration, mitigations are based on an assessment of the severity of the participant's status, potential underlying etiologies, and GI symptoms. See Sections 6.6.2 and 10.11 for the recommended steps to follow.

5.3.5. Blood Donation

Study participants should be instructed not to donate blood or blood products during the study, starting at Visit 1, and up to Visit 801 following the study.

5.4. Screen Failures

A screen failure occurs when a participant who consents to participate in the clinical study is not subsequently assigned to study intervention. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the CONSORT publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, and any SAE.

Individuals who do not meet the criteria for participation in this study (screen failure) may be considered for rescreening once at the investigator's discretion and with medical monitor approval. If rescreening is performed, the individual must sign a new ICF and will be assigned a new identification number.

5.5. Criteria for Temporarily Delaying Enrollment of a Participant

Not applicable.

6. Study Intervention(s) and Concomitant Therapy

Study intervention is defined as any medicinal product(s) or medical device(s) intended to be administered to or used by a study participant according to the study protocol.

6.1. Study Intervention(s) Administered

This table lists the interventions used in this clinical study.

Intervention Name	Retatrutide	Placebo
Dosage Level(s)	2 mg 4 mg ^a 6 mg 9 mg ^a 12 mg ^a	Not applicable
Route of Administration	Subcutaneous injection QW	Subcutaneous injection QW

Abbreviation: QW = once weekly.

^a Retatrutide 4 mg, 9 mg, and 12 mg are planned maintenance doses for the treatment period.

Packaging and labeling

Study interventions will be supplied by the sponsor or its designee in accordance with current Good Manufacturing Practice. Study interventions will be labeled as appropriate for country requirements.

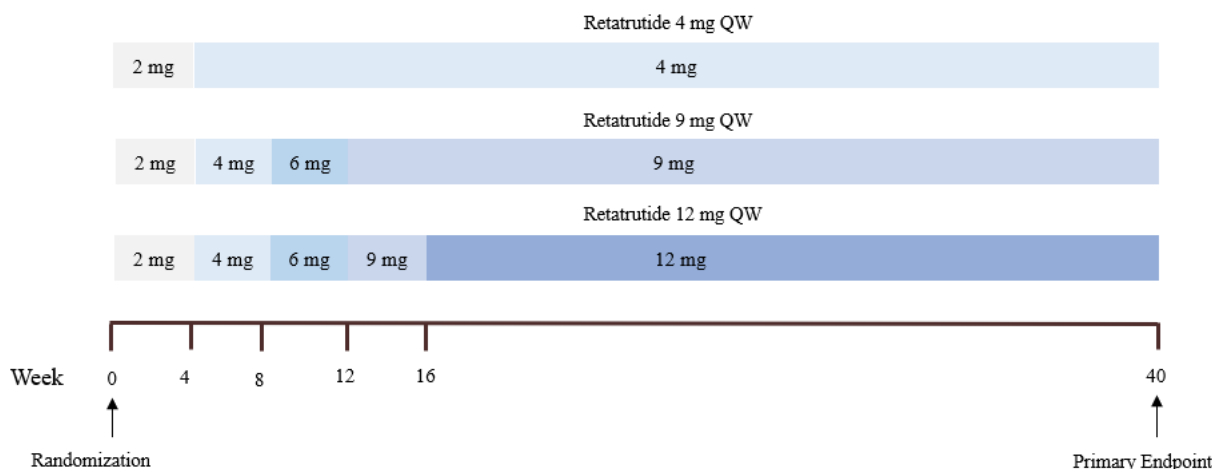
6.1.1. Dose Escalation Schema

Participants randomly assigned to retatrutide

Participants randomly assigned to retatrutide, will start with a 2 mg QW dose and increase dose every 4 weeks until the randomly assigned maintenance dose (4 mg or 9 mg or 12 mg) is reached as outlined in the figure below.

To maintain blinding, participants randomly assigned to 4 mg and 9 mg will follow a sham dose escalation schedule to match the retatrutide 12 mg treatment group. The maintenance dose (4 mg or 9 mg or 12 mg) of retatrutide will be continued for the remainder of the treatment period. For management of participants with intolerable GI symptoms, see Section 6.6.1.

Investigators, site staff, clinical monitors, and participants will remain blinded to the assigned maintenance dose of retatrutide, that is, 4 mg, 9 mg or 12 mg until the study is complete.



Participants randomly assigned to placebo

To maintain blinding, participants randomly assigned to the placebo treatment group will also receive study intervention every 4 weeks during the dose escalation period, following a sham dose escalation schedule to match the retatrutide 12 mg group.

Timing of dose administration

Participants should try to administer the SC injections on the same day of the week and similar time of day, but there are no restrictions on the time of day for each weekly dose.

The actual date and time of all dose administrations will be recorded by the participant in the participant diary. The day of weekly administration can be changed, if necessary, if the last dose was administered 72 or more hours before. If a dose is missed, the participant should take it as soon as possible unless it is within 72 hours of the next dose, in which case that dose should be skipped, and the next dose should be taken at the appropriate time. Participant should record the date and the reason the dose was not taken in the diary.

Anatomical location of injections

Acceptable locations for injection by the participant include the abdomen or thigh, or upper arm if another person gives the injection.

Participants should rotate injection sites when injecting within the same region.

The participant will record the injection site location of all dose administrations in the diary.

6.1.2. Medical Devices, Investigational Devices, or Device Constituents

KwikPen device

The Lilly manufactured medical device provided for use in this study is the multi-dose, single-patient use PFP. The PFP is a mechanical, prefilled, disposable pen injector system with a 3.0-mL cartridge for SC injection of retatrutide. The PFP is designed for once-weekly SC injection of study intervention for adult participants with T2D for doses listed in Section 6.1.

Instructions for device use and risks will be provided.

All PCs including malfunction shall be documented and reported by the investigator throughout the study (see Section 8.3) and appropriately managed by the sponsor.

6.2. Preparation, Handling, Storage, and Accountability

The investigator or designee must confirm appropriate storage conditions have been maintained during transit for all study intervention received and any discrepancies are reported and resolved before use of the study intervention.

Only participants enrolled in the study may receive study intervention. Only authorized study personnel may supply, prepare, or administer study intervention.

All study intervention must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions with access limited to the investigator and authorized study personnel.

The investigator or authorized study personnel are responsible for study intervention accountability, reconciliation, and record maintenance, (that is, receipt, reconciliation, and final disposition records).

Further guidance and information for the final disposition of unused study interventions are provided in the Pharmacy Manual.

6.3. Assignment to Study Intervention

Method of treatment assignment

Participants who meet all criteria for enrollment will be randomly assigned to study intervention using an IWRS.

Study intervention will be dispensed at the study visits summarized in the SoA. Returned study intervention should not be re-dispensed to the participants.

Randomization ratios and stratification factors

Participants will be randomly assigned in a 1:1:1:1 ratio to retatrutide 4 mg, 9 mg, 12 mg, or placebo.

Participants will be stratified based on

- country
- HbA1c ($\leq 8.0\%$, $>8.0\%$) at Visit 1, and
- prior use of any antihyperglycemic medication (yes or no).

6.4. Blinding

This is a double-blind study. Investigators, site staff, clinical monitors, and participants will remain blinded to retatrutide or placebo until the final database lock.

To maintain the blind, a minimum number of Lilly personnel will see the randomization table and treatment assignments before the study is complete.

Unblinding details are specified in the blinding and unblinding plan.

Emergency unblinding

In case of an emergency, the investigator has the sole responsibility for determining if unblinding of a participant's treatment assignment is warranted for medical management of the event. The participant's safety must always be the first consideration in making such a determination. If a participant's treatment assignment is unblinded, Lilly must be notified immediately.

If the investigator decides that unblinding is warranted, it is the responsibility of the investigator to promptly document the decision and rationale and notify Lilly as soon as possible.

Emergency unblinding may be performed through the IWRS. This option may be used ONLY if the participant's well-being requires knowledge of the participant's treatment assignment. All unblinding events are recorded and reported by the IWRS.

Discontinuation from the study intervention in case of unblinding

If an investigator, site personnel performing assessments, or participant is unblinded, the participant must be permanently discontinued from study intervention (Section 7.1.1) but should be continued in the study to be evaluated for efficacy and safety endpoints and monitored for all visits and testing.

6.5. Study Intervention Compliance

Participant compliance with study intervention will be assessed as indicated in the SoA (Section 1.3). Compliance will be assessed by direct questioning and reviewing the eDiary and documented in the source documents and CRF.

Participants will be instructed to return any unused study intervention at the times specified in the SoA.

Treatment compliance for each visit interval is defined as taking at least 75% of the required doses of study intervention.

In addition to the assessment of a participant's compliance with the study intervention administration, other aspects of compliance with the study will be assessed at each visit based on the participant's adherence to the visit schedule, completion of study eDiaries, and any other parameters the investigator considers necessary.

Participants considered to be poorly compliant with their medication and/or the study procedures will receive additional training and instruction, as required, and will be reminded of the importance of complying with the protocol.

6.6. Dose Modification

All dose adjustments for retatrutide and placebo will be managed through IWRS in a blinded fashion.

Dose modification is only allowed for the management of

- intolerable GI symptoms (Section 6.6.1)
- poor nutrition or dehydration (Sections 6.6.2 and 10.11)
- participant reaching a BMI ≤ 20 kg/m² (Section 6.6.3), or

- perceived excessive weight reduction (Section 6.6.3).

6.6.1. Management of Participants with GI Symptoms

Dose modifications for intolerable GI symptoms can be initiated only during dose escalation (the first 20 weeks of the study).

Follow these steps if a participant reports intolerable GI symptoms, such as nausea, vomiting, or diarrhea:

1. Counsel the participant on dietary behaviors that may help mitigate nausea and vomiting, for example,
 - a. eating smaller meals
 - b. splitting 3 daily meals into 4 or more smaller ones
 - c. stopping eating when they feel full, and
 - d. avoiding foods that trigger symptoms.See Sections 5.3.3 and 5.3.4 for additional details.
2. If symptoms persist, prescribe symptomatic medication, for example, antiemetic, or antidiarrheal medication, per local country availability and individual participant needs. Record these medications as concomitant medications in the CRF.
3. If Steps 1 and 2 do not alleviate intolerable GI symptoms, temporary interruption of study intervention for 1 dose is permitted, if the participant has taken the last 3 weekly injections (at any dose). Resume study intervention immediately, when possible, either alone or in combination with symptomatic medication.
4. If Steps 1, 2, and 3 do not alleviate intolerable GI symptoms during dose escalation, follow guidance for temporary dose de-escalation and dose re-escalation. Only 1 de-escalation and re-escalation attempt for intolerable GI symptoms will be permitted.

In the event of intolerable GI symptoms that occur after the dose escalation period, the investigator can advise on Steps 1 and 2 above (in other words, changes in dietary behaviors or consider prescribing symptomatic medication to keep the participant on study intervention) before stopping the treatment permanently.

6.6.1.1. Temporary Dose De-escalation

If intolerable GI symptoms persist despite the measures outlined in Steps 1 to 3 above (Section 6.6.1), during dose escalation, the investigator should contact the sponsor to consider de-escalating the dose to the next lower dose for 4 weeks in a blinded fashion. Dose de-escalation will occur as outlined in the following table.

If a participant is currently taking...	Then dose will be de-escalated for 4 weeks to...
4 mg/Placebo	2 mg/Placebo
6 mg/Placebo	4 mg/Placebo
9 mg/Placebo	6 mg/Placebo
12 mg/Placebo	9 mg/Placebo

Note: If intolerable GI symptoms occur and the participant is currently taking retatrutide 2 mg, then participant will discontinue the study intervention and continue in the study.

6.6.1.2. Dose Re-escalation

If intolerable GI symptoms persist with dose de-escalation described above, the participant should discontinue study intervention and continue in the study. If the participant is able to tolerate the de-escalated dose, the investigator must resume the dose escalation.

If the dose re-escalation results in the return of intolerable GI symptoms then the next step is dependent on whether or not the participant has reached the escalation or sham escalation to 6 mg dose.

If subsequent re-escalation is not tolerated and the participant is currently taking...	Then dose will be de-escalated to...
6 mg/Placebo	4 mg or Placebo in a blinded fashion
9 mg/Placebo	6 mg or Placebo in a blinded fashion
12 mg/Placebo	9 mg or Placebo in a blinded fashion

Note: If a participant does not tolerate retatrutide 4 mg, then the participant will discontinue the study intervention and continue in the study. No dose re-escalation is allowed. Participants in the 4-mg treatment group reaching a sham dose escalation to 6 mg or higher will switch to placebo to maintain the blind.

If intolerable GI symptoms persist despite dose de-escalation, the participant should discontinue study intervention. All participants who discontinue study intervention are encouraged to continue to attend the remaining scheduled study visits (see Section 7.1.1). The study participants who stop study intervention permanently, may receive another antihyperglycemic medication as allowed per protocol (see Section 6.9.4). The new glucose-lowering intervention will be recorded on the CRF specified for collecting antihyperglycemic medications.

6.6.2. Management of Participants with Poor Nutrition or Dehydration

Follow these steps (and refer to Section 10.11) if a participant is at risk of poor nutrition or dehydration based on clinical judgement and criteria described in Sections 5.3.3, 5.3.4, and 10.2.2:

1. Assess if there is an intercurrent illness (for example, systemic infection, cholecystitis, etc) that could be causing or contributing to the inadequate oral intake.
 - a. If yes, consider a temporary interruption of study intervention per Section 6.6.4 until the illness has resolved.
 - b. If no, go to Step 2.

2. Assess if increased oral intake may likely trigger/worsen GI symptoms.
 - a. If no,
 - i. provide nutritional counseling and consider strategies to increase oral intake. Consider consultation with a registered dietitian (or local equivalent) if not already done. Refer to Sections 5.3.3 and 5.3.4.
 - ii. assess concomitant medications and consider adjustment as appropriate; consider laboratory tests as appropriate (see Sections 10.10 and 10.2.2).
 - b. If yes,
 - i. temporarily interrupt the study intervention for 1 to 2 doses
 - ii. provide nutritional counseling and consider strategies to increase oral intake. Consider consultation with a registered dietitian (or local equivalent) if not already done. Refer to Sections 5.3.3 and 5.3.4
 - iii. assess concomitant medications and consider adjustment as appropriate; consider laboratory tests as appropriate (see Sections 10.10 and 10.2.2).
 - iv. on reassessment, if nutritional and hydration concerns have resolved, restart study intervention per Section 6.6.4.
3. If nutritional and hydration concerns persist, contact the sponsor for possible dose reduction to the next lower dose or placebo in a blinded fashion. Dose reduction to retatrutide 2 mg is not allowed. If a participant is currently taking 4 mg retatrutide or 2 mg retatrutide, then the participant may discontinue the study intervention (retatrutide or placebo) permanently and continue in the study. No dose re-escalation is allowed.
4. Assess if the condition persists despite dose reduction.
 - a. If yes, discontinue the study intervention.
 - b. If no, continue the study intervention at the reduced dose.

Only 1 dose reduction for the management of participants with poor nutrition or dehydration, BMI ≤ 20 kg/m², or perceived excessive weight reduction is allowed.

6.6.3. Management of Participants with BMI ≤ 20 kg/m² or Perceived Excessive Weight Reduction

If ...	Then ...
a participant has a BMI ≤ 20 kg/m ² at any time during the study	the investigator may contact the sponsor to consider a dose reduction to the next lower dose or placebo in a blinded fashion or discontinue the study intervention
a participant has a BMI ≤ 18.5 kg/m ² at any time during the study	the investigator must contact the sponsor to consider dose reduction to the next lower dose or placebo in a blinded fashion or discontinue the study intervention
a participant with perceived excessive weight loss decides and wishes to discontinue from study intervention	the investigator must contact the sponsor to consider dose reduction to the next lower dose or placebo in a blinded fashion or discontinue the study intervention

Note: Dose reduction to retatrutide 2 mg is not allowed. If a participant is currently taking 4 mg retatrutide or 2 mg retatrutide, then the participant may discontinue the study intervention (retatrutide or placebo) permanently and continue in the study. Participants in the 4-mg treatment group reaching a sham dose escalation to 6 mg or higher will switch to placebo to maintain the blind.

The decision of study intervention discontinuation or dose reduction will depend on several factors including timing of when these criteria are met, whether the participant is in weight reduction or weight maintenance phase, other comorbid conditions, concomitant medication, and AEs, and will be mutually agreed between sponsor and investigator to ensure safety of participants.

Only 1 dose reduction for the management of participants with poor nutrition or dehydration, BMI ≤ 20 kg/m², or perceived excessive weight reduction is allowed. No dose re-escalation is allowed.

Participants with low BMI (≤ 20 kg/m²) should be assessed at least every 2 to 4 weeks for weight stabilization with either an in-office weight check or self-report via telehealth.

All participants who discontinue study intervention should be encouraged to continue to attend the remaining scheduled study visits and continue the study (see Section 7.1.1). Study participants who stop study intervention permanently should receive another antihyperglycemic medication as allowed per protocol (see Section 6.9.4). The new glucose-lowering intervention will be recorded on the CRF specified for collecting antihyperglycemic medications.

6.6.4. Temporary Interruption of Study Intervention Unrelated to Gastrointestinal Tolerability

In certain situations, the investigator may need to temporarily interrupt study intervention, for example, if the participant experiences acute illness, surgery, requires hospitalization, due to an AE unrelated to GI tolerability, or a clinically significant laboratory value. Every effort should be made by the investigator to maintain participants on study intervention and to restart study intervention after any temporary interruption, as soon as it is safe to do so.

When restarting study intervention, notify the sponsor at least 2 weeks prior to the dosing visit to ensure adequate study intervention supply at the site.

1. If the number of consecutive missed doses is ≤ 2 , and the dose was well-tolerated prior to the interruption, then restart retatrutide at the same dose the participant received before the interruption or if during the first 20 weeks, the escalation will continue as planned.
2. If the number of consecutive missed doses is ≥ 3 ,
 - a. in the first 4 weeks of the study, restart retatrutide at 2 mg and subsequently escalate the dose as required by protocol.
 - b. after the first 4 weeks of the study, restart retatrutide at 4 mg, regardless of the dose the participant received before the interruption and subsequently escalate the dose as required by protocol.
3. The investigator will use the IWRS to receive the appropriate study intervention dispensing information to preserve blinding of the retatrutide dose.

The data related to temporary interruption of study intervention will be documented in source documents and entered on the CRF.

6.7. Continued Access to Study Intervention after the End of the Study

Study intervention will not be made available to participants after the conclusion of the study.

6.8. Treatment of Overdose

For this study, any total dose of study intervention within a 72-hour period for retatrutide, greater than the dose assigned by IWRS for that participant will be considered a potential overdose and should be assessed for AE reporting as per criteria described in Section 10.3. In the event of an overdose, refer to the IB for retatrutide.

In the event of an overdose, the investigator should

- initiate supportive treatment according to the participant's clinical signs and symptoms
- contact the medical monitor immediately
- evaluate the participant to determine, in consultation with the medical monitor, whether study intervention should be interrupted
- closely monitor the participant for any AE or SAE and laboratory abnormalities that could be associated with the overdose and initiate supportive treatment according to the participant's clinical signs and symptoms, and
- obtain a plasma sample for PK analysis within 5 days from the date of the last dose of study intervention if requested by the medical monitor. This is determined on a case-by-case basis.

6.9. Prior and Concomitant Therapy

6.9.1. General Considerations

Data on prior treatment with GLP-1 RA or GIP/GLP-1 RA will be collected in CRF per the SoA (Section 1.3). Prior treatment refers to the treatment that was received by the participant before enrollment in the study and discontinued prior to Visit 1.

Participants are permitted to use concomitant medications that they require during the study except medications that may interfere with the assessment of efficacy and safety characteristics of the study interventions.

Participants must consult with the investigator or a designated site staff member if they are prescribed any new medications during the study. If this is not possible due to treatment of medical emergencies, the participant should inform the investigator or a designated site staff member as soon as possible.

The initiation of weight promoting medications is discouraged and alternate therapies not associated with weight gain should be considered whenever possible.

In this study population, many study participants will be taking medications to treat comorbidities, including antihypertensive, diuretic, and lipid-lowering agents, or medications to

mitigate GI tolerability, including laxatives and anti-emetics, or medications with weight-based dosing, such as thyroid hormone.

Investigators are responsible for closely monitoring the need to adjust these medications throughout the study. See Section 10.10 for concomitant medication considerations. Discuss medication adjustment as soon as possible with primary care physician or specialist, as applicable.

Recording concomitant medications

Site staff will record changes in intensity of medications of special interest on the level of medication intensity CRF at intervals specified in the SoA. These changes should also be recorded in the concomitant medication CRF. Medications of special interest that should be recorded in both CRFs include

- antihypertensive, and
- lipid lowering medications.

Any medication or vaccine, over-the-counter or prescription medicine (including hormonal contraceptives), vitamins or herbal supplements, or other specific categories of interest that the participant is receiving at the time of enrollment or receives during the study must be recorded in the CRF along with

- reason for use
- dates of administration including start and end dates, and
- dosage information including dose and frequency only for concomitant medications of special interest, for example antihypertensive, antihyperglycemic, and lipid lowering medications.

6.9.2. Prohibited or Restricted Use Medications

The following medications are prohibited throughout the study (up to Visit 801):

- antihyperglycemic medications
 - GLP-1 RA or other incretin-based therapies (for example tirzepatide), pramlintide (amylin analogs), DPP-4 inhibitors (Section 10.9.1.1).
 - Use of other antihyperglycemic medications regardless of indication (for example, heart failure and CKD) is prohibited except:
 - permanent discontinuation of the study intervention (Section 7.1.1)
 - rescue therapy due to severe persistent hyperglycemia (Section 6.9.4), or
 - additional glycemic control during the safety follow-up period.
- subcutaneous/intramuscular or implantable/injectable contraceptives such as Depo-Provera[®], Nexplanon[®], are prohibited except when initiated and used continuously for at least 18 months prior to study entry, and
- medications intended to promote weight reduction, including prescribed, over the counter, or alternative remedies. Examples are provided in Section 10.9.1.2.

Note: Short-term mealtime insulin use for up to 14 days is allowed for certain clinical situations, for example, elective surgery, during hospitalization, and hyperosmolar states, and must be differentiated from mealtime insulin use as rescue therapy when reported in the CRF.

6.9.3. Prohibited Surgical Treatments or Procedures for Weight Management

Surgical treatments, endoscopic therapy, and/or device-based therapy for weight management are not permitted during the study. Refer to Section 5.2 for details.

6.9.4. Rescue Therapy for Severe Persistent Hyperglycemia

Criteria for severe persistent hyperglycemia

Investigators will be trained on how to apply decision criteria for the timing and method of intervention in participants who do not reach glycemic targets during the treatment period. An additional therapeutic intervention should be considered in participants who are fully compliant with the assigned therapeutic regimen in the absence of any acute condition that raises BG, AND who meet any of the following criteria:

Criteria for severe persistent hyperglycemia are described in the table below.

If a participant has...	
An average BG from the once-weekly 4-point SMBG profile of	
>270 mg/dL (15.0 mmol/L)	over at least a consecutive 2-week period any time during the first 8 weeks postrandomization
>240 mg/dL (13.3 mmol/L)	over at least a consecutive 2-week period at any time 9 to 16 weeks postrandomization
>200 mg/dL (11.1 mmol/L)	over at least a consecutive 2-week period at any time beyond the first 16 weeks postrandomization
HbA1c $\geq$ 8.5% (69 mmol/mol)	at or after Week 24 with inadequate response to the existing regimen defined as improvement in HbA1c over the last 3 months that is <0.3%.

Abbreviations: BG = blood glucose; HbA1c = hemoglobin A1c; SMBG = self-monitoring of blood glucose.

During the weeks when 7-point SMBG profiles are collected, the once-weekly 4-point SMBG values are included in the 7-point profiles.

6.9.5. Rescue Medication

For participants who meet criteria for severe persistent hyperglycemia, the investigator will decide on an appropriate antihyperglycemic intervention (rescue therapy) that is not a prohibited medication listed in Sections 6.9.2 and 10.9.

If a new antihyperglycemic medication is introduced, metformin should be the first choice of therapy unless contraindicated.

Initiation of insulin as the first rescue intervention for hyperglycemia should be reserved for

- participants with severe persistent hyperglycemia with an average BG from the once-weekly 4-point SMBG profile ≥ 300 mg/dL (≥ 16.7 mmol/L), or
- participants with symptoms of hyperglycemia, if there is evidence of ongoing catabolism or in other clinical situations where the investigator believes more rapid glycemic control is warranted (ADA 2023).

Participants who receive rescue medicine for hyperglycemia management should also continue administering study intervention for the remaining period in the trial. Record the use of rescue medication as concomitant medication in the CRF.

If any new antihyperglycemic medication is initiated during the safety follow-up period, it will not be classified as rescue therapy.

6.9.6. Symptomatic Medication for Gastrointestinal Symptoms

The investigator may prescribe symptomatic medication for management of GI symptoms as needed during the study, for example, antiemetic or antidiarrheal medication, per local country availability and standard of care (see Section 6.6.1). Record the use of symptomatic medication as concomitant medication in the CRF.

7. Discontinuation of Study Intervention and Participant Discontinuation/Withdrawal

Discontinuation of specific sites or of the study as a whole are handled as part of Section 10.1.

7.1. Discontinuation of Study Intervention

7.1.1. Permanent Discontinuation from Study Intervention

When necessary, a participant may be permanently discontinued from study intervention. If so, the participant will discontinue the study intervention and will remain in the study and follow procedures for all remaining study visits, as shown in the SoA.

If the participant is unwilling or unable to return for all applicable study visits, the site should attempt to collect as much follow-up information as possible, especially data collection pertaining to primary and key secondary efficacy endpoints at Week 40 of the treatment period.

A participant should be permanently discontinued from study intervention if

- the participant requests to discontinue study intervention
- the participant is diagnosed with acute or chronic pancreatitis, confirmed by adjudication. See Section 8.3.3.7
- the participant is diagnosed with MTC or MEN2 syndrome
- the participant develops significant elevation of serum calcitonin. See Section 8.3.3.8
- the participant is diagnosed with an active or untreated malignancy (other than basal or squamous cell skin cancer, in situ carcinomas of the cervix, or in situ prostate cancer, for example, Gleason 6 or lower)
- the participant is diagnosed with T1D
- in the opinion of the investigator, for safety reasons
- the participant has intolerable GI symptoms despite management as described in Section 6.6.1
- the participant initiates any other GLP-1 RA or other incretin-based therapies (for example, tirzepatide), pramlintide (amylin analogs), or DPP-4 inhibitors and will not or cannot discontinue the medication
- the participant develops any other TEAE, SAE, or clinically significant laboratory value for which the investigator believes that permanent study intervention discontinuation is the appropriate measure to be taken, or
- the participant becomes pregnant.

Other possible reason which may lead to permanent discontinuation of study intervention include

- if the participant is unable to maintain adequate nutrition or hydration despite management, as described in Section 6.6.2.
- if the participant has a BMI ≤ 20 kg/m² (see Section 6.6.3)

- the participant has bariatric surgery, body weight reduction procedures, or medications for weight management described in Section 6.9.3
- if the investigator determines that a systemic hypersensitivity reaction has occurred related to study intervention administration, the participant may be permanently discontinued from the study intervention, and the sponsor's designated medical monitor should be notified. If the investigator is uncertain about whether a systemic hypersensitivity reaction has occurred and whether discontinuation of study intervention is warranted, the investigator may consult the sponsor
- PHQ-9 score ≥ 15
- C-SSRS
 - answered "yes" to Question 4 or Question 5 on the "Suicidal Ideation" portion of the C-SSRS, **or**
 - answered "yes" to any of the suicide-related behaviors on the "Suicidal Behavior" portion of the C-SSRS.

Note: In the event either the PHQ-9 score or C-SSRS conditions above are met, participants should be referred to a psychiatrist or appropriately trained professional for consultation. If a participant's psychiatric disorder can be adequately treated with psycho- and/or pharmacotherapy, then the participant, at the discretion of the investigator (in agreement with the psychiatrist or appropriately trained professional), may be continued in the study on study intervention.

Antihyperglycemic therapy after permanent discontinuation of study intervention

Following permanent discontinuation of study intervention for any reason, participants may require additional antihyperglycemic therapy to maintain glycemic control. Any locally approved antihyperglycemic therapy that is not a prohibited/restricted medication per Section 10.9 and is not contraindicated for use in the participant may be prescribed in accordance with local prescribing information.

The new antihyperglycemic intervention will be recorded on the concomitant medications CRF.

7.1.2. Hepatic Criteria for Study Intervention Interruption or Discontinuation

See Section 8.2.6.3 for hepatic criteria for study intervention interruption or discontinuation.

7.1.3. Temporary Interruption of Study Intervention

Refer to Sections 6.6 and 8.2.6.3 for information on temporary dose interruptions.

7.2. Participant Discontinuation/Withdrawal from the Study

Discontinuation is expected to be uncommon. To minimize the amount of missing data and to enable assessment of study objectives as planned in the study protocol, every attempt will be made to keep participants in the study regardless of study intervention use.

Participants will be discontinued from the study if the participant becomes pregnant.

A participant may withdraw from the study

- at any time at the participant's own request for any reason or without providing any reason
- at the request of the participant's designee, for example, parents or legal guardian
- at the discretion of the investigator for safety, behavioral, compliance, or administrative reasons, or
- if enrolled in any other clinical study involving an investigational product or enrolled in any other type of medical research judged not to be scientifically or medically compatible with this study.

At the time of discontinuing from the study, if possible, the participant will complete procedures for an ED visit and safety follow-up, as shown in the SoA.

Participants who have discontinued study intervention early and are only willing to come back for Visit 11 (Week 40) are not required to complete a safety follow-up visit if they have been off of study intervention for at least 28 days.

If the participant has not already discontinued the study intervention, the participant will be permanently discontinued from the study intervention at the time of the decision to discontinue the study.

If the participant withdraws consent for disclosure of future information, the sponsor may retain and continue to use any data collected before such a withdrawal of consent.

If a participant withdraws from the study, the participant may request destruction of any samples taken and not tested, and the investigator must document this in the site study records.

7.3. Lost to Follow up

A participant will be considered lost to follow-up if they repeatedly fail to return for scheduled visits and are unable to be contacted by the study site. Site personnel or designee are expected to make diligent attempts to contact participants who fail to return for a scheduled visit or were otherwise unable to be followed up by the site.

8. Study Assessments and Procedures

Study procedures and their timing are summarized in the SoA.

Section 10.2 lists the laboratory tests that will be performed for this study.

Immediate safety concerns should be discussed with the sponsor immediately upon occurrence or awareness to determine if the participant should continue or discontinue study intervention.

Adherence to the study design requirements, including those specified in the SoA, is essential and required for study conduct.

All screening evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria. The investigator will maintain a screening log to record details of all participants screened and to confirm eligibility or record reasons for screening failure, as applicable.

8.1. Efficacy Assessments

8.1.1. Primary Efficacy Assessment

The primary efficacy assessment in this study is mean change in HbA1c values from baseline to 40 weeks, as determined by the central laboratory (Section 10.2).

Blood samples for HbA1c measurements will be collected at visits specified in the SoA (Section 1.3).

8.1.2. Secondary Efficacy Assessment

- HbA1c (measured through central laboratory; Section 10.2)
- fasting serum glucose (measured through central laboratory; Section 10.2)
- body weight (measuring method is described in Section 10.12)
- lipid parameters (measured through central laboratory; Section 10.2)
- BP (measuring method is described in Section 10.12)
- 7-point SMBG (see Section 5.3.2)
- waist circumference (measuring method is described in Section 10.12)
- BMI (derived using body weight in kilograms divided by the square of height in meters)
- patient-reported outcomes (see Section 8.1.3)
- population PK and PD parameters (see Sections 8.4 and 8.5), and
- safety parameters

8.1.3. Patient-Reported Outcomes

It is possible that availability of the PROs may vary in different countries. The PROs will be administered if available in the language needed for the participant.

If a PRO, other than DTSQc, is not available at baseline, for example, no valid translation exists, then the PRO will not be administered during the study.

8.1.3.1. EQ-5D-5L

EQ-5D-5L (EuroQol Research Foundation 2019) is a standardized 5-item, self-administered instrument for use as a measure of health outcome. It provides a simple descriptive profile and a single index value for health status that can be used in the clinical and economic evaluation of healthcare as well as population health surveys. The EQ-5D-5L assesses 5 dimensions of health:

- mobility
- self-care
- usual activities
- pain/discomfort, and
- anxiety/depression.

Each dimension has 5 response levels: no problems, slight problems, moderate problems, severe problems, and extreme problems.

The visual analogue scale records the respondent's self-rated health on a vertical visual analogue scale where the endpoints are labeled as "best imaginable health state" (100) and "worst imaginable health state" (0).

8.1.3.2. Ability to Perform Physical Activities of Daily Living (APPADL)

The APPADL (Hayes et al. 2012) is a patient-reported outcome instrument that assesses the "current" ability of people living with T2D and obesity to perform physical activities of daily living.

The APPADL contains 7 items that assess how difficult it is for participants to engage in certain activities considered to be integral to normal daily life, such as walking, standing, and climbing stairs (Hayes et al. 2012). Items are scored on a 5-point numeric rating scale, where 5 = "not at all difficult" and 1 = "unable to do." A raw overall score is calculated by simply summing the scores of the 7 items and dividing by the number of items, and then a transformed overall score is obtained by linearly transforming the raw overall score to a 0 to 100 scale. A higher raw overall score and a higher transformed overall score are indicative of better self-reported ability to perform physical activities of daily living.

8.1.3.3. Impact of Weight on Quality of Life-Lite Clinical Trials Version (IWQOL-Lite-CT)

The IWQOL-Lite-CT (Kolotkin et al. 2017, 2019) is a 20-item, obesity-specific, patient-reported outcomes instrument developed for use in weight management clinical studies.

The IWQOL-Lite-CT assesses 2 primary domains of obesity-related, health-related quality of life: Physical (7 items) and Psychosocial (13 items). A 5-item subset of the Physical domain – the Physical Function composite – is also supported. Items in the Physical Function composite describe physical impacts related to general and specific physical activities.

All items are rated on either a 5-point frequency ("never" to "always") scale or a 5-point truth ("not at all true" to "completely true") scale. The 3 domain scores and total score range from 0 to 100 with higher scores indicating greater functioning.

8.1.3.4. Diabetes Treatment Satisfaction Questionnaire-Status Version (DTSQs)

The DTSQs (Bradley and Lewis 1990; Bradley 1994) is an 8-item, diabetes-specific patient-reported outcome instrument that assesses the overall treatment satisfaction and perceived frequency of hyperglycemia and hypoglycemia over the past few weeks, prior to the visit. It is appropriate for use in both T1D and T2D.

Each item is rated on a 7-point Likert scale. Items 1, 4 to 7, and 8 are rated from 0 to 6. These 6 items can be summed up to produce a treatment satisfaction score, with higher scores indicating greater satisfaction. Items 2 and 3 evaluate the perceived frequency of hyperglycemia and hypoglycemia and are rated from 0 (none of the time) to 6 (most of the time), with lower scores indicating better glycemic control.

8.1.3.5. Diabetes Treatment Satisfaction Questionnaire-Change Version (DTSQc)

The DTSQc (Bradley 1999) has the same 8 items as the status version but is reworded slightly to measure the change in treatment satisfaction rather than absolute treatment satisfaction.

Each item is scored on a 7-point scale ranging from -3 to +3. For most items (items 1, 4-7, and 8), higher scores indicate greater improvement in treatment satisfaction, lower scores indicate greater deterioration in treatment satisfaction, and a score of 0 represents no change. For item 2 (perceived frequency of hyperglycemia) and item 3 (perceived frequency of hypoglycemia), lower scores indicate better glycemic control.

8.1.3.6. Diabetes Injection Device – Experience Questionnaire (DID-EQ)

The DID-EQ (Matza et al. 2018) is a self-administered, 10 item questionnaire designed to assess participants' perceptions of non-insulin diabetes injection delivery systems for T2D. Each item is rated on a 4-point scale with higher scores indicating more positive perceptions of the injection device. Items 1 to 7 focus on specific characteristics of non-insulin injection device, and these 7 items comprise the Device Characteristics subscale. In addition, there are 3 global items assessing overall satisfaction, ease of use, and convenience of non-insulin injection devices (items 8, 9, and 10). These 3 global items are each scored separately.

8.1.3.7. Control of Eating Questionnaire (CoEQ)

The CoEQ (Dalton et al. 2015) is a 19-item, participant-completed questionnaire that assesses the intensity of food cravings, food types craved, appetite, and mood over the past 7 days. The 11-point numeric rating scale ranges from 0 (not at all) to 10 (extremely) for each of the 19 items.

Seventeen of the 19 items are grouped into the 4 domains of

- craving control
- positive mood
- craving for savory foods, and
- craving for sweet foods.

The remaining 2 items, which assess the degree of hunger and fullness, are scored individually.

Higher scores represent higher levels of the concept measured in each domain or individual item (for the 2 standalone items).

8.1.3.8. Food Craving Questionnaire-Trait reduced (FCQ-T-r)

The FCQ-T-r (Meule et al. 2014) is a 15-item, participant-completed questionnaire that measures the frequency and intensity of food craving experiences in general.

The questionnaire assesses these 5 domains:

- lack of control overeating
- thoughts or preoccupation with food
- intentions and plans to consume food
- emotions before or during food craving, and
- cues that may trigger food craving.

Items are scored on a 6-point scale from 1 to 6, where

1 = Never/Not applicable, and

6 = Always.

Domain scores are the sum of items scores in each domain.

The total score is the sum of all 15 item scores. Higher scores indicate greater food cravings.

8.2. Safety Assessments

Planned time points for all safety assessments are provided in the SoA.

8.2.1. Physical Examinations

Complete physical examination

A complete physical examination will include, at a minimum, assessments of these systems:

- CV
- thyroid
- respiratory
- GI, and
- neurological.

A complete physical examination excludes pelvic, rectal, and breast examinations unless clinically indicated.

Investigators should pay special attention to clinical signs related to previous serious illnesses.

Symptom-directed physical assessment may be conducted at the discretion of the principal investigator or qualified personnel as indicated per local regulations based on participant status and standard of care.

Additional assessments

Height, weight, waist circumference, and vital signs will also be measured and recorded. See Section [10.12](#) for further details.

8.2.2. Vital Signs

For each participant, vital signs measurements (triplicate) should be conducted according to the SoA (Section 1.3) and following the study-specific recommendations included in Section 10.12.

Any clinically significant findings from vital signs measurement that result in a diagnosis and occur during the study should be reported to the sponsor as an AE via CRF.

8.2.3. Electrocardiograms

Triplicate 12-lead ECGs will be obtained as outlined in the SoA (Section 1.3) using centrally provided ECG equipment. Participants should be supine for approximately 5 to 10 minutes before ECG collections and remain supine but awake during the ECG collection. The recording of each ECG should occur at approximately 1-minute intervals. The collection of all replicates at a time point should not exceed 5 minutes. Collect ECG before blood samples for laboratory testing and PK sampling.

ECGs will be interpreted by the investigator or qualified designee at the site as soon after the time of ECG collection as possible, and ideally while the participant is still present for immediate participant management, should any clinically relevant findings be identified. Any clinically relevant findings from ECGs that result in a diagnosis should be reported as an AE via CRF.

All digital ECGs will be electronically transmitted to a designated central ECG laboratory. The central ECG laboratory will perform a basic quality control check (for example, demographics and study details) and then store the ECGs in a database. At a future time, the stored ECG data may be overread by a cardiologist at the central ECG laboratory for further evaluation of machine-read measurements or to meet regulatory requirements. The machine-read ECG intervals and heart rate may be used for data analysis and report-writing purposes, unless a cardiologist overreading of the ECGs is conducted prior to completion of the final study report (in which case, the overread data would be used).

The investigator, or qualified designee, must document their review of the ECG printed at the time of evaluation and any alert reports.

8.2.4. Clinical Safety Laboratory Tests

See Section 10.2 for the list of clinical laboratory tests to be performed and the SoA for the timing and frequency.

The investigator must review the laboratory results, document this review, and report any clinically relevant changes occurring during the study as an AE. The laboratory results must be retained with source documents unless a Source Document Agreement or comparable document cites an electronic location that accommodates the expected retention duration. Clinically significant abnormal laboratory findings are those which are not associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.

All laboratory tests with values considered clinically significantly abnormal during participation in the study should be repeated until the values return to normal or baseline or are no longer considered clinically significant by the investigator or medical monitor.

- If such values do not return to normal/baseline within a period of time judged reasonable by the investigator, the etiology should be identified, and the sponsor notified.
- All protocol-required laboratory assessments, as defined in Section 10.2, must be conducted in accordance with the SoA, standard collection requirements.

If laboratory values from non-protocol specified laboratory assessments performed at an investigator-designated local laboratory require a change in participant management or are considered clinically significant by the investigator, for example, SAE or AE or dose modification, then report the information as an AE.

8.2.5. Pregnancy Testing

Pregnancy testing will occur as outlined in the SoA.

Participants who become pregnant during the study should be permanently discontinued from the study (Section 7.2). Participants who become pregnant will complete procedures for an ED visit and safety follow-up, as shown in the SoA.

Details of all pregnancies in female participants and, if indicated, female partners of male participants will be collected as outlined in Sections 8.3.1 and 8.3.2.

8.2.6. Hepatic Safety Monitoring, Evaluation, and Criteria for Study Intervention Interruption or Discontinuation

The following tables summarize actions to take based on abnormal hepatic laboratory or clinical changes.

Participants with normal or near normal baseline (ALT, AST, or ALP <1.5x ULN)

If this laboratory value is observed...	Then...		
	Initiate or continue close hepatic monitoring (see Section 8.2.6.1)	Initiate comprehensive evaluation (see Section 8.2.6.2)	Interrupt or discontinue study intervention (see Section 8.2.6.3)
ALT or AST ≥ 3 x ULN	X		
ALP ≥ 2 x ULN	X		
TBL ≥ 2 x ULN ^b	X		
ALT or AST ≥ 5 x ULN	X	X	
ALP ≥ 2.5 x ULN	X	X	
ALT or AST ≥ 3 x ULN with hepatic signs or symptoms ^a	X	X	X
ALT or AST ≥ 5 x ULN for more than 2 weeks	X	X	X
ALT or AST ≥ 8 x ULN	X	X	X
ALT or AST ≥ 3 x ULN and TBL ≥ 2 x ULN ^b or INR ≥ 1.5	X	X	X
ALP ≥ 3 x ULN	X	X	X
ALP ≥ 2.5 x ULN and TBL ≥ 2 x ULN ^b	X	X	X
ALP ≥ 2.5 x ULN with hepatic signs or symptoms ^a	X	X	X

^a Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

^b In participants with Gilbert's syndrome, the threshold for TBL may be higher.

Participants with elevated baseline (ALT, AST, or ALP ≥ 1.5 x ULN)

If this laboratory value is observed...	Then...		
	Initiate or continue close hepatic monitoring (see Section 8.2.6.1)	Initiate comprehensive evaluation (see Section 8.2.6.2)	Interrupt or discontinue study intervention (see Section 8.2.6.3)
ALT or AST ≥ 2 x baseline	X		
ALP ≥ 2 x baseline	X		
TBL ≥ 2 x ULN ^b	X		
ALT or AST ≥ 3 x baseline or ≥ 250 U/L (whichever occurs first)	X	X	
ALP ≥ 2.5 x baseline	X	X	
ALT or AST ≥ 2 x baseline or ≥ 250 U/L (whichever occurs first) with hepatic signs or symptoms ^a	X	X	X
ALT or AST ≥ 3 x baseline or ≥ 250 U/L (whichever occurs first) for more than 2 weeks	X	X	X
ALT or AST ≥ 4 x baseline or ≥ 400 U/L (whichever occurs first)	X	X	X
ALT or AST ≥ 2 x baseline or ≥ 250 U/L (whichever occurs first) and TBL ≥ 2 x ULN ^b or INR ≥ 1.5	X	X	X
ALP ≥ 3 x baseline	X	X	X
ALP ≥ 2.5 x baseline and TBL ≥ 2 x ULN ^b	X	X	X
ALP ≥ 2.5 x baseline with hepatic signs or symptoms ^a	X	X	X

^a Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

^b In participants with Gilbert's syndrome, the threshold for TBL may be higher.

8.2.6.1. Close Hepatic Monitoring

If a participant develops any one of these changes, initiate close hepatic monitoring:

Participants with normal or near-normal baseline (ALT, AST, or ALP <1.5x ULN)	Participants with elevated baseline (ALT, AST, or ALP ≥ 1.5 x ULN)
ALT or AST ≥ 3 x ULN or	ALT or AST ≥ 2 x baseline
ALP ≥ 2 x ULN or	ALP ≥ 2 x baseline
TBL ≥ 2 x ULN ^a	TBL ≥ 2 x ULN ^a

^a In participants with Gilbert's syndrome, the threshold for TBL may be higher.

Close hepatic monitoring should include these actions:

- Laboratory tests (Section 10.6) including ALT, AST, ALP, TBL, D. Bil, GGT, CK, and CBC with differential, should be checked within 48 to 72 hours of the detection of

elevated liver tests to confirm the abnormality and to determine if it is increasing or decreasing.

- If the abnormality persists, clinical and laboratory monitoring should continue at a frequency of 2 to 3 times weekly until levels normalize or return to approximate baseline values. Subsequently, the frequency of monitoring may be lowered to once every 1 to 2 weeks if the participant's clinical condition and lab results stabilize.
- In addition to lab tests, basic evaluation for possible causes of abnormal liver tests should be initiated by the investigator in consultation with the Lilly-designated medical monitor. At a minimum, this evaluation should include physical examination and a thorough medical history, including current symptoms, recent illnesses (for example, heart failure, systemic infection, hypotension, or seizures), recent travel, concomitant medications (including over-the-counter), herbal and dietary supplements, history of alcohol drinking and other substance abuse.

8.2.6.2. Comprehensive Hepatic Evaluation

If a participant develops any one of the following laboratory or clinical changes, initiate a comprehensive hepatic evaluation:

Participants with normal or near-normal baseline (ALT, AST, or ALP <1.5x ULN)	Participants with elevated baseline (ALT, AST, or ALP ≥1.5x ULN)
ALT or AST ≥5x ULN or	ALT or AST ≥3x baseline or ≥250 U/L (whichever occurs first) or
ALP ≥2.5x ULN or	ALP ≥2.5x baseline or
ALT or AST ≥3x ULN with hepatic signs or symptoms ^a or	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) with hepatic signs or symptoms ^a or
ALT or AST ≥5x ULN for more than 2 weeks or	ALT or AST ≥3x baseline or ≥250 U/L (whichever occurs first) for more than 2 weeks or
ALT or AST ≥8x ULN or	ALT or AST ≥4x baseline or ≥400 U/L (whichever occurs first) or
ALT or AST ≥3x ULN and TBL ≥2x ULN ^b or INR ≥1.5	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) and TBL ≥2x ULN ^b or INR ≥1.5

^a Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

^b In participants with Gilbert's syndrome, the threshold for TBL may be higher.

Comprehensive hepatic evaluation should include these actions:

- At a minimum, comprehensive hepatic evaluation should include
 - physical examination and a thorough medical history, as outlined above, as well as tests for PT-INR; tests for viral hepatitis A, B, C, and E; tests for autoimmune hepatitis; and an abdominal imaging study (for example, ultrasound or CT scan).
- Based on the participant's history and initial results, further testing should be considered in consultation with the Lilly-designated medical monitor, including tests for
 - hepatitis D virus
 - cytomegalovirus

- Epstein-Barr virus
- acetaminophen levels
- acetaminophen protein adducts
- urine toxicology screen
- Wilson's disease
- blood alcohol levels
- urinary ethyl glucuronide, and
- blood phosphatidylethanol.
- Based on the circumstances and the investigator's assessment of the participant's clinical condition, the investigator should consider referring the participant for a hepatologist or gastroenterologist consultation, and additional tests including magnetic resonance cholangiopancreatography (MRCP), endoscopic retrograde cholangiopancreatography (ERCP), cardiac echocardiogram, or a liver biopsy.
- Clinical and laboratory monitoring should continue at a frequency of 1 to 3 times weekly until levels normalize or return to approximate baseline values.
- All the medical information and tests results related to the hepatic monitoring and comprehensive hepatic evaluation should be collected and recorded in a hepatic safety CRF.

8.2.6.3. Study Intervention Interruption or Discontinuation

If a participant develops any one of the following laboratory or clinical changes, interrupt the study intervention and continue close monitoring and comprehensive hepatic evaluation as described in Sections 8.2.6.1 and 8.2.6.2.

Participants with normal or near normal baseline (ALT, AST, or ALP <1.5x ULN)	Participants with elevated baseline (ALT, AST, or ALP ≥1.5x ULN)
ALT or AST ≥3x ULN with hepatic signs or symptoms ^a or	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) with hepatic signs or symptoms ^a or
ALT or AST ≥5x ULN for more than 2 weeks or	ALT or AST ≥3x baseline or ≥250 U/L (whichever occurs first) for more than 2 weeks or
ALT or AST ≥8x ULN or	ALT or AST ≥4x baseline or ≥400 U/L (whichever occurs first) or
ALT or AST ≥3x ULN and TBL ≥2x ULN ^b or INR ≥1.5 or	ALT or AST ≥2x baseline or ≥250 U/L (whichever occurs first) and TBL ≥2x ULN ^b or INR ≥1.5 or
ALP ≥3x ULN or	ALP ≥3x baseline or
ALP ≥2.5x ULN and TBL ≥2x ULN ^b or	ALP ≥2.5x baseline and TBL ≥2x ULN ^b or
ALP ≥2.5x ULN with hepatic signs or symptoms ^a	ALP ≥2.5x baseline with hepatic signs or symptoms ^a

^a Examples of hepatic signs or symptoms: severe fatigue, nausea, vomiting, right upper quadrant abdominal pain, fever, rash, and/or eosinophilia >5%.

^b In participants with Gilbert's syndrome, the threshold for TBL may be higher.

Interruption or discontinuation of study intervention should include these actions:

- While the participant is not receiving the study intervention, clinical and laboratory monitoring should continue at a frequency of 1 to 3 times weekly until liver tests normalize or return to approximate baseline values.
- If the hepatic event continues past the anticipated end of the study (that is, data lock) the investigator should consult with the Lilly-designated medical monitor to determine the need for further data collection beyond the end date of the study (that is, data lock date).
- All the medical information and tests results related to the close hepatic monitoring and comprehensive hepatic evaluation should be collected and recorded in a hepatic safety CRF.
- Resumption of the study intervention after interruption for a hepatic reason can be considered only in consultation with the Lilly-designated medical monitor and only if the liver test results returned to near baseline and if a self-limited, non-study-drug etiology is identified. Otherwise, the study drug should be permanently discontinued.

8.2.7. Hypersensitivity Reactions

Many drugs, including oral agents and biologic agents, carry the risk of systemic hypersensitivity reactions. If such a reaction occurs, additional data should be provided to the sponsor in the designated CRFs.

Sites should have appropriately trained medical staff and appropriate medical equipment available when study participants are receiving study intervention. It is recommended that participants who experience a systemic hypersensitivity reaction be treated per national and international guidelines.

In the case of a suspected systemic hypersensitivity event, additional blood samples should be collected as described in Section 10.2.1. Laboratory results are provided to the sponsor via the central laboratory.

8.2.8. Diabetic Retinopathy Screening and Evaluation

Baseline fundus photography should be performed on participants who remain eligible based on screening activities performed at Visit 1, including laboratory results. The overall ophthalmological evaluation of the baseline fundus photographs must be completed prior to randomization to confirm eligibility. Participants with any degree of diabetic retinopathy or macular edema on the baseline assessment that warrants treatment based on the clinical judgment of the ophthalmologist or other qualified healthcare provider will be excluded (see exclusion criterion 13).

All participants will have follow-up fundus photography performed according to the SoA (Section 1.3). Only participants with any degree of retinopathy present at baseline will repeat the fundus photography at Week 24 (± 4 weeks), and all participants will repeat the fundus photography at Week 40 (-4 weeks to $+2$ weeks) prior to start of new antihyperglycemic medication. Fundus photography may be performed more frequently if warranted based on the severity of retinopathy or macular edema at baseline or during the study consistent with screening guidelines for diabetic retinopathy (Solomon et al. 2017; Flaxel et al. 2020).

Fundus photography should be performed in both eyes (unless precluded due to severe cataract, prosthesis, or other conditions) using the broadest field of view available to the site or associated qualified eye care professional, including 7-standard views, 4-wide field views, or at a minimum

2-field views with at least 60-degree field of view. Photographs may be performed at the study site if equipment and staff trained in collecting the required views are available; otherwise, they should be referred to an ophthalmologist or other suitable qualified healthcare provider.

All fundus photographs will be evaluated by an ophthalmologist or other qualified healthcare provider experienced in the assessment of diabetic retinopathy and macular edema. The evaluation will include whether diabetic retinopathy and/or macular edema are present and whether treatment is recommended. Fundus photographs deemed unevaluable for any reason will be repeated as recommended by the ophthalmological evaluator. Whenever possible, the same qualified evaluator should review all photographs for an individual participant throughout the study. Any evidence of diabetic retinopathy, macular edema, or other clinically significant findings on the fundus photographs will require additional ophthalmological follow-up per local standards of care. The use of optical coherence tomography should be encouraged to confirm cases of macular edema.

Results from the overall ophthalmological evaluation, including final diagnoses of nonproliferative retinopathy, proliferative retinopathy, or diabetic macular edema, will be provided to the investigator and recorded on a specific retinopathy CRF. Any interventions initiated for diabetic retinopathy or macular edema during the study will be captured on the concomitant medication CRF.

8.2.9. Suicidal Ideation and Behavior Risk Monitoring

Baseline assessment of SIB/intervention emergent SIB will be monitored using the C-SSRS and PHQ-9. See Section 8.3.3.20 for more details.

All factors contributing to SIB should be evaluated, and consideration should be given to discontinuation of the study intervention (Section 7.1.1).

Columbia-Suicide Severity Rating Scale (C-SSRS)

The C-SSRS captures the occurrence, severity, and frequency of suicidal ideation and behavior during the assessment period via a semi-structured interview by a trained rater.

For this study, the C-SSRS is adapted for the assessment of the ideation and behavior categories only. The Intensity of Ideation and Lethality of Behavior sections are removed.

Timing of collection and AE monitoring

Nonleading AE collection should occur prior to the collection of the C-SSRS.

Only report a suicide-related event discovered during collection of the C-SSRS on the AE form if it is an AE that leads to discontinuation or an SAE.

Follow standard procedures for reporting SAEs.

8.2.10. Depression Monitoring

Monitor participants receiving study intervention for depression or any other unusual changes in behavior, especially at the beginning and end of the course of treatment, or at the time of dose changes, either increases or decreases. In this study, depression will be monitored using PHQ-9.

Patient Health Questionnaire-9 (PHQ-9)

The PHQ-9 (Spitzer et al. 1999; Moriarty et al. 2015) is a validated, participant-reported instrument that assesses the specific diagnostic symptoms that determine the presence of a clinical depressive disorder per the Diagnosis and Statistical Manual for Mental Disorders, 5th Edition (DSM-5).

The questionnaire assesses the previous 2 weeks.

The PHQ-9 assesses 9 diagnostic symptoms:

- mood
- anhedonia
- appetite change
- sleep disturbance
- psychomotor agitation or retardation
- loss of energy
- feelings of worthlessness or guilt
- diminished concentration, and
- suicidal thoughts or attempts.

Each question has 4 response options, with scores ranging from 0 to 3. Higher numbers indicate greater dysfunction.

This table describes the interpretation of results.

Interpretation of Depression	Total Score
Minimal to none	0-4
Mild	5-9
Moderate	10-14
Moderately severe	15-19
Severe	20-27

8.3. Adverse Events, Serious Adverse Events, and Product Complaints

The definitions of the following events can be found in Section 10.3:

- AEs
- SAEs, and
- PCs.

These events will be reported by the participant, or, when appropriate, by a caregiver, surrogate, or the participant's legally authorized representative.

The investigator and any qualified designees are responsible for detecting, documenting, and recording events that meet these definitions and remain responsible for following up events that are serious, considered related to the study intervention or study procedures, or that caused the participant to discontinue the study intervention or study.

Care will be taken not to introduce bias when detecting events. Open-ended and non-leading verbal questioning of the participant is the preferred method to inquire about event occurrences.

After the initial report, the investigator is required to proactively follow each participant at subsequent visits/contacts. All SAEs and AEs of special interest as defined in Section 8.3.3 will be followed until resolution, stabilization, the event is otherwise explained, or the participant is lost to follow-up (as defined in Section 7.3).

For PCs, the investigator is responsible for ensuring that follow-up includes any supplemental investigations as indicated to elucidate the nature or causality. Further information on follow-up procedures is provided in Section 10.2.1.

8.3.1. Timing and Mechanism for Collecting Events

This table describes the timing, deadlines, and mechanism for collecting events.

Event	Collection Start	Collection Stop	Timing for Reporting to Sponsor or Designee	Mechanism for Reporting	Back-up Method of Reporting
Adverse Event					
AE	Signing of ICF	Participation in study has ended	As soon as possible upon site awareness	AE CRF	N/A
Serious Adverse Event					
SAE and SAE updates – prior to start of study intervention and deemed reasonably possibly related to study procedures	Signing of the ICF	Start of intervention	Within 24 hours of awareness	SAE CRF	SAE paper form
SAE and SAE updates – after start of study intervention	Start of intervention	Participation in study has ended	Within 24 hours of awareness	SAE CRF	SAE paper form
SAE ^a – after participant's study participation has ended and the investigator becomes aware	After participant's study participation has ended	N/A	Promptly	SAE paper form	N/A

Event	Collection Start	Collection Stop	Timing for Reporting to Sponsor or Designee	Mechanism for Reporting	Back-up Method of Reporting
Pregnancy					
Pregnancy in female participants and female partners of male participants	After the start of study intervention	For female participants until 30 days after the last dose of study intervention For male participants, it is 90 days + at least 5 terminal half-lives	Within 24 hours (see Section 8.3.2)	Pregnancy CRF	Pregnancy paper form
Product Complaints					
PC associated with an SAE or might have led to an SAE	Start of study intervention	End of study intervention	Within 24 hours of awareness	PC form	N/A
PC not associated with an SAE	Start of study intervention	End of study intervention	Within 1 business day of awareness	PC form	N/A
Updated PC information	—	—	As soon as possible upon site awareness	Originally completed PC form with all changes signed and dated by the investigator	N/A
PC (if investigator becomes aware)	Participation in study has ended	N/A	Promptly	PC form	

Abbreviations: AE = adverse event; CRF = case report form; ICF = informed consent form; N/A = not applicable; PC = product complaint; SAEs = serious adverse events.

^a SAEs should not be reported unless the investigator deems them to be possibly related to study treatment or study participation.

8.3.2. Pregnancy

Collection of pregnancy information

Male participants with partners who become pregnant

The investigator will attempt to collect pregnancy information on any male participant's female partner who becomes pregnant while the male participant is in this study. This applies only to male participants who receive study intervention.

After learning of a pregnancy in the female partner of a study participant, the investigator

- will obtain a consent to release information from the pregnant female partner directly, and
- within 24 hours after obtaining this consent will record pregnancy information on the appropriate form and submit it to the sponsor.

The female partner will also be followed to determine the outcome of the pregnancy. Information on the status of the mother and child will be forwarded to the sponsor. Generally, the follow-up will be no longer than 6 to 8 weeks following the estimated delivery date. Any termination of the pregnancy will be reported regardless of gestational age, fetal status (presence or absence of anomalies) or indication for the procedure.

Female participants who become pregnant

The investigator will collect pregnancy information on any female participant who becomes pregnant while participating in this study. The initial information will be recorded on the appropriate form and submitted to the sponsor within 24 hours of learning of a participant's pregnancy.

The participant will be followed to determine the outcome of the pregnancy. The investigator will collect follow-up information on the participant and the neonate, and the information will be forwarded to the sponsor. Generally, follow-up will not be required for longer than 6 to 8 weeks beyond the estimated delivery date. Any termination of pregnancy will be reported, regardless of gestational age, fetal status (presence or absence of anomalies) or indication for the procedure.

While pregnancy itself is not considered to be an AE or SAE, any pregnancy complication or elective termination of a pregnancy for medical reasons will be reported as an AE or SAE.

A spontaneous abortion (occurring at <20 weeks gestational age) or still birth (occurring at ≥20 weeks gestational age) is always considered to be an SAE and will be reported as such.

Any post-study pregnancy related SAE considered reasonably related to the study intervention by the investigator will be reported to the sponsor as described in protocol Section 8.3.1. While the investigator is not obligated to actively seek this information in former study participants, he or she may learn of an SAE through spontaneous reporting.

Any female participant who becomes pregnant while participating in the study will discontinue study intervention and will be withdrawn from the study. If the participant is discontinued from the study, follow the standard discontinuation process and continue directly to the follow-up phase. The follow-up on the pregnancy outcome should continue independent of intervention or study discontinuation.

8.3.3. Adverse Events of Special Interest and Other Safety Topics

8.3.3.1. Major Adverse Cardiovascular Events

Nonfatal CV AEs and all deaths will be adjudicated by a committee of physicians external to Lilly with cardiology expertise. This committee will be blinded to treatment assignment. The nonfatal CV AEs to be adjudicated include

- myocardial infarction
- hospitalization for unstable angina
- hospitalization or urgent visit for heart failure
- coronary interventions, such as coronary artery bypass graft or percutaneous coronary intervention, and
- cerebrovascular events, including cerebrovascular accident (stroke) and transient ischemic attack.

Case adjudication and data entry

An independent CEC with cardiology expertise will adjudicate all suspected cases of major adverse cardiovascular events. The investigator must first report these events as an AE as described in Section 8.3.1 and then report them as an endpoint on the CRF with all required source documents provided for adjudication to the CEC (see Section 10.1.5). If a single AE results in 2 or more endpoints, complete an endpoint CRF for each unique endpoint. In the case where an event is reported but does not meet a prespecified event definition detailed in the CEC charter, as reviewed by the independent CEC, no further action will be taken by the study site. Clinical event reporting begins after randomization.

8.3.3.2. Arrhythmias and Cardiac Conduction Disorders

Treatment-emergent cardiac arrhythmias and conduction disorders will be further evaluated. Participants who develop any event from these groups of disorders should undergo an ECG, which should be submitted to the central ECG vendor. Additional diagnostic tests to determine exact diagnosis should be performed, as needed. The specific diagnosis will be recorded as an AE. Events that meet criteria for serious conditions as described in Section 10.3.2 must be reported as SAEs.

8.3.3.3. Hypoglycemia

Participants will be trained by authorized study personnel about signs and symptoms of hypoglycemia and how to treat hypoglycemia and instructed to record relevant information for each episode of hypoglycemia in the study eDiary.

All hypoglycemic episodes that are recorded in the study eDiary, do not need to be recorded on the AE CRF unless the event meets criteria of severe hypoglycemia which should then be recorded as serious on the AE CRF and reported to the sponsor as an SAE.

To avoid duplicate reporting, all consecutive BG values <70 mg/dL (3.9 mmol/L) occurring within a 1-hour period may be considered a single hypoglycemic event (Weinberg et al. 2010; Danne et al. 2013).

In this study, increased risk of hypoglycemia is defined as having a single episode of severe hypoglycemia or having more than 1 episode of documented symptomatic hypoglycemia within a 1-week period at any time during the treatment period. In cases where a participant experiences hypoglycemia as described above, to confirm the increased risk, the study sites must ensure that the participant has been fully compliant with the assigned study intervention and also that there is no evidence of other possible causes of hypoglycemia, for example, omission of meal and unexpected increase in exercise.

Investigators should use the following classification of hypoglycemia:

Level 1 hypoglycemia:

Glucose <70 mg/dL (3.9 mmol/L) and $\geq$ 54 mg/dL (3.0 mmol/L): Level 1 hypoglycemia can alert a person to take action such as treatment with fast-acting carbohydrates. Providers should continue to counsel participants to treat hypoglycemia at this glucose alert value.

Level 2 hypoglycemia:

Glucose <54 mg/dL (3.0 mmol/L): Level 2 hypoglycemia is also referred to as documented or blood glucose confirmed hypoglycemia with glucose <54 mg/dL (3.0 mmol/L). This glucose threshold is clinically relevant regardless of the presence or absence of symptoms of hypoglycemia.

Level 3 hypoglycemia:

Severe hypoglycemia (in adults): A severe event characterized by altered mental and/or physical status requiring assistance for treatment of hypoglycemia.

Examples of severe hypoglycemia events are

- a participant with altered mental status that could not assist in their own care, or
- a participant who becomes semiconscious or unconscious, or
- a participant who experiences coma with or without seizures, and
- the assistance of another person was needed to actively administer carbohydrate, glucagon, or other resuscitative actions.

Glucose measurements may not be available during such an event, but neurological recovery attributable to the restoration of glucose concentration to normal is considered sufficient evidence that the event was induced by a low glucose concentration.

The determination of a hypoglycemic event as an episode of severe hypoglycemia, as defined above, is made by the investigator based on the medical need of the participant to have required assistance and is not predicated on the report of a participant simply having received assistance.

If a hypoglycemic event meets the criteria of severe hypoglycemia, the investigator must record the event as serious on the AE CRF and report it to Lilly as an SAE.

Nocturnal hypoglycemia:

Nocturnal hypoglycemia is a hypoglycemia event (including severe hypoglycemia) that occurs at night and presumably during sleep.

8.3.3.4. Severe Persistent Hyperglycemia

Criteria for severe persistent hyperglycemia are provided in Section 6.9.4.

Occurrences of severe persistent hyperglycemia that meet these criteria for rescue therapy should be reported as an AE on the CRF.

8.3.3.5. Severe/Serious Gastrointestinal Adverse Events

Retatrutide may cause severe GI AEs, such as nausea, vomiting, and diarrhea. Information about severe GI AEs, as well as antiemetic or antidiarrheal use, will be collected in the AE and concomitant medications CRFs, respectively. For detailed information concerning the management of GI AEs, please refer to Sections 6.6.1.

8.3.3.6. Acute Renal Events

Renal safety will be assessed based on repeated renal functional assessment as well as assessment of AEs suggestive of acute renal failure or worsening of preexisting chronic renal failure.

The GI AEs that have been reported with retatrutide including nausea, diarrhea, and vomiting, are similar to those of marketed GLP-1RAs and the GIP/GLP-1 RA, tirzepatide. These events may lead to dehydration, which could cause a deterioration in renal function, including acute renal failure.

Participants should be advised to notify investigators in case of severe nausea, frequent vomiting, or symptoms of dehydration.

8.3.3.7. Pancreatitis

Diagnosis of acute pancreatitis

Acute pancreatitis is an AE of interest in all studies with retatrutide, including this study. The diagnosis of acute pancreatitis requires 2 of the following 3 features (Banks and Freeman 2006; Koizumi et al. 2006):

- abdominal pain, characteristic of acute pancreatitis, that is, epigastric pain radiating to the back, often associated with nausea and vomiting
- serum pancreatic amylase and/or lipase ≥ 3 x ULN, and
- characteristic findings of acute pancreatitis on CT scan or MRI.

If acute pancreatitis is suspected, the investigator should

- obtain appropriate laboratory tests, including pancreatic amylase and lipase
- perform imaging studies, such as abdominal CT scan with or without contrast, or abdominal MRI

Note: Abdominal ultrasound may be used as an alternative method only if CT and MRI cannot be performed.

and

- evaluate for possible causes of acute pancreatitis, including alcohol use, gallstone or gall bladder disease, hypertriglyceridemia, and concomitant medications.

Discontinuation for acute pancreatitis

If acute pancreatitis is suspected by the investigator,

- the participant must temporarily discontinue use of the study intervention (Section 7.1.3). In this case, the participant needs to receive an appropriate alternative antihyperglycemic regimen. If acute pancreatitis is ruled out, the participant should resume study intervention based on investigator's clinical judgment.

If acute pancreatitis is confirmed by the adjudication committee,

- the study intervention must be permanently discontinued (Section 7.1), and the participant should be followed throughout the duration of the study. If the case is not confirmed, then the participant can restart the study intervention if the investigator deems as clinically appropriate as described in Section 7.1.3.

Case adjudication and data entry

An independent CEC will adjudicate all suspected cases of acute pancreatitis. This committee will be blinded to treatment assignment. The investigator must first report these events as an AE as described in Section 8.3.1 and then report them as an endpoint on the CRF with all required source documents provided for adjudication to the CEC (see Section 10.1.5). Clinical event reporting begins after randomization.

Asymptomatic elevation of serum amylase and/or lipase

Serial measures of pancreatic enzymes have limited clinical value for predicting episodes of acute pancreatitis in asymptomatic participants (Nauck et al. 2017; Steinberg et al. 2017a, 2017b). Therefore, further diagnostic follow-up of cases of asymptomatic elevation of pancreatic enzymes (lipase and/or pancreatic amylase $\geq 3\times$ ULN) is not mandated but may be performed based on the investigator's clinical judgment and assessment of the participant's overall clinical condition.

Cases of pancreatic hyperenzymemia with symptoms or asymptomatic cases of pancreatic hyperenzymemia that undergo additional diagnostic follow-up will be submitted for adjudication.

8.3.3.8. Thyroid Malignancies and C-Cell Hyperplasia

Participants who are diagnosed with MTC and/or MEN2 during the study will have study intervention stopped (Section 7.1.1) and should continue follow-up with an endocrinologist.

The assessment of thyroid safety during the trial will include reporting of any case of thyroid neoplasms (including MTC, papillary carcinoma, and others) and measurements of calcitonin. The purpose of calcitonin measurements is to assess the potential of retatrutide to affect thyroid C-cell function, which may indicate development of C-cell hyperplasia and neoplasms.

Calcitonin measurements

If an increased calcitonin value is observed in a participant who received a medication that is known to increase serum calcitonin, then stop the medication, and repeat the calcitonin measurement after an appropriate washout period.

For participants who require additional endocrine assessment because of increased calcitonin value, record data from the follow-up assessment in the specific section of the CRF.

If a participant's laboratory serum calcitonin results meet the criteria described below, then record the clinically significant laboratory results as an AE.

Calcitonin measurements in participants with eGFR ≥ 60 mL/min/1.73 m²

Serum calcitonin value ≥ 20 pg/mL (≥ 5.9 pmol/L) and < 35 pg/mL (< 10.2 pmol/L) AND $\geq 50\%$ increase from the screening value

If a participant has this value, repeat the calcitonin measurement within 1 month.

If the repeat value increases $\geq 10\%$, then permanently discontinue study intervention.

Site staff should encourage the participant to undergo additional endocrine assessment and longer-term follow-up by an endocrinologist to exclude any SAEs on the thyroid.

Serum calcitonin value ≥ 35 pg/mL (≥ 10.2 pmol/L) AND $\geq 50\%$ increase from the screening value

If the increased concentration of calcitonin is confirmed, permanently discontinue study intervention.

Participants should immediately undergo additional endocrine assessments and longer-term follow-up by an endocrinologist to exclude any SAEs on the thyroid.

Calcitonin measurement in participants with eGFR < 60 mL/min/1.73 m²

Serum calcitonin value ≥ 35 pg/mL (≥ 10.2 pmol/L) AND $\geq 50\%$ increase from the screening value

If the increased concentration of calcitonin is confirmed, permanently discontinue study intervention.

Participants should immediately undergo additional endocrine assessments and longer-term follow-up by an endocrinologist to exclude any SAEs on the thyroid.

8.3.3.9. Diabetic Retinopathy Complications

Diabetic retinopathy and macular edema will be monitored using retinal fundus photography and additional ophthalmological follow-up as warranted throughout the study (Section 8.2.8).

Diabetic retinopathy will be assessed as nonproliferative (mild, moderate, or severe), proliferative, and diabetic macular edema as present or absent. All TEAEs of new or worsening diabetic retinopathy, macular edema, and other related complications will also be assessed.

8.3.3.10. Malignancies

All events of malignancy or other suspected events related to malignancy should be evaluated and additional diagnostic tests performed as needed.

8.3.3.11. Hepatic Disorders

All events of hepatic disorders or other suspected events related to hepatic disorders should be evaluated and additional diagnostic tests performed as needed. In cases of elevated liver markers, hepatic monitoring should be initiated as outlined in Section 8.2.6.

8.3.3.12. Gallbladder and Biliary Tract Disorders

All events of TE biliary colic, cholecystitis, cholelithiasis, or other suspected events related to acute gallbladder disease should be evaluated and additional diagnostic tests performed, as needed. A CRF will be used to capture additional information about events of symptomatic acute gallbladder disease (including gallstones and cholecystitis).

8.3.3.13. Hypersensitivity Reactions

Refer to Section 8.2.7.

8.3.3.14. Injection Site Reactions

Symptoms and signs of a local injection site reaction may include erythema, induration, pain, pruritus, and edema.

If an ISR is reported by a participant or site staff, it should be recorded as an AE, and the ISR CRF will be used to capture additional information about this reaction (for example, injection-site pain, degree and area of erythema, induration, pruritus, and edema).

At the time of AE occurrence, for severe or recurrent injection site reactions, samples will be collected for measurement of retatrutide ADA and retatrutide concentration.

8.3.3.15. Skin Burning Sensation and Related Adverse Events

If skin burning sensation and related AEs, for example, hyperesthesia and sensitive skin, are reported, it should be recorded as an AE and a skin assessment CRF should be used to capture additional information about these events.

8.3.3.16. Abuse Potential

All events of abuse potential should be evaluated, and additional investigations performed as needed.

8.3.3.17. Ketoacidosis Due to Inadequate Nutritional Intake and Other Types of Metabolic Acidosis and Ketosis

Metabolic disturbances, likely due to nutrient deprivation, have been observed in retatrutide clinical trials, the extreme being ketoacidosis. Cases of starvation ketoacidosis have been reported in completed and ongoing retatrutide clinical trials. This is a serious, potentially life-threatening condition requiring urgent hospitalization.

Investigators should instruct participants that, if they have severe, intractable vomiting, they should seek urgent medical advice, as this could be a sign of ketoacidosis. Inadequate nutritional intake, in particular of carbohydrates, increases the risk. All participants, especially those with markedly reduced appetite, should be monitored for adequate nutritional intake, including carbohydrates.

Participants who present with signs and symptoms consistent with severe metabolic acidosis should be assessed for ketoacidosis regardless of presenting BG levels, as ketoacidosis may be present even if BG levels are <250 mg/dL.

Decreased oral intake may also increase the risk for other forms of metabolic acidosis (for example, diabetic ketoacidosis, euglycemic diabetic ketoacidosis, and metabolic acidosis due to acute renal injury). The risk of ketoacidosis may also be increased with excessive alcohol consumption and dehydration.

Treatment of ketoacidosis may require insulin, fluid, and carbohydrate replacement. Routine bicarbonate assessment will be performed during the course of the study.

8.3.3.18. Excessive Weight Loss

Any AEs resulting to excessive weight loss should be reported as an AE.

If a participant has a BMI ≤ 20 kg/m² at any time during the study the investigator may contact the sponsor to consider a dose reduction or discontinuation of study intervention (see Section 6.6.3 for additional details).

8.3.3.19. Malnutrition and Inadequate Fluid Intake

Serious events related to malnutrition and inadequate fluid intake (resulting in dehydration) have been reported in ongoing blinded retatrutide clinical trials. Investigators should monitor participants for signs and symptoms of malnutrition and dehydration and provide nutritional guidance. By recognizing early signs of poor oral intake, preventive actions can be implemented to reduce the risk of potential complications. Supplementation and treatment of deficiencies should be provided as clinically indicated. Refer to Sections 5.3, 10.10 and 10.11 more information.

8.3.3.20. Depression and Suicidal Ideation or Behavior Monitoring

Participants will be monitored for depression and suicidal ideation or behavior through AE collection and by using the C-SSRS and the PHQ-9 questionnaires. Scores of the questionnaires must be reviewed by the investigator at the time of each visit and appropriate actions as described in Sections 7.1.1, 8.2.9, and 8.2.10 should be taken.

8.4. Pharmacokinetics

At the visits and times specified in the SoA, PK blood samples will be collected for all participants. Only samples from participants assigned to treatment with retatrutide will be analyzed for drug concentration.

Instructions for the collection and handling of blood samples will be provided by the sponsor.

The actual date and time (24-hour clock time) of each sampling will be recorded.

Drug concentration information that would unblind the study will not be reported to study sites or blinded personnel while the study is blinded.

Samples may be used for exploratory analyses, such as metabolism, protein binding, or bioanalytical cross-validation purposes.

8.4.1. Bioanalytical

Samples will be analyzed at a laboratory designated by the sponsor and stored at a facility designated by the sponsor. Concentrations of retatrutide will be assayed using a validated bioanalytical method.

Sample retention is described in Section [10.1.12](#).

8.5. Pharmacodynamics

Efficacy measures will be used as indicators of PD response.

8.6. Genetics

A whole-blood sample will be collected from participants to enable DNA isolation for exploratory pharmacogenetics analysis as specified in the SoA, where local regulations allow.

Samples will not be used to conduct unspecified disease or population genetic research either now or in the future.

Samples may be used to investigate variable exposure or response to retatrutide and to investigate genetic variants thought to play a role in diabetes mellitus, obesity, and related clinical traits or complications, including nonalcoholic steatohepatitis and CKD. Assessment of variable response may include evaluation of AEs or differences in PD, mechanistic, safety, or efficacy measures.

See Section [10.5](#) for information regarding genetic research and Section [10.1.12](#) for details about sample retention and custody.

8.7. Biomarkers

Stored serum and plasma samples for biomarker research will be collected at times specified in the SoA, where local regulations allow. Sample test type is specified in Section [10.2](#).

Samples will be used for research on the drug targets, disease process, variable response to retatrutide, pathways associated with diabetes mellitus, obesity, and related clinical traits, or comorbidities, including nonalcoholic steatohepatitis and CKD, mechanism of action of retatrutide or research method, or for validating diagnostic tools or assay(s) related to diabetes mellitus, obesity, or related clinical traits, or comorbidities, including nonalcoholic steatohepatitis and CKD.

All samples will be coded with the participant number. These samples and any data generated can be linked back to the participant only by the investigative site staff.

See Section [10.1.12](#) for details about sample retention.

8.8. Immunogenicity Assessments

At the visits and times specified in the SoA (Section [1.3](#)), venous blood samples will be collected to determine antibody production against retatrutide. Antibodies may be further characterized for cross-reactive binding to endogenous counterparts (native GIP, GLP-1, and glucagon), and their ability to neutralize the activity of retatrutide and endogenous counterparts.

To interpret the results of immunogenicity, a venous blood sample will be collected at the same time points to determine the concentrations of retatrutide. All samples for immunogenicity should be taken predose when applicable and possible.

Immunogenicity will be assessed by a validated assay designed to detect ADA in the presence of retatrutide at a laboratory approved by the sponsor.

TE ADA are defined in Section 9.3.8. In the event of drug hypersensitivity reactions (immediate or nonimmediate), additional samples may be collected as described in Section 10.2.1.

See Section 10.1.12 for details about sample retention.

8.9. Health Economics

Health economics parameters are not evaluated in this study.

9. Statistical Considerations

The statistical analysis plan will be finalized prior to study unblinding following the final database lock, and it will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints, including primary and key secondary endpoints.

9.1. Statistical Hypotheses

The null hypotheses corresponding to the primary objective are as follows:

- $H_{1,12,0}$, $H_{1,9,0}$, and $H_{1,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the mean change from baseline in HbA1c at Week 40, respectively.

The null hypotheses corresponding to the key secondary objectives are as follows:

- $H_{2,12,0}$, $H_{2,9,0}$, and $H_{2,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants achieving HbA1c $<7.0\%$ at Week 40, respectively.
- $H_{3,12,0}$, $H_{3,9,0}$, and $H_{3,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants achieving HbA1c $\leq 6.5\%$ at Week 40, respectively.
- $H_{4,12,0}$, $H_{4,9,0}$, and $H_{4,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants achieving HbA1c $<5.7\%$ at Week 40, respectively.
- $H_{5,12,0}$, $H_{5,9,0}$, and $H_{5,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the mean change from baseline in fasting serum glucose at Week 40, respectively.
- $H_{6,12,0}$, $H_{6,9,0}$, and $H_{6,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the mean percentage change from baseline in body weight at Week 40, respectively.
- $H_{7,12,0}$, $H_{7,9,0}$, and $H_{7,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the mean change from baseline in body weight at Week 40, respectively.
- $H_{8,12,0}$, $H_{8,9,0}$, and $H_{8,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants who achieved weight reduction of $\geq 5\%$ at Week 40, respectively.
- $H_{9,12,0}$, $H_{9,9,0}$, and $H_{9,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants who achieved weight reduction of $\geq 10\%$ at Week 40, respectively.
- $H_{10,12,0}$, $H_{10,9,0}$, and $H_{10,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants who achieved weight reduction of $\geq 15\%$ at Week 40, respectively.

- $H_{11,12,0}$, $H_{11,9,0}$, and $H_{11,4,0}$: No difference in retatrutide 12 mg, 9 mg, and 4 mg compared to placebo with respect to the proportion of participants who achieved $HbA_{1c} \leq 6.5\%$ and $\geq 10\%$ weight reduction at Week 40, respectively.
- $H_{12,12,0}$ and $H_{12,9,0}$: No difference in retatrutide 12 mg and 9 mg compared to placebo with respect to the mean percentage change from baseline in non-HDL cholesterol at Week 40, respectively.
- $H_{13,12,0}$ and $H_{13,9,0}$: No difference in retatrutide 12 mg and 9 mg compared to placebo with respect to the mean percentage change from baseline in triglycerides at Week 40, respectively.
- $H_{14,12,0}$ and $H_{14,9,0}$: No difference in retatrutide 12 mg and 9 mg compared to placebo with respect to the mean change from baseline in systolic blood pressure at Week 40, respectively.

9.1.1. Multiplicity Adjustment

A prespecified graphical scheme (Bretz et al. 2009, 2011) will be implemented to control the family-wise error rate at a 2-sided alpha level of 0.05 for testing the null hypotheses stated in Section 9.1. This graphical approach is a closed testing procedure; hence, it strongly controls the family-wise error rate across all endpoints (Bretz et al. 2009, 2011; Alosh et al. 2014).

The testing scheme will be fully detailed in the SAP. Unless otherwise specified, there will be no adjustment for multiple comparisons for any other analyses outside the primary and key secondary endpoints. In addition, no multiplicity adjustments will be made for conducting separate analyses relative to the 2 estimands, as they are intended for different purposes.

9.2. Analyses Sets

The following participant analysis sets are defined:

Participant Analysis Set	Description
Entered participants	All participants who sign informed consent.
Randomized participants	All participants who are randomly assigned a study intervention.
Safety participants	All participants who are randomly assigned a study intervention and who take at least 1 dose of study intervention.

The following data points sets are defined:

Data Points Sets	Description
Treatment regimen estimand data points set	All data points obtained during the treatment period defined as at or after baseline and up to the last visit within the treatment period, regardless of adherence to study intervention or initiation of additional antihyperglycemic medications ^a .

Data Points Sets	Description
Efficacy estimand data points set	All data points obtained during the treatment period at or after baseline and up to the earliest date of permanent discontinuation of study intervention or initiation of additional antihyperglycemic medications ^a .
Safety data points set	All data points obtained during the treatment period and the follow-up period defined as at or after baseline and up to the date of study withdrawal/study completion including the follow-up period regardless of adherence to study intervention or initiation of additional antihyperglycemic medications ^a .

^a “Additional antihyperglycemic medications” refers to any antihyperglycemic therapy that is used for more than 2 weeks (14 days).

9.3. Statistical Analyses

9.3.1. General Considerations

Statistical analysis will be the responsibility of Lilly or its designee.

Any change to the data analysis methods described in the protocol will require an amendment ONLY if it changes a principal feature of the protocol. Any other change to the data analysis methods described in the protocol, and the justification for making the change, will be described in the SAP or the CSR. Additional exploratory analyses of data will be conducted, as deemed appropriate.

Unless otherwise noted, all tests for superiority will be conducted at a 2-sided alpha level of 0.05, and the CI will be calculated at 95%, 2-sided. For objectives controlled for type 1 error, the alpha will be allocated by the graphical testing scheme.

Baseline is defined as the last available non-missing measurement prior to the first dose of study intervention, unless otherwise specified.

General descriptions of analyses

For continuous measures, summary statistics will include sample size, mean, SD, median, minimum, and maximum for both the actual and the change from baseline measurements. LS means and standard errors derived from the analysis models will also be displayed for the change from baseline measurements. Treatment comparisons will be displayed showing the treatment difference LS means and the 95% CIs for the treatment differences, along with the p-values for the treatment comparisons. All baseline measures will be analyzed using an analysis of variance (ANOVA) model that has treatment group as the fixed effect.

The analysis model to make comparisons between treatment groups relative to continuous efficacy measurements will be an ANCOVA with a variance estimator that is robust to model misspecification and heteroscedasticity (Ye et al. 2022). The model will include treatment group, strata, and the continuous covariate of baseline values and their interactions with treatment group (strata*treatment group and baseline*treatment group).

For measures evaluated at multiple postbaseline visits, a restricted maximum likelihood (REML) based MMRM analysis (Wang and Du 2022) will be used. The analysis model to make

comparisons between treatment groups relative to continuous measurements assessed over time will include an MMRM, with fixed effects of visit, and treatment, stratification factors, and baseline measurement all nested within visits. The variance-covariance matrix will be estimated using sandwich variance estimators (Tsiatis 2007).

For categorical measures, summary statistics will include sample size, frequency, and percentages. For safety categorical measures, risk difference and its 95% CI will be provided. If applicable, Fisher's exact test or Pearson's chi-square test will be used for treatment comparisons in other categorical outcomes.

Unless otherwise specified, safety and tolerability assessments will be based on the safety analysis data points for the safety population.

Other statistical methods may be used, as appropriate, and details will be described in the SAP.

Missing data should be minimized at the best precaution. Handling of missing, unused, and spurious data is addressed prospectively in the overall statistical methods described in the protocol and in the SAP, where appropriate. Adjustments to the planned analyses will be described in the final CSR.

Participants will be analyzed according to the treatment group to which they were randomly assigned.

9.3.2. Treatment Group Comparability

9.3.2.1. Participant Disposition

A detailed description of participant disposition will be provided.

Frequency counts and percentages of all randomized participants will be presented by treatment groups. A listing of randomized participants not receiving study intervention will be provided. All participants who discontinue the study will be identified, and the extent of their participation in the study will be reported. If known, a reason for their discontinuation will be given. The primary reasons for discontinuation will be listed and will be summarized by treatment groups. The percentage of participants discontinuing from treatment groups will be compared using Fisher's exact test. Kaplan-Meier analyses of time from randomization to discontinuation from study intervention by treatment group will be provided.

9.3.2.2. Participant Characteristics

Demographics and other baseline characteristics will be summarized by treatment group for all randomized participants.

9.3.2.3. Concomitant Therapy

Concomitant medications, including previous therapy, will be summarized by treatment group for the randomized participants.

9.3.2.4. Treatment Compliance

Treatment compliance is defined as taking at least 75% of scheduled study intervention during the treatment period. Frequency counts and percentages of participants compliant to study

intervention will be summarized by treatment group using the safety participants during the treatment period.

9.3.3. Primary Endpoint(s)/Estimand(s) Analysis

The null hypotheses corresponding to the primary objective can be found in Section 9.1.

The primary objective will be evaluated relative to 2 estimands, “treatment regimen” and “efficacy” (see Section 3).

The primary objective aligned to the “treatment regimen” estimand will be evaluated using the randomized participants and the treatment regimen estimand data points set as described in Section 9.2.

For the assessment of the primary objective aligned to the “treatment regimen” estimand, data for participants with missing values at Week 40 will be imputed and then analyzed using an ANCOVA with a variance estimator that is robust to model misspecification and heteroscedasticity (Ye et al. 2022). The model will include treatment group, strata, and the continuous covariate of baseline values and their interactions with treatment group (strata*treatment group and baseline*treatment group).

If there are occurrences of missing data despite the best precautions, missing data will be imputed in a manner consistent with what the values would likely have been had they been collected. In principle, missing data due to permanent discontinuation of study will be imputed using retrieved dropouts, namely using multiple imputation based on data retrieved from participants who permanently discontinued study intervention but continued in the study with non-missing measurements from the same treatment group. More details regarding the imputation for missing values will be described in the SAP.

The primary objective aligned to the “efficacy” estimand will be evaluated using the randomized participants and the efficacy estimand data points set (Section 9.2). The MMRM model described in Section 9.3.1 will be used for the assessment of the primary objective guided by the “efficacy” estimand. All the longitudinal observations at each scheduled postbaseline visit will be included in the analysis. For the “efficacy” estimand, the hypothetical strategy is used to handle the ICEs, so only data collected before the occurrence of any ICEs will be used in the MMRM analysis. Through the MMRM, the potential efficacy measures (after the ICEs) will be implicitly imputed as if participants did not have ICEs.

A multiple imputation-based tipping-point analysis is planned as a sensitivity analysis to assess the robustness of primary efficacy results. Additional details of the statistical modeling will be provided in the SAP.

9.3.4. Secondary Endpoint(s)/Estimand(s) Analysis

The null hypotheses corresponding to the key secondary objectives can be found in Section 9.1.

Key secondary objectives will be evaluated relative to both the “treatment regimen” and “efficacy” estimands, as described in Section 9.3.3. Analyses for key secondary endpoints will be controlled for type 1 error through the graphical approach.

The assessment of key secondary objectives for continuous variables will be evaluated by an ANCOVA for the “treatment regimen” estimand and an MMRM for the “efficacy” estimand as described in Section 9.3.3.

For the assessment of key secondary objectives for binary variables, a logistic regression model with treatment group and strata as fixed effects and the continuous baseline value as a covariate will be used to examine the treatment difference with missing endpoints imputed. The unconditional treatment group effect will be assessed by risk difference and relative risk using the marginal standardization method, where the treatment group-specific risk will be derived from the counterfactual risks for each participant that are predicted with the fitted logistic model (Steingrimsson et al. 2017; FDA 2023; Ye et al. 2023). The estimated treatment group-specific risk, risk difference, relative risk, p-value, and 95% CI will be presented.

For binary efficacy measures derived from a continuous variable, the missing value in the underlying continuous variable will be imputed first and then the corresponding binary variable will be derived. The details about imputation under “treatment regimen” estimand is described in Section 9.3.3. Under the “efficacy” estimand, missing values for continuous variables at Week 40 will be imputed through multiple imputation using all non-missing data (excluding data collected after ICEs) from the same treatment group under the missing-at-random assumption.

Details for additional secondary analyses will be provided in the SAP.

9.3.5. Pharmacokinetics/Pharmacodynamic Analyses

Retatrutide concentration data will be analyzed using a population PK approach using nonlinear mixed-effects modeling techniques implemented on the NONMEM® software.

The relationships between retatrutide dose and/or concentration and selected efficacy, tolerability, and safety endpoints will be characterized. Additionally, the impact of intrinsic and extrinsic participant factors such as age, weight, gender, and renal function on retatrutide PK and/or PD parameters may be examined. If ADA titers are detected from immunogenicity testing, then the impact of immunogenicity titers on retatrutide PK or any relevant PD parameters may also be examined.

9.3.6. Safety Analyses

Safety analyses will be conducted using the safety participants and the safety data points set, unless otherwise specified.

AEs will be coded from the actual term using the Medical Dictionary for Regulatory Activities and reported by preferred terms within system organ class. Selected notable AEs of interest may be reported using high-level terms or Standardized Medical Dictionary for Regulatory Activities Queries. Summary statistics will be provided for incidence of TEAEs, SAEs, study discontinuation due to AEs, study intervention discontinuation due to AEs, deaths, and other CV endpoints. Counts and proportions of participants experiencing AEs will be reported for each treatment group, and Fisher’s exact test will be used to compare the treatment groups.

9.3.6.1. Adverse Events of Special Interest and Other Safety Topics

The analysis details for the AEs of special interest and other safety topics (as defined in Section 8.3.3) will be provided in the SAP.

9.3.6.2. Central Laboratory Measures, Vital Signs, and Electrocardiograms

Actual and change from baseline to postbaseline values of selected central laboratory measures, vital signs, and selected ECG parameters will be summarized at each scheduled visit. Continuous variables, as well as the change from baseline for these variables, will be analyzed by MMRM models as described in Section 9.3.1. The percentages of participants with elevated or low values meeting specified levels, including laboratory, vital, and ECG parameters, will be summarized and compared between treatment groups using Fisher's exact test or Pearson's chi-squared test.

9.3.6.3. Depression, Suicidal Ideation, and Behavior

In addition to the summary of TEAEs, suicide-related thoughts and behaviors occurring during treatment will be summarized based on responses to the C-SSRS consistent with the C-SSRS Scoring and Data Analysis Guide [C-SSRS]. Depression-related symptoms will be summarized based on responses to the PHQ-9.

The analysis details will be provided in the SAP.

9.3.7. Other Analyses

Subgroup analyses

Subgroup analyses to assess consistency of the effect of retatrutide across subgroups for the primary endpoint (HbA1c) will be detailed in the SAP. The following subgroups will be considered but not limited to:

- age group (<65, ≥65 years)
- sex (female, male)
- baseline HbA1c stratum (≤8.0%, >8.0%)
- baseline BMI (<30 kg/m², ≥30 kg/m²)
- duration of T2D (<median, ≥median)
- race
- ethnicity, and
- country/region.

Additional subgroup analyses on the key secondary endpoints, for example, weight, may also be performed. Further details on the statistical analysis will be provided in the SAP.

9.3.8. Evaluation of Immunogenicity

The frequency and percentage of participants with preexisting ADA and with retatrutide TE ADA may be reported.

TE ADA are defined as

- Treatment-induced ADA - Those with a titer 2-fold (1 dilution) greater than the minimum required dilution if no ADA were detected at baseline, or
- Treatment-boosted ADA - Those with a 4-fold (2 dilutions) increase in titer compared to baseline if ADA were detected at baseline.

For the TE ADA+ participants, the distribution of maximum titers may be reported.

The frequency of neutralizing antibodies against retatrutide, if performed, may be reported in TE ADA+ participants.

If cross-reactivity to native GLP-1, GIP, and glucagon, or neutralizing antibodies against native GLP-1, GIP, and glucagon assays are performed, the frequency of each may be reported.

The relationship between the presence of antibodies and the PK parameters and PD response, including safety and efficacy of retatrutide, may be assessed.

Additional details may be provided in the SAP.

9.4. Interim Analysis

No interim analyses are planned for this study. If an unplanned interim analysis is deemed necessary for reasons other than a safety concern, the protocol must be amended.

9.5. Sample Size Determination

This trial is powered to assess superiority of retatrutide 4 mg, 9 mg, and/or 12 mg, each tested in parallel, against placebo using a 2-sample t-test at a 2-sided significance level of 0.0167, relative to the primary endpoint (change from baseline in HbA1c at 40 weeks), under the following assumptions:

- 0.6% greater mean reduction in HbA1c from baseline for 4 mg, 9 mg, and 12 mg of retatrutide compared with placebo, and
- common SD of 1.2%,

Under the assumptions above, randomly assigning 480 participants in a 1:1:1:1 ratio to retatrutide 4 mg (n=120), retatrutide 9 mg (n=120), retatrutide 12 mg (n=120), and placebo (n=120) provides more than 90% power to demonstrate the superiority of retatrutide 4 mg, 9 mg, and/or 12 mg to placebo.

These assumptions are considered to ensure the power for the primary endpoint under both efficacy and treatment estimands. The sample size was calculated using the software nQuery version 8.

10. Supporting Documentation and Operational Considerations

10.1. Appendix 1: Regulatory, Ethical, and Study Oversight Considerations

10.1.1. Regulatory and Ethical Considerations

This study will be conducted in accordance with the protocol and with the following:

- Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines.
- Applicable ICH GCP Guidelines.
- International Organization for Standardization (ISO) 14155.
- Applicable laws and regulations.

The protocol, protocol amendments, ICF, IB, and other relevant documents (for example, advertisements) must be submitted to an IRB/IEC by the investigator and reviewed and approved by the IRB/IEC before the study is initiated.

Any amendments to the protocol will require IRB/IEC approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants.

Protocols and any substantial amendments to the protocol will require health authority approval prior to initiation except for changes necessary to eliminate an immediate hazard to study participants.

The investigator will be responsible for the following:

- providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC
- notifying the IRB/IEC of SAEs or other significant safety findings as required by IRB/IEC procedures
- providing oversight of study conduct for participants under their responsibility and adherence to requirements of 21 Code of Federal Regulations (CFR), ICH guidelines, the IRB/IEC, European regulation 536/2014 for clinical studies (if applicable), and all other applicable local regulations
- reporting to the sponsor or designee significant issues related to participant safety, participant rights, or data integrity.

Investigator sites are compensated for participation in the study as detailed in the clinical trial agreement.

10.1.2. Financial Disclosure

Investigators and sub-investigators will provide the sponsor with sufficient, accurate financial information as requested to allow the sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

10.1.3. Informed Consent Process

The investigator or the investigator's representative will explain the nature of the study, including the risks and benefits, to the potential participant (or, when appropriate, a caregiver, surrogate, or the participant's legally authorized representative) and answer all questions regarding the study.

Potential participants must be informed that their participation is voluntary. Participants (or, when appropriate, a caregiver, surrogate, or the participant's legally authorized representative) will be required to sign a statement of informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, privacy and data protection requirements, where applicable, and the IRB/IEC or study center.

The medical record must include a statement that written informed consent was obtained before the participant was entered in the study and the date the written consent was obtained. The authorized person obtaining the informed consent must also sign the ICF.

Revised consents must be appropriately obtained using the correct approved ICFs for applicable study participants in accordance with sponsor and Ethical Review Board consenting guidance.

A copy of the ICF(s) must be provided to the participant (or, when appropriate, a caregiver, surrogate, or the participant's legally authorized representative) and is kept on file.

10.1.4. Data Protection

Participants will be assigned a unique identifier by the sponsor to protect the participant's personal data. Any participant information, such as records, datasets or tissue samples that are transferred to the sponsor will contain the identifier only. Participant names or any information which would make the participant identifiable will not be transferred.

The participant must be informed that the participant's personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the participant who will be required to give consent for their data to be used as described in the informed consent. This is done by the site personnel through the informed consent process.

The participant must be informed through the informed consent by the site personnel that their medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.

The sponsor has processes in place to ensure information security, data integrity, and data protection. These processes address management of data transfer, and prevention and management of unauthorized access, disclosure, dissemination, alteration or loss of information

or personal data. These processes include appropriate contingency plan(s) for appropriate and timely response in the event of a data security breach.

The transfer of personal data is subject to appropriate safeguards through contractual agreements and processes. The sponsor's processes are compliant with local privacy laws and relevant legislations including the General Data Protection Regulation (GDPR).

10.1.5. Committees Structure

10.1.5.1. Clinical Endpoint Committee

An independent CEC with membership external to the sponsor will be responsible for event adjudication in a blinded fashion.

The independent CEC will adjudicate events listed in Section 8.3.3. The CEC charter will contain the final detailed event definitions used for adjudication.

10.1.6. Dissemination of Clinical Study Data

Reports

The sponsor will disclose a summary of study information, including tabular study results, on publicly available websites where required by local law or regulation.

The summary of results will be posted within the time frame specified by local law or regulation. If the study remains ongoing in some countries and a statistical analysis of an incomplete dataset would result in analyses lacking scientific rigor (for example, underpowered) or compromise the integrity of the overall analyses (for example, trial not yet unblinded), the summary of results will be submitted within 1 year after the end of the study globally or as soon as available, whichever is earlier.

Data

The sponsor provides access to all individual participant data collected during the trial, after anonymization, with the exception of PK or genetic data.

Data are available to request 6 months after the indication studied has been approved in the US and EU and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available.

Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data-sharing agreement.

Data and documents, including the study protocol, SAP, CSR, and blank or annotated case report forms, will be provided in a secure data-sharing environment for up to 2 years per proposal.

For details on submitting a request, see the instructions provided at www.vivli.org.

10.1.7. Data Quality Assurance

All participant data relating to the study will be recorded on printed or electronic CRFs unless transmitted to the sponsor or designee electronically (for example, laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the CRF.

The investigator must maintain accurate documentation (source data) that supports the information entered in the CRF. This includes laboratory tests, medical records, and clinical notes.

The investigator must review and confirm that data entries are accurate and complete throughout the duration of the study, by physically or electronically signing the CRF, as instructed by the sponsor. All completed CRFs must be signed prior to archival.

The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source documents.

Data monitoring and management

Quality tolerance limits will be predefined to identify systematic issues that can impact participant safety and/or reliability of study results. These predefined parameters will be monitored during the study and important excursions from the quality tolerance limits and remedial actions taken will be summarized in the CSR.

Monitoring details describing strategy, for example, risk-based initiatives in operations and quality such as risk management and mitigation strategies and analytical risk-based monitoring, methods, responsibilities and requirements, including handling of noncompliance issues and monitoring techniques are provided in the Monitoring Plan.

The sponsor or designee is responsible for the data management of this study including quality checking of the data.

The sponsor assumes accountability for actions delegated to other individuals, for example, contract research organizations.

The sponsor or designee will perform monitoring to confirm that data transcribed into the CRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for the time period outlined in the Clinical Trial Agreement (CTA) unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

In addition, the sponsor or its representatives will periodically check a sample of the participant data recorded against source documents at the study site. The study may be audited by the sponsor or its representatives, or regulatory agencies at any time. Investigators will be given notice before an audit occurs.

Data capture system

The investigator is responsible for ensuring the accuracy, completeness, legibility, and timeliness of the data reported to the sponsor.

Electronic data capture system

An EDC system will be used in this study for the collection of CRF data. The investigator maintains a separate source for the data entered by the investigator or designee into the sponsor-provided EDC system. The investigator is responsible for the identification of any data to be considered source and for the confirmation that data reported are accurate and complete by signing the CRF.

Clinical outcome assessments

The eCOA data (participant-focused outcome instrument) for all other assessments will be directly recorded by the participant into an instrument (for example, hand-held smart phone or tablet). The eCOA data will serve as the source documentation, and the investigator does not maintain a separate written or electronic record of these data.

Data storage and access

Data collected via the sponsor-provided data capture system(s) will be stored at third parties. The investigator will have continuous access to the data during the study and until decommissioning of the data capture system(s). Prior to decommissioning, the investigator will receive or access an archival copy of pertinent data for retention.

Data managed by a central vendor, such as laboratory test data, will be stored electronically in the central vendor's database system and reports will be provided to the investigator for review and retention. Data will subsequently be transferred from the central vendor to the sponsor data warehouse.

Data from complaint forms submitted to the sponsor will be encoded and stored in the global PC management system.

10.1.8. Source Documents

Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.

Data reported on or entered in the CRF and are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.

Definition of what constitutes source data can be found in the site confirmation of source data.

10.1.9. Study and Site Start and Closure**First act of recruitment**

The study start date and the first act of recruitment is the date on which the clinical study will be open for recruitment of participants.

Study or site termination

The sponsor or sponsor's designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the sponsor. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to

- for study termination due to discontinuation of further study intervention development
- for site termination due to
 - failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines
 - inadequate recruitment, evaluated after a reasonable amount of time of participants by the investigator, or
 - total number of participants included earlier than expected.

If the study is prematurely terminated or suspended, the sponsor shall promptly inform the investigators, the IECs/IRBs, the regulatory authorities, and any contract research organization(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

10.1.10. Publication Policy

In accordance with the sponsor's publication policy, the results of this study will be submitted for publication by a peer-reviewed journal.

10.1.11. Investigator Information

Researchers with appropriate education, training, and experience, as determined by the sponsor, will participate as investigators in this clinical trial.

10.1.12. Sample Retention

Sample retention enables use of new technologies, response to regulatory questions, and investigation of variable response that may not be observed until later in the development of retatrutide or after retatrutide become(s) commercially available.

Sample Type	Custodian	Maximum Retention Period After the Last Participant Visit ^a
Exploratory biomarkers	Sponsor or designee	Up to 15 years
Pharmacokinetic	Sponsor or designee	1 year
Genetics/PD	Sponsor or designee	Up to 15 years
Immunogenicity	Sponsor or designee	Up to 15 years

^a Retention periods may differ locally.

10.2. Appendix 2: Clinical Laboratory Tests

The tests detailed in the table below will be performed by the central laboratory.

Local laboratory results are only required in the event that the central laboratory results are not available in time for either study intervention administration and/or response evaluation. If a local sample is required, it is important that the sample for central analysis is obtained at the same time. Additionally, if the local laboratory results are used to make either a study intervention decision or response evaluation, the results must be recorded.

In circumstances where the sponsor approves local laboratory testing in lieu of central laboratory testing (in the table below), the local laboratory must be qualified in accordance with applicable local regulations.

Protocol-specific requirements for inclusion or exclusion of participants are detailed in Section 5 of the protocol.

Additional tests may be performed at any time during the study as determined necessary by the investigator or required by local regulations.

Investigators must document their review of the laboratory safety results.

Laboratory/analyte results that could unblind the study will not be reported to investigative sites or other blinded personnel until the study has been unblinded.

Clinical Laboratory Tests	Comments
Hematology	Assayed by Lilly-designated laboratory.
Hemoglobin	
Hematocrit	
Erythrocyte count (RBCs - red blood cells)	
Mean cell volume	
Mean cell hemoglobin	
Mean cell hemoglobin concentration	
Leukocytes (WBCs - white blood cells)	
Differential	
Absolute count of	
Neutrophils	
Lymphocytes	
Monocytes	
Eosinophils	
Basophils	
Platelets	
Clinical chemistry panel	Assayed by Lilly-designated laboratory.
Sodium	
Potassium	
Magnesium	
Chloride	

Clinical Laboratory Tests	Comments
Bicarbonate	
Total bilirubin	
Direct bilirubin	
Alkaline phosphatase (ALP)	
Alanine aminotransferase (ALT)	
Aspartate aminotransferase (AST)	
Gamma-glutamyl transferase (GGT)	
Blood urea nitrogen (BUN)	
Creatinine	
Creatine kinase (CK)	
Uric acid	
Total protein	
Albumin	
Calcium	
Phosphorus	
Lipid panel	
High-density lipoprotein cholesterol (HDL-C)	Assayed by Lilly-designated laboratory.
Total cholesterol	
Triglycerides	
Low-density lipoprotein cholesterol (LDL-C)	Calculated by Lilly-designated laboratory. LDL will be directly measured if triglycerides exceed maximum value of calculation.
Very-low-density lipoprotein cholesterol (VLDL-C)	Calculated by Lilly-designated laboratory.
Non-high density lipoprotein cholesterol (non-HDL-C)	
Hepatitis serology	Assayed by Lilly-designated laboratory.
Hepatitis C virus (HCV) testing:	
HCV antibody	
HCV RNA	Performed only for participants who test positive for HCV antibody.
Hepatitis B virus (HBV) testing:	
HBV DNA	Performed only for participants who test positive for anti-HBc.
Hepatitis B core antibody (HBc-Ab)	
Hepatitis B surface antigen (HBsAg)	
Hepatitis B surface antibody (anti-HBs)	
Hormones (female)	Assayed by Lilly-designated laboratory unless stated otherwise.
Serum pregnancy	
Urine pregnancy	Evaluated locally.
Follicle-stimulating hormone (FSH)	
Urine chemistry	Assayed by Lilly-designated laboratory.
Albumin	

Clinical Laboratory Tests	Comments
Creatinine	
Calculations	Generated by Lilly-designated laboratory.
eGFR (CKD-EPI)	CKD-EPI cystatin-C equation (2012). Results will be used for eligibility criteria.
Urinary albumin/creatinine ratio (UACR)	
Other testing	Assayed by Lilly-designated laboratory.
HbA1c	
Calcitonin	
Pancreatic amylase	
Lipase	
Glucose	
Cystatin-C	
Insulin	Results will not be provided to investigative sites.
C-peptide	Results will not be provided to investigative sites.
Anti-glutamic acid decarboxylase (GAD) antibodies	Results will not be provided to investigative sites.
Apolipoprotein B (Apo B)	Results will not be provided to investigative sites.
C-reactive protein, high-sensitivity (hsCRP)	Results will not be provided to investigative sites.
Pharmacokinetic samples	Assayed by Lilly-designated laboratory.
Retatrutide concentration	Results will not be provided to investigative sites.
Genetics sample	Assayed by Lilly-designated laboratory.
Whole blood (EDTA)	Results will not be provided to investigative sites.
Exploratory biomarker storage samples	Assayed by Lilly-designated laboratory.
Serum	Results will not be provided to investigative sites.
Plasma (EDTA)	
Immunogenicity samples	Assayed by Lilly-designated laboratory. Results will not be provided to investigative sites.
Anti-retatrutide antibodies	
Anti-retatrutide antibodies neutralization	

10.2.1. Laboratory Samples to Be Obtained at the Time of a Systemic Hypersensitivity Event

Purpose of collecting samples after a systemic hypersensitivity event

The samples listed in this appendix are not collected for acute study participant management. The sponsor will use the laboratory tests results from these samples to characterize hypersensitivity events across the clinical development program.

When to collect samples after a systemic hypersensitivity event occurs

Collect the samples listed below if a systemic hypersensitivity event is suspected. The timing should be as designated in the table, assuming the participant has been stabilized.

Obtain follow-up predose samples at the next regularly scheduled laboratory sample collection (ideally prior to the next dose after the event) to assess post-event return-to-baseline values.

Timing	Laboratory Test ^a
Collect from 30 min to 4 hr after the start of the event. Note: The optimal collection time is from 1 to 2 hours after the start of event.	total tryptase
Collect only if not already collected on the same day as the event. Note: If collecting, collect up to 12 hr after the start of the event.	retatrutide antidrug antibodies (ADA)
	Retatrutide concentration

^a All samples for hypersensitivity testing will be assayed by Lilly-designated laboratory. Results will not be provided to the study site. If samples are not collected or are collected outside the specified time period, this will not be considered a protocol deviation.

What information to record

Record the date and time when the samples are collected.

Allowed additional testing for participant management

The investigator may perform additional tests locally, if clinically indicated, for acute study participant management.

10.2.2. Centrally Available Laboratory Tests Related to the Assessment of Nutritional Status and Micronutrient Levels

The following are additional tests available through the central laboratory that may be assessed per investigator's discretion if there is concern for poor nutrition or specific micronutrient deficiencies.

Category	Laboratory Test
Special tests	Vitamin A
	Vitamin B1 (Thiamine)
	Vitamin B9 (Folate)
	Vitamin B12
	Vitamin D (25-hydroxy vitamin D)
Hematology	Complete blood count ^a
Clinical chemistry	Clinical chemistry panel ^b
	Zinc
	Pre-Albumin

Category	Laboratory Test
Iron panel	iPTH
	TSH reflex to free T4
	Serum iron
	TIBC
	Transferrin saturation (%)
	Ferritin

Abbreviations: iPTH = intact parathyroid hormone; TIBC = total iron binding capacity; TSH = thyroid stimulating hormone.

^a Refer to Section 10.2 for contents of complete blood count panel.

^b Refer to Section 10.2 for contents of clinical chemistry panel.

Note: Tests will be assayed by a Lilly-designated laboratory, and results will be provided to the investigative sites where allowed (or appropriate).

What information to record

Record the date and time when the samples are collected.

Allowed additional testing for participant management

The investigator may perform additional tests locally if

- clinically indicated for acute study participant management, or
- a test is not available centrally.

10.3. Appendix 3: Adverse Events and Serious Adverse Events: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

The definitions and procedures detailed in this appendix are in accordance with ISO 14155.

Both the investigator and the sponsor will comply with all local medical device reporting requirements.

The detection and documentation procedures described in this protocol apply to all sponsor medical devices provided for use in the study. See Section 6.1.2 for the list of sponsor medical devices.

10.3.1. Definition of AE

- An AE is any untoward medical occurrence in a participant administered a pharmaceutical product and which does not necessarily have a causal relationship with the study intervention. An AE can therefore be any unfavourable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a medicinal (investigational) product, or investigational combination product, whether or not related to the medicinal (investigational) product or investigational combination product.
- An AE is any untoward medical occurrence, unintended disease or injury, or untoward clinical signs (including abnormal laboratory finding) in study participants, users, or other persons, whether or not related to the investigational device.

Events meeting the AE definition

- Any abnormal laboratory test results (hematology, clinical chemistry, or urinalysis) or other safety assessments, for example, ECG, radiological scans, and vital signs measurements, including those that worsen from baseline, considered clinically significant in the medical and scientific judgment of the investigator, that is, not related to progression of underlying disease.
- Exacerbation of a chronic or intermittent pre-existing condition including either an increase in frequency and/or intensity of the condition.
- New condition detected or diagnosed after study intervention administration even though it may have been present before the start of the study.
- Signs, symptoms, or the clinical sequelae of a suspected drug-drug interaction.
- Medication error, misuse, or abuse of IMP, including signs, symptoms, or clinical sequelae.
- Lack of efficacy or failure of expected pharmacological action per se will not be reported as an AE or SAE. Such instances will be captured in the efficacy assessments. However, the signs, symptoms, and/or clinical sequelae resulting from lack of efficacy will be reported as AE or SAE if they fulfill the definition of an AE or SAE.

Events NOT meeting the AE definition

- Any clinically significant abnormal laboratory findings or other abnormal safety assessments that are associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.
- The disease or disorder being studied or expected progression, signs, or symptoms of the disease or disorder being studied, unless more severe than expected for the participant's condition.
- Medical or surgical procedure, for example, endoscopy, appendectomy. The condition that leads to the procedure is the AE.
- Situations in which an untoward medical occurrence did not occur (social and/or convenience admission to a hospital).
- Anticipated day-to-day fluctuations of preexisting disease(s) or condition(s) present or detected at the start of the study that do not worsen.

10.3.2. Definition of SAE

An SAE is defined as any untoward medical occurrence that, at any dose, meets 1 or more of the criteria listed:

- Results in death
- Is life-threatening
 - The term *life-threatening* in the definition of *serious* refers to an event in which the participant was at risk of death at the time of the event. It does not refer to an event, which hypothetically might have caused death, if it were more severe.
- Requires inpatient hospitalization or prolongation of existing hospitalization
 - In general, hospitalization signifies that the participant has been admitted to hospital or emergency ward (usually involving at least an overnight stay) for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AEs. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether hospitalization occurred or was necessary, the AE should be considered serious.
 - Hospitalization for elective treatment of a preexisting condition that did not worsen from baseline is not considered an AE.
- Results in persistent disability/incapacity
 - The term disability means a substantial disruption of a person's ability to conduct normal life functions.
 - This definition is not intended to include experiences of relatively minor medical significance, such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma, for example, sprained ankle, which may interfere with or prevent everyday life functions but do not constitute a substantial disruption.

- Is a congenital anomaly/birth defect
 - Abnormal pregnancy outcomes, for example, spontaneous abortion, fetal death, stillbirth, congenital anomalies, and ectopic pregnancy, are considered SAEs.
- Other situations:
 - Medical or scientific judgment should be exercised by the investigator in deciding whether SAE reporting is appropriate in other situations such as important medical events that may not be immediately life-threatening or result in death or hospitalization but may jeopardize the participant or may require medical or surgical intervention to prevent one of the other outcomes listed in the above definition. These events should usually be considered serious.
 - Examples of such events include invasive or malignant cancers, intensive treatment in an emergency room or at home for allergic bronchospasm, blood dyscrasias or convulsions that do not result in hospitalization, or development of drug dependency or drug abuse.
- Resulted in medical or surgical intervention to prevent life-threatening illness or injury or permanent impairment to a body structure or a body function.

10.3.3. Definition of Product Complaints

Product complaint

A PC is any written, electronic, or oral communication that alleges deficiencies related to the identity, quality, durability, reliability, safety, effectiveness or performance of a study intervention. When the ability to use the study intervention safely is impacted, the following are also PCs:

- deficiencies in labeling information, and
- use errors for device or drug-device combination products due to ergonomic design elements of the product.

PCs related to study interventions used in clinical trials are collected to ensure the safety of participants, monitor quality, and to facilitate process and product improvements.

Investigators will instruct participants to contact the site as soon as possible if he or she has a PC or problem with the study intervention so that the situation can be assessed.

An event may meet the definition of both a PC and an AE/SAE. In such cases, it should be reported as both a PC and as an AE/SAE.

10.3.4. Recording and Follow-Up of AE and/or SAE and Product Complaints

AE, SAE, and PC recording

When an AE/SAE/PC occurs, it is the responsibility of the investigator to review all documentation (for example, hospital progress notes, laboratory reports, and diagnostics reports) related to the event.

The investigator will then record all relevant AE/SAE/PC information in the participant's medical records, in accordance with the investigator's normal clinical practice. AE/SAE

information is reported on the appropriate CRF page and PC information is reported on the Product Complaint Form.

Note: An event may meet the definition of both a PC and an AE/SAE. In such cases, it should be reported as both a PC and as an AE/SAE.

It is **not** acceptable for the investigator to send photocopies of the participant's medical records to the sponsor or designee in lieu of completion of the CRF page for AE/SAE and the Product Complaint Form for PCs.

There may be instances when copies of medical records for certain cases are requested by the sponsor or designee. In this case, all participant identifiers, with the exception of the participant number, will be redacted on the copies of the medical records before submission to the sponsor or designee.

The investigator will attempt to establish a diagnosis of the event based on signs, symptoms, and/or other clinical information. Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the AE/SAE.

Assessment of intensity

The investigator will make an assessment of intensity for each AE and SAE reported during the study and assign it to one of the following categories:

- Mild: A type of AE that is usually transient and may require only minimal treatment or therapeutic intervention. The event does not generally interfere with usual activities of daily living.
- Moderate: A type of AE that is usually alleviated with additional specific therapeutic intervention. The event interferes with usual activities of daily living, causing discomfort but poses no significant or permanent risk of harm to the research participant.
- Severe: A type of AE that interrupts usual activities of daily living, or significantly affects clinical status, or may require intensive therapeutic intervention. An AE that is assessed as severe should not be confused with an SAE. Severe is a category utilized for rating the intensity of an event, and both AEs and SAEs can be assessed as severe.

An event is defined as “serious” when it meets at least 1 of the predefined outcomes as described in the definition of an SAE, NOT when it is rated as severe.

Assessment of causality

The investigator is obligated to assess the relationship between study intervention and each occurrence of each AE/SAE. The investigator will use clinical judgment to determine the relationship.

A “reasonable possibility” of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out.

Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study intervention administration will be considered and investigated.

The investigator will also consult the IB in their assessment.

The investigator **must** review and provide an assessment of causality for each AE/SAE and document this in the medical notes.

There may be situations in which an SAE has occurred and the investigator has minimal information to include in the initial report to the sponsor or designee. However, it is very important that the investigator always make an assessment of causality for every event before the initial transmission of the SAE data to the sponsor or designee.

The investigator may change their opinion of causality in light of follow-up information and send an SAE follow-up report with the updated causality assessment.

The causality assessment is one of the criteria used when determining regulatory reporting requirements.

Follow-up of AEs and SAEs

- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by the sponsor or designee to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other healthcare professionals.
- If a participant dies during participation in the study or during a recognized follow-up period, the investigator will provide the sponsor or designee with a copy of any postmortem findings including histopathology.

10.3.5. Reporting of SAEs

SAE reporting via an electronic data collection tool

The primary mechanism for reporting an SAE will be the electronic data collection tool.

If the electronic system is unavailable, then the site will use the SAE paper form (see next section) in order to report the event within 24 hours.

The site will enter the SAE data into the electronic system as soon as it becomes available.

After the study is completed at a given site, the electronic data collection tool will be taken off-line to prevent the entry of new data or changes to existing data.

If a site receives a report of a new SAE from a study participant or receives updated data on a previously reported SAE after the electronic data collection tool has been taken off-line, then the site can report this information on an SAE paper form (see next section) or to the SAE coordinator by telephone.

Contacts for SAE reporting can be found in in study training materials.

SAE reporting via paper form

Facsimile transmission of the SAE paper form is the preferred method to transmit this information to the SAE coordinator.

Initial notification via telephone does not replace the need for the investigator to complete and sign the SAE CRF pages within the designated reporting time frames.

Contacts for SAE reporting can be found in study training materials.

10.3.6. Regulatory Reporting Requirements**SAE regulatory reporting**

Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities toward the safety of participants and the safety of a study intervention under clinical investigation are met.

The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The sponsor will evaluate the reported SAEs, including confirmation of relatedness and assessment of expectedness. The sponsor has processes for safety reports for identification, recording, and expedited reporting of SUSARs according to local regulatory requirements. The sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRBs/IECs, and investigators.

An investigator who receives an investigator safety report describing an SAE or other specific safety information (for example, summary or listing of SAEs) from the sponsor will review and then file it along with the IB and will notify the IRB/IEC, if appropriate according to local requirements.

10.4. Appendix 4: Contraceptive and Barrier Guidance

10.4.1. Definitions

Word/Phrase	Definition
Women of childbearing potential (WOCBP)	Adult females are considered WOCBP unless they are WNOCBP.
Women not of childbearing potential (WNOCBP)	Females are considered WNOCBP if they • have a congenital anomaly such as Müllerian agenesis resulting in confirmed infertility • are infertile due to surgical sterilization, or • are postmenopausal. Acceptable surgical sterilization include hysterectomy, bilateral salpingo-oophorectomy, bilateral salpingectomy, or bilateral oophorectomy.
Postmenopausal state	The postmenopausal state is defined as a woman: • at any age at least 6 weeks post-surgical bilateral oophorectomy with or without hysterectomy, confirmed by operative note; or • aged at least 40 years and up to 55 years with an intact uterus, not on hormone therapy^a, who has had cessation of menses for at least 12 consecutive months without an alternative medical cause, AND with a follicle-stimulating hormone ≥ 40 mIU/mL; or • 55 years or older not on hormone therapy, who has had at least 12 months of spontaneous amenorrhea, or • aged at least 55 years with a diagnosis of menopause prior to starting hormone replacement therapy. ^a Women should not be taking medications during amenorrhea such as oral contraceptives, hormonal replacement therapy (HRT), gonadotropin-releasing hormone, anti-estrogens, selective estrogen receptor modulators (SERMs), or chemotherapy that could induce transient amenorrhea. Females on HRT and those whose menopausal status cannot be confirmed will be required to comply with protocol contraception requirements if they wish to continue their HRT during the study. Otherwise, they must discontinue HRT to allow confirmation of postmenopausal status before study enrollment.

10.4.2. Contraception Guidance

10.4.2.1. Females

WOCBP who are completely abstinent as their preferred and usual lifestyle, or in a same-sex relationship as their preferred and usual lifestyle:

Must...	Must not...
agree to either remain abstinent or stay in a same-sex relationship without sexual relationships with males, and not plan a pregnancy during the study	• use periodic abstinence methods ○ calendar ○ ovulation ○ symptothermal, or ○ post-ovulation • declare abstinence just for the duration of a trial, or • use the withdrawal method

WOCBP who are NOT completely abstinent as their preferred and usual lifestyle, or NOT in a same-sex relationship as their preferred and usual lifestyle, must do the following:

Topic	Condition
Contraception	Agree to use 2 methods of effective contraception, where at least 1 method must be highly effective. These methods of contraception must be used during the study and for at least 60 days after the last dose of the study intervention.

10.4.2.2. Males

The table below describes contraception guidance for all men.

Participant Population	Contraception Guidance
All men...	should refrain from sperm donation for the duration of the study and for 4 months after the last dose of the study intervention.
Men with partner(s) who are WOCBP	must • remain abstinent (if this is their preferred and usual lifestyle), or • use condoms and at least 1 additional effective method of contraception for the duration of the study and for 4 months after the last dose of the study intervention.
Men with partner(s) who are pregnant	must • remain abstinent (if this is their preferred and usual lifestyle), or • use condoms for the duration of the study and for 4 months after the last dose of the study intervention.

Men in exclusively same-sex relationships, as their preferred and usual lifestyle...	are not required to use contraception.
--	--

10.4.2.3. Examples of Different Methods of Contraception

Methods	Examples
Highly effective contraception (less than 1% failure rate)	• fallopian tubal sterilization methods other than bilateral salpingectomy (laparoscopic bipolar electrocoagulation, plastic ring application on the uterine tubes, fallopian tube ligation, hysteroscopic sterilization). Note: Bilateral salpingectomy is indicative of permanent sterilization. Please see the WNOCBP definition above. • combination oral contraceptive pill • progestin-only contraceptive pill (mini-pill) • contraceptive patch (only women <198 pounds or 90 kg) • total abstinence • vasectomy (for men in clinical trials and for female partner if only sexual partner) • fallopian tube implants (if confirmed by hysterosalpingogram) • combined contraceptive vaginal ring, or • intrauterine devices
Effective contraception	• male or female condoms with spermicide • diaphragms with spermicide or cervical sponges • barrier method with use of a spermicide ○ condom with spermicide ○ diaphragm with spermicide, or ○ female condom with spermicide Note: Male and female condoms should not be used in combination.
Ineffective methods of contraception whether used alone or in any combination	• spermicide alone • periodic abstinence • fertility awareness (calendar method, temperature method, cervical mucus, or symptothermal) • withdrawal • postcoital douche, or • lactational amenorrhea

10.5. Appendix 5: Genetics

Use/Analysis of DNA

Genetic variation may impact a participant's response to study intervention, susceptibility to, and severity and progression of disease. Variable response to study intervention may be due to genetic determinants that impact drug absorption, distribution, metabolism, and excretion; mechanism of action of the drug; disease etiology; and/or molecular subtype of the disease being treated. Therefore, where local regulations and IRB/IEC allow, a blood sample will be collected for DNA analysis from consenting participants.

DNA samples may be used for research related to study intervention or diabetes mellitus, obesity, and related clinical traits, complications, or comorbidities, including nonalcoholic steatohepatitis and CKD. They may also be used to develop tests/assays including diagnostic tests related to study intervention and/or interventions related to this drug class and/or diabetes mellitus, obesity, and related clinical traits, complications, or comorbidities, including nonalcoholic steatohepatitis and CKD. Genetic research may consist of the analysis of 1 or more candidate genes or the analysis of genetic markers throughout the entire genome (as appropriate).

The samples may be analyzed as part of a multi-study assessment of genetic factors involved in the response to study intervention or study interventions related to this class to understand diabetes mellitus, obesity, and related clinical traits, complications, or comorbidities, including nonalcoholic steatohepatitis and CKD or related conditions.

The results of genetic analyses may be reported in the CSR or in a separate study summary.

The sponsor will store the DNA samples in a secure storage space with adequate measures to protect confidentiality.

The samples will be retained while research on study intervention or study interventions related to this class or indications continues but no longer than 15 years or other period as per local requirements.

10.6. Appendix 6: Liver Safety: Suggested Actions and Follow-up Assessments

Hepatic Evaluation Testing

See Section 8.2.6 above for guidance on appropriate test selection.

The Lilly-designated central laboratory should complete the analysis of all selected testing except for testing listed in the investigator-designated local laboratory table. The central laboratory will report results if a validated test or calculation is available. Note that testing for some analytes may not be available from the central lab or local lab in certain regions due to local availability of assay or regulatory requirements. If testing is not available, this will not be considered a protocol deviation.

Local testing may be performed *in addition to central testing* when necessary for immediate participant management. If local testing is performed in lieu of central testing, the results should be recorded in the appropriate local lab CRF. For example, if immediate participant management circumstances preclude collection of central lab samples – e.g, emergency room visit or hospitalization. The local laboratory must be qualified in accordance with applicable local regulations. If testing is not available in certain regions based on local requirements, consult with Lilly-designated medical monitor.

Tests assayed by Lilly-designated central laboratory	
Hepatic Hematology Panel	Hepatitis A virus (HAV) testing:
Hemoglobin	HAV total antibody ^a
Hematocrit	HAV IgM antibody
Erythrocytes (RBCs - red blood cells)	Hepatitis B virus (HBV) testing:
Leukocytes (WBCs - white blood cells)	Hepatitis B surface antigen (HBsAg)
Differential:	Hepatitis B surface antibody (anti-HBs)
Neutrophils	Hepatitis B core total antibody (anti-HBc)
Lymphocytes	Hepatitis B core IgM antibody
Monocytes	HBV DNA ^b
Basophils	Hepatitis C virus (HCV) testing:
Eosinophils	HCV total antibody ^a
Platelets	HCV RNA ^b
Cell morphology (RBC and WBC)	Hepatitis E virus (HEV) testing:
Hepatic Clinical Chemistry Panel	HEV IgG antibody
Total bilirubin	HEV IgM antibody
Direct bilirubin	HEV RNA ^b
Alkaline phosphatase (ALP)	Anti-nuclear antibody (ANA)
Alanine aminotransferase (ALT)	Anti-smooth muscle antibody (ASMA) or anti-actin antibody
Aspartate aminotransferase (AST)	Immunoglobulin IgA (quantitative)
Gamma-glutamyl transferase (GGT)	Immunoglobulin IgG (quantitative)
Creatine kinase (CK)	Immunoglobulin IgM (quantitative)
Hepatic Coagulation Panel	
Prothrombin time, INR (PT-INR)	
Urine Chemistry	
Drug screen	
Haptoglobin	

Tests assayed by investigator-designated local laboratory	
Acetaminophen	Cytomegalovirus (CMV) testing:
Acetaminophen protein adducts^c	CMV antibody
Alkaline phosphatase isoenzymes	CMV DNA ^b
Ceruloplasmin	Herpes simplex virus (HSV) testing:
Copper	HSV (Type 1 and 2) antibody
Ethyl alcohol (ethanol, EtOH)	HSV (Type 1 and 2) DNA ^b
Phosphatidylethanol (PEth)	Liver kidney microsomal type 1 (LKM-1) antibody
Urine Chemistry	Microbiology Culture:
Ethyl glucuronide (EtG)	Blood
Epstein-Barr virus (EBV) testing:	Urine
EBV antibody	Hepatitis D virus (HDV) testing:
EBV DNA ^b	HDV total antibody ^a
	HDV IgM antibody
	HDV RNA ^b

Abbreviations: IgA = immunoglobulin A, IgG = immunoglobulin G; IgM = immunoglobulin M; INR = international normalized ratio; RNA = ribonucleic acid.

^a If lab does not offer total antibody testing, IgG and/or IgM are acceptable substitutes.

^b Reflex/confirmation dependent on regulatory requirements, testing availability, or both.

^c If HDV testing is not available, HBV testing may be sufficient. If HBV testing is positive, consult with the Lilly-designated medical monitor.

10.7. Appendix 7: Medical Device Adverse Events (AEs), Adverse Device Effects (ADEs), Serious Adverse Events (SAEs) and Device Deficiencies: Definition and Procedures for Recording, Evaluating, Follow-up, and Reporting

Refer to Section [10.3](#) for definitions and procedures for recording, evaluating, follow-up, and reporting of all events.

10.8. Appendix 8: World Health Organization Classification of Diabetes and Diagnostic Criteria

Type 1 diabetes

T1D is judged to be present when the classical symptoms of diabetes (thirst, polyuria, wasting and stupor, or coma) are associated with readily detectable concentrations of glucose and ketone bodies in the blood and urine. Insulin treatment is necessary not only to control hyperglycemia but also to prevent spontaneous ketosis and death.

Type 2 diabetes

T2D, although often asymptomatic, may also present with classical hyperglycemic symptoms (thirst, polyuria, and weight loss), but despite hyperglycemia, ketone bodies are present in only low concentrations in the blood and urine. Coma is rare in T2D but may result from extreme hyperglycemia and hyperosmolarity; lactic acidosis or ketoacidosis can also occur in fulminating illness (for example, severe infection or mesenteric artery thrombosis) due to an acute increase in insulin requirements, but spontaneous ketosis does not occur. Some patients with T2D later progress to a state of absolute insulin deficiency (Alberti and Zimmet 1998).

10.9. Appendix 9: Prohibited Medications or Medications with Special Use Restrictions

10.9.1. Excluded/Prohibited or Restricted Use Medications

Medications within the following categories are strictly prohibited during the study. The lists below provide examples of each category of medication, but the examples are not exhaustive. As stated in Section 6.9.2, participants who have to initiate certain prohibited concomitant medications during this study will be discontinued from the study intervention (Section 7.1.1).

10.9.1.1. Antihyperglycemic Medications

The following antihyperglycemic medications are prohibited within 90 days of Visit 1 or between Visit 1 and Visit 3, as well as at any time during the study.

- alogliptin
- dulaglutide
- exenatide
- linagliptin
- liraglutide
- pramlintide
- saxagliptin
- sitagliptin
- semaglutide
- tirzepatide, and
- other incretin-based therapy.

10.9.1.2. Weight Reduction Medications

Weight reduction medications are prohibited within 90 days of Visit 1, or between Visit 1 and Visit 3, as well as at any time during the study.

- ingested material that transiently occupies space in the stomach, for example, Plenity[®]
- liraglutide
- lorcaserin
- mazindol
- naltrexone/bupropion
- orlistat
- over-the-counter medications, for example, alli[®]
- phentermine
- phentermine/topiramate
- phenylpropanolamine
- semaglutide
- sibutramine, and
- other incretin-based therapies.

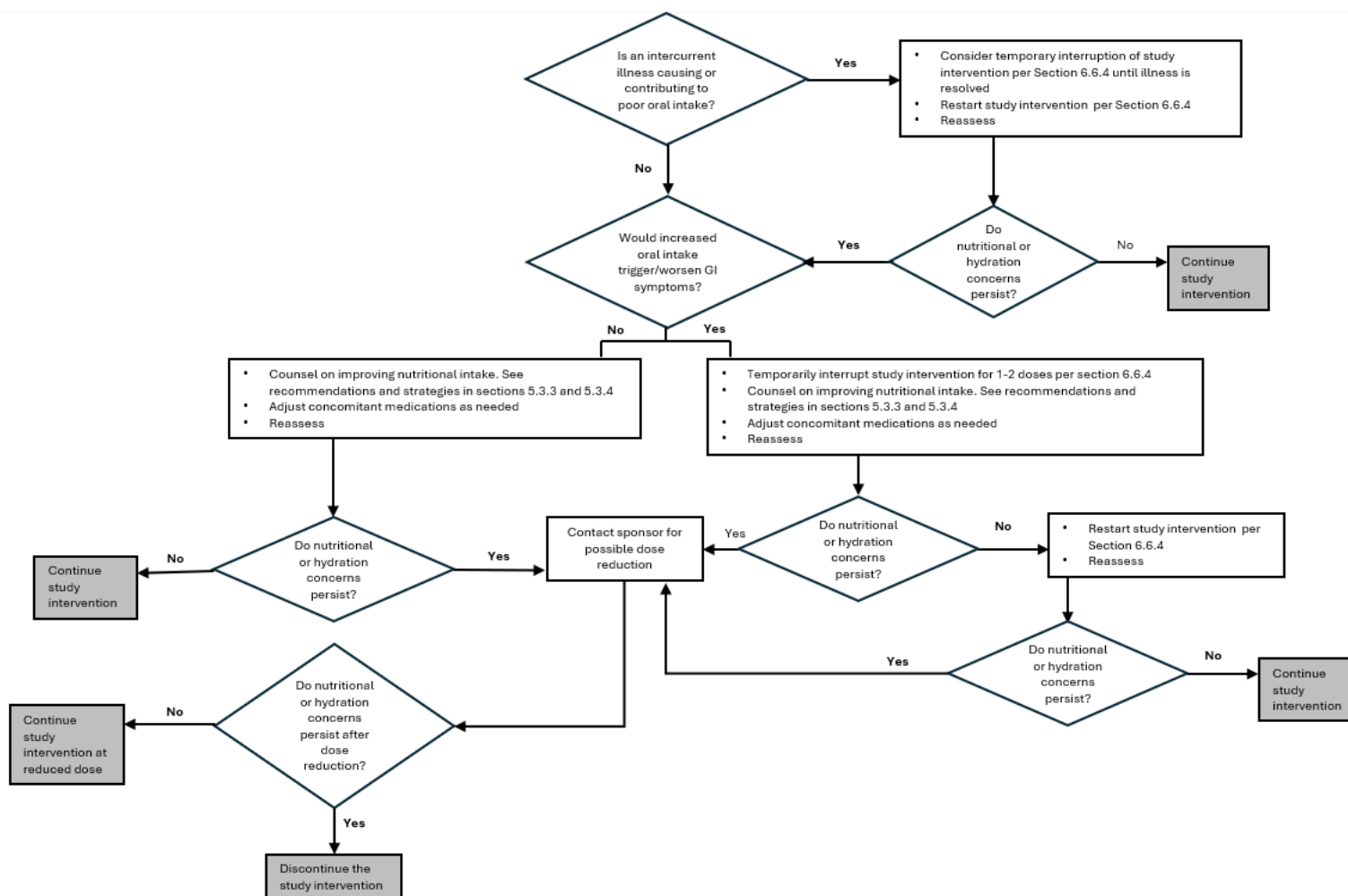
10.10. Appendix 10: Considerations for Concomitant Medications

The table below provides considerations for adjusting concomitant medications. See Section 6.9.1 for general considerations on documenting changes in concomitant medications.

Concomitant medication	Consider adjusting or discontinuing if ...
antihypertensives (including diuretics)	the participant is at risk of low blood pressure (for example, in the setting of ongoing weight reduction or signs/symptoms of dehydration, if blood pressure is within or below normal range)
diuretics and laxatives	the participant is at risk of dehydration (for example, based on decreased fluid intake, increased fluid losses, or clinical signs or symptoms of dehydration)
stimulant and osmotic laxatives	the participant has hyperuricemia or electrolyte abnormalities, such as hypernatremia or hypokalemia
anti-emetics, such as domperidone and those with serotonin receptor agonism, such as ondansetron, granisetron, palonosetron, and dolasetron	the participant has electrolyte abnormalities, such as hypokalemia or hypomagnesemia
medications with weight-based dosing, such as thyroid medications	the participant is experiencing weight changes

10.11. Appendix 11: Management of Poor Nutrition and Dehydration

Follow the guidance in the flow chart and see Section 6.6.2 if there are concerns for poor nutrition or dehydration. Consider consultation with a registered dietitian (or local equivalent) if not already done. Note that this flow chart does not preclude taking any other necessary steps for participant safety per investigator's discretion.



10.12. Appendix 12: Standardized Protocols for the Measurement of Height, Weight, Waist Circumference, and Vital Signs

The following information has been adapted from standardized physical measurement protocols for the WHO's STEPwise approach to Surveillance (STEPS) (WHO 2017).

Measuring height

Step 1. Ask the participant to remove their footwear and any headgear (light headgear worn for religious reasons can remain, but this should be worn by the participant at every clinic visit when their height is measured).

Step 2. Ask the participant to stand on the calibrated height-measuring board (stadiometer) or against a wall with their feet together and their knees straight with their heels against the backboard, the stadiometer, or the wall.

Step 3. Ask the participant to look straight ahead without tilting their head up.

Step 4. Ask the participant to breathe in and stand tall. Measure and record the participant's height in centimeters to 1 decimal place.

Measuring weight

- Body weight measurements should be done in a consistent manner using a calibrated electronic scale capable of measuring weight in kilograms to 1 decimal place.

All weights for a given participant should be measured using the same scale, whenever possible, at approximately the same time in the morning after evacuation of bladder contents.

Step 1. Ask the participant to empty their pockets, remove their footwear, outerwear (coat, jacket, etc.), and any headgear (light headgear worn for religious reasons can remain, but this should be worn by the participant at every visit when weight is measured).

Step 2. Make sure the scale is placed on a firm, flat, even surface (not on carpet, on a sloping surface, or a rough, uneven surface).

Step 3. Ask the participant to step onto the scale with 1 foot on each side of the scale.

Step 4. Ask the participant to stand still with arms by sides and then record weight in kilograms to the nearest one-tenth kilogram.

Measuring waist circumference

- Waist circumference should be measured in the horizontal plane and at the midpoint between the lower margin of the last palpable rib and the top of the iliac crest.
- Measurements should be taken at the end of a normal expiration using a non-stretchable measuring tape. The tape should lie flat against the skin without compressing the soft tissue.

- The waist circumference should be measured twice, rounded to the nearest 0.5 cm. The measuring tape should be removed between the 2 measurements. Both measurements will be recorded in the CRF. If the difference between the 2 measurements exceeds 1 cm, this set of measurements should be discarded and the 2 measurements repeated.

Step 1. Ask the participant to wear little clothing (if available, patient gowns could also be used).

Step 2. Ask the participant to stand with their feet close together, arms at their side, body weight evenly distributed.

Step 3. Ask the participant to relax and measure the participant's waist circumference.

Vital sign measurements (blood pressure and heart rate)

- Vital sign measurements (BP and HR, measured by pulse) should be taken before obtaining an ECG tracing and before collection of blood samples for laboratory testing and PK sampling.
- The participant should sit quietly for 5 minutes before vital signs measurements are taken.
- For each parameter, 3 measurements will be taken using the same arm, preferably the nondominant arm.
- The recordings should be taken at least 1 minute apart. Each measurement of sitting pulse and BP needs to be recorded in the CRF.
- BP must be taken with an automated BP instrument.
- If BP and pulse measurements are taken separately, pulse should be taken prior to BP.

Note: In the event pulse measurement cannot be taken via an automated BP instrument, the preferred location for measurement of pulse is the radial artery.

Calculating BMI

Round to 1 decimal place. For example, a BMI of 24.6 kg/m² should **not** be rounded to 25.0 kg/m².

10.13. Appendix 13: Country-specific Requirements

Country-specific requirements for non-EU countries will be described in a separate protocol addendum, if any.

10.14. Appendix 14: Provisions for Changes in Study Conduct During Exceptional Circumstances

Implementation of this appendix

The changes to procedures described in this appendix are temporary measures intended to be used only during specific time periods as directed by the sponsor in partnership with the investigator.

Exceptional circumstances

Exceptional circumstances are rare events that may cause disruptions to the conduct of the study. Examples include pandemics or natural disasters. These disruptions may limit the ability of the investigators, participants, or both to attend on-site visits or to conduct planned study procedures.

Implementing changes under exceptional circumstances

In an exceptional circumstance, after receiving the sponsor's written approval, sites may implement changes if permitted by local regulations.

After approval by local Ethical Review Boards, regulatory bodies and any other relevant local authorities, implementation of these exceptional circumstance changes will not typically require additional notification to these groups, unless they have specific requirements in which notification is required (for example, upon implementation and suspension of changes). All approvals and notifications must be retained in the study records.

If the sponsor grants written approval for changes in study conduct, the sponsor will also provide additional written guidance, if needed.

Considerations for making a change

The prevailing consideration for making a change is ensuring the safety of study participants. Additional important considerations for making a change are compliance with GCP, enabling participants to continue safely in the study and maintaining the integrity of the study.

Informed consent

Additional consent from the participant will be obtained, if required, for:

- participation in remote visits, as defined in Section "Remote Visits"
- a change in the method of study intervention administration
- dispensation of additional study intervention during an extended treatment period
- alternate delivery of study intervention and ancillary supplies, and
- provision of their personal or medical information required prior to implementation of these activities.

Changes in study conduct during exceptional circumstances

Changes in study conduct not described in this appendix, or not consistent with applicable local regulations, are not allowed.

The following changes in study conduct will not be considered protocol deviations.

Remote visits***Types of remote visits***

Telemedicine - Telephone or technology-assisted virtual visits, or both, are acceptable to complete appropriate assessments. Assessments to be completed in this manner include, but are not limited to,

- collection of AEs
- administer C-SSRS Since Last Assessed
- review of SMBG values, and
- concomitant medications.

Mobile healthcare - Healthcare visits may be performed by a mobile healthcare provider at locations other than the study site when participants cannot travel to the site due to an exceptional circumstance if written approval is provided by the sponsor. Procedures performed at such visits include, but are not limited to,

- concomitant medications
- vital signs (BP and HR)
- body weight
- collection of blood samples
- physical assessments, and
- collection of health information.

Data capture

In source documents and the CRF, the study site should capture the visit method, with a specific explanation for any data missing because of missed in-person site visits.

Safety reporting

Regardless of the type of remote visits implemented, the protocol requirements regarding the reporting of AEs, SAEs, and PCs remain unchanged.

Return to on-site visits

Every effort should be made to enable participants to return to on-site visits as soon as reasonably possible, while ensuring the safety of both the participants and the site staff.

Local laboratory testing option

Local laboratory testing may be conducted in lieu of central laboratory testing. However, central laboratory testing must be retained for: Visit 1, Visit 3, Visit 9, Visit 11, ED, and V801. The local laboratory must be qualified in accordance with applicable local regulations.

Study intervention and ancillary supplies (including participant diaries)

When a participant is unable to go to the site to receive study supplies during normal on-site visits, the site should work with the sponsor to determine appropriate actions. These actions may include

- asking the participant to go to the site and receive study supplies from site staff without completion of a full study visit

- asking the participant's designee to go to the site and receive study supplies on a participant's behalf, and
- arranging delivery of study supplies.

These requirements must be met before action is taken:

- Alternate delivery of study intervention should be performed in a manner that does not compromise treatment blinding and ensures product integrity. The existing protocol requirements for product accountability remain unchanged, including verification of participant's receipt of study supplies.
- When delivering supplies to a location other than the study site (for example, participant's home), the investigator, sponsor, or both should ensure oversight of the shipping process to ensure accountability and product quality (that is, storage conditions maintained and intact packaging upon receipt).
- Instructions may be provided to the participant or designee on the final disposition of any unused or completed study supplies.

Screening period guidance

To ensure safety of study participants, laboratory values and other eligibility assessments taken at screening visit are valid for a maximum of 45 days.

The following rules will be applied for active, nonrandomized participants whose participation in the study must be interrupted due to exceptional circumstances.

- If screening is interrupted for less than 45 days from Visit 1 to randomization visit, the participant will proceed to the next study visit per the usual SoA, provided that randomization visit must be conducted within 45 days from Visit 1.
 - The site should conduct the next visit if the participant's eligibility criteria are confirmed, and the site should document the reason for delay.
 - The site should also reconfirm the impacted participant's consent and document this confirmation in the source documentation.
- If screening is interrupted for more than 45 days from screening visits to randomization visit, the participant must be discontinued because of screening interruption due to an exceptional circumstance. This is documented as a screen failure in the CRF. The participant can reconsent and be rescreened as a new participant. The screening procedures per the usual SoA should be followed, starting at Visit 1 to ensure participant eligibility by randomization visit.

Adjustments to visit windows

Whenever possible and safe to do so, as determined by the investigator's discretion, participants should complete the usual SoA. To maximize the possibility that these visits can be conducted as on-site visits, the windows for visits may be adjusted, upon further guidance from the sponsor. This minimizes missing data and preserves the intended conduct of the study.

This table describes the allowed adjustments to visit windows.

Visit Number	Tolerance
Visit 1 through 3	No adjustments to visit window described in the SoA
Visit 4 through 7	Within 6 days before or after the intended date per the SoA
Visits 8 through 11	Within 10 days before or after the intended date per the SoA
Visit 801	Within 10 days before the intended date or up to 10 days after the intended date

For participants whose visits have extended windows, additional study intervention may need to be provided to avoid interruption and maintain overall integrity of the study.

Documentation

Changes to study conduct will be documented

Sites will identify and document the details of how participants, visit types, and conducted activities were affected by exceptional circumstances. Dispensing/shipment records of study intervention and relevant communications, including delegation, should be filed with site study records.

Source documents at alternate locations

Source documents generated at a location other than the study site should be part of the investigator's source documentation and should be transferred to the site in a secure and timely manner.

10.15. Appendix 15: Abbreviations and Definitions

Term	Definition
abuse	use of a study intervention for recreational purposes or to maintain an addiction or dependence
ADA	antidrug antibodies
AE	adverse event
ALP	alkaline phosphatase
ALT	alanine aminotransferase
ANCOVA	analysis of covariance
AST	aspartate aminotransferase
BG	blood glucose
blinding/masking	A single-blind study is one in which the investigator and/or the investigator's staff are aware of the treatment but the participant is not, or vice versa, or when the sponsor is aware of the treatment but the investigator and/the investigator's staff and the participant are not. A double-blind study is one in which neither the participant nor any of the investigator or sponsor staff who are involved in the treatment or clinical evaluation of the subjects are aware of the treatment received.
BMI	body mass index
BP	blood pressure
CEC	Clinical Endpoint Committee
CI	confidence interval
CK	creatine kinase
CKD	chronic kidney disease
complaint	A complaint is any written, electronic, or oral communication that alleges deficiencies related to the identity, quality, purity, durability, reliability, safety or effectiveness, or performance of a drug or drug delivery system.
compliance	adherence to all study-related, good clinical practice (GCP), and applicable regulatory requirements
CONSORT	Consolidated Standards of Reporting Trials
CRF	case report form; a printed, optical, or electronic document designed to record all of the protocol-required information to be reported to the sponsor for each trial participant

Term	Definition
C-SSRS	Columbia-Suicide Severity Rating Scale
CSR	clinical study report
CT	computerized tomography
CV	cardiovascular
D. Bil	direct bilirubin
DNA	deoxyribonucleic acid
DPP-4	dipeptidyl peptidase IV
DTSQ	Diabetes Treatment Satisfaction Questionnaire
eCOA	electronic Clinical Outcome Assessment
ECG	electrocardiogram
ED	early discontinuation
eGFR	estimated glomerular filtration rate
enroll	The act of assigning a participant to a treatment. Participants who are enrolled in the study are those who have been assigned to a treatment.
enter	Participants entered into a study are those who sign the informed consent form directly or through their legally acceptable representatives.
GCGR	glucagon receptor
GCP	Good Clinical Practice
GGT	gamma-glutamyl transferase
GI	gastrointestinal
GIP	glucose-dependent insulintropic polypeptide
GIPR	gastric inhibitory polypeptide receptor
GLP-1	glucagon-like peptide-1
GLP-1R	glucagon-like peptide-1 receptor
GLP-1 RA	glucagon-like peptide-1 receptor agonist
HbA1c	hemoglobin A1c
HBV	hepatitis B virus

Term	Definition
HCV	hepatitis C virus
HDL	high-density lipoprotein
HR	heart rate
IB	Investigator's Brochure
IEC	Independent Ethics Committee
ICE	intercurrent event
ICF	informed consent form
ICH	International Council for Harmonisation
IMP	Investigational Medicinal Product (see also "investigational product") A medicinal product which is being tested or used as a reference, including as a placebo, in a clinical trial.
informed consent	A process by which a participant voluntarily confirms their willingness to participate in a particular study, after having been informed of all aspects of the study that are relevant to the participant's decision to participate. Informed consent is documented by means of a written, signed, and dated informed consent form.
INR	international normalized ratio
interim analysis	An interim analysis is an analysis of clinical study data, separated into treatment groups, that is conducted before the final reporting database is created/locked.
investigational product	A pharmaceutical form of an active ingredient or placebo being tested or used as a reference in a clinical trial, including products already on the market when used or assembled (formulated or packaged) in a way different from the authorized form, or marketed products used for an unauthorized indication, or marketed products used to gain further information about the authorized form. See also "IMP."
IRB	Institutional Review Board
ISR	injection site reaction
IWQOL-Lite-CT	Impact of Weight on Quality of Life-Lite-Clinical Trials
IWRS	interactive web-response system
LDL	low-density lipoprotein
LS	least squares
MASLD	metabolic dysfunction-associated steatotic liver disease

Term	Definition
medication error	Errors in the prescribing, dispensing, or administration of a study intervention, regardless of whether or not the medication is administered to the participant or the error leads to an AE. Medication error generally involve a failure to uphold one or more of the five “rights” of medication use: 1. the right participant 2. the right drug 3. the right dose 4. the right route, 5. at the right time.
MEN2	multiple endocrine neoplasia type 2
misuse	Use of a study intervention for self-treatment that either is inconsistent with the prescribed dosing regimen or indication, or both, or is obtained without a prescription
MMRM	mixed-model for repeated measures
MRI	magnetic resonance imaging
MTC	medullary thyroid cancer
participant	equivalent to CDISC term “subject”: an individual who participates in a clinical trial, either as recipient of an investigational medicinal product or as a control
PC	product complaint
PFP	prefilled pen
PHQ-9	Patient Health Questionnaire-9
PK/PD	pharmacokinetics/pharmacodynamics
PRO	Patient-Reported Outcome
PT	prothrombin time
QW	once weekly
RNA	ribonucleic acid
SAE	serious adverse event
SAP	statistical analysis plan
SC	subcutaneous
screen	The act of determining if an individual meets minimum requirements to become part of a pool of potential candidates for participation in a clinical study.

Term	Definition
SD	standard deviation
SF-36v2	Short-Form 36 Version 2
SGLT2	sodium-glucose cotransporter-2
SIB	suicidal ideation and behavior
SMBG	self-monitoring of blood glucose
SoA	Schedule of Activities
T1D	type 1 diabetes
T2D	type 2 diabetes
TBL	total bilirubin level
TE	treatment emergent
TEAE	Treatment-emergent adverse event: An untoward medical occurrence that emerges during a defined treatment period, having been absent pretreatment, or worsens relative to the pretreatment state, and does not necessarily have to have a causal relationship with this treatment.
ULN	upper limit normal
WHO	World Health Organization
WNOCBP	women not of childbearing potential
WOCBP	women of childbearing potential

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